

Your Personalized Medicine Report

Physician's Guide



For informational purposes only, not for medical use

Table of Contents

Section / Drug	Page
1. Introduction	
Table of Contents	1
About Personalized Medicine	5
About This Report	7
How to Understand Your Results	10
Note to Doctor	15
2. Summary of Medication Insights	
Summary of Medication Insights	17
Coverage Statement	19
3. Drug-Specific Recommendations	
3.1 MUSCULO-SKELETAL SYSTEM DRUGS	
Allopurinol	20
Celecoxib	21
Flurbiprofen	23
Ibuprofen	25
Lornoxicam	28
Meloxicam	29
Pegloticase	30
Piroxicam	31
Succinylcholine	33
Tenoxicam	34
3.2 CARDIOVASCULAR SYSTEM DRUGS	
Atorvastatin	35
Fluvastatin	36
Hydralazine	38
Ibuprofen	26
Lovastatin	39
Mavacamten	40
Pitavastatin	41
Pravastatin	42
Procainamide	43
Rosuvastatin	44
Simvastatin	46

Amifampridine Phosphate	47
Methylene Blue	48
Toluidine Blue	49
3.4 ALIMENTARY TRACT AND METABOLISM DRUGS	
Dexlansoprazole	50
Dronabinol	51
Lansoprazole	52
Omeprazole	53
Pantoprazole	54
Sulfasalazine	55
3.5 NERVOUS SYSTEM DRUGS	
Amifampridine	56
Amitriptyline	57
Brivaracetam	58
Citalopram	59
Clobazam	60
Clomipramine	61
Desflurane	62
Doxepin	63
Enflurane	64
Escitalopram	65
Fosphenytoin	66
Halothane	68
Imipramine	69
Isoflurane	70
Methoxyflurane	71
Phenytoin	72
Sertraline	74
Sevoflurane	76
Trimipramine	77
3.6 RESPIRATORY SYSTEM DRUGS	
Flurbiprofen	23
Ivacaftor	78
3.7 ANTIINFECTIVES FOR SYSTEMIC USE	
Atazanavir	79
Dapsone	80

Isoniazid	84
Nitrofurantoin	85
Sulfamethoxazole / Trimethoprim	86
Voriconazole	87
3.8 GENITO URINARY SYSTEM AND SEX HORMONES	
Ibuprofen	26
3.9 BLOOD AND BLOOD FORMING ORGAN DRUGS	
Acenocoumarol	88
Avatrombopag	89
Clopidogrel	90
Phenprocoumon	92
Warfarin	93
3.10 DERMATOLOGICALS	
Dapsone	80
Flucytosine	83
Tacrolimus	95
3.11 ANTINEOPLASTIC AND IMMUNOMODULATING AGENTS	
Azathioprine	97
Capecitabine	99
Celecoxib	21
Fluorouracil	100
Mercaptopurine	101
Siponimod	103
Tacrolimus	96
Thioguanine	104
3.12 SENSORY ORGAN DRUGS	
Flurbiprofen	24
Piroxicam	31
3.13 VARIOUS DRUG CLASSES IN ATC	
Rasburicase	106
3.14 ANTIPARASITIC PRODUCTS, INSECTICIDES AND REPELLENTS	
Primaquine	107
Tafenoquine	108
4. Additional Information	
No Guideline Available	109
Other Evaluated Medications	111

5. Appendix	
Genes Requiring Specialized Testing	113
Genotype Summary	116
Disclaimer	118

About Personalized Medicine

What Is Personalized Medicine and Why Does It Matter?

Medicine has traditionally been prescribed based on population averages: a standard drug at a standard dose for everyone with the same diagnosis. But we are not all the same. Your genes, lifestyle, age, and other biological factors shape how your body responds to a medication. Personalized medicine, sometimes called precision medicine, is an approach that uses these individual differences to guide treatment choices.

One of the most important drivers of this variability is pharmacogenomics (PGx): the study of how your genetic makeup influences how your body processes and responds to drugs. Even small differences in specific genes can change how quickly you break down a drug, how much of it reaches your bloodstream, and how strongly you respond to it.

Knowing this in advance helps your doctor make smarter choices — potentially avoiding drugs that won't work well for you, choosing safer alternatives, or adjusting doses before problems arise.

Where the Science Stands

Pharmacogenomics is not new; it has been studied for decades, but clinical adoption has accelerated dramatically. Today, major health institutions around the world use PGx testing to guide prescribing in oncology, psychiatry, cardiology, infectious disease, and more.

Authoritative bodies like CPIC (Clinical Pharmacogenetics Implementation Consortium), ClinPGx (formerly PharmGKB), and the FDA have published evidence-based guidelines that translate genetic findings directly into prescribing recommendations.

At the same time, the field is still growing. Not every drug has robust PGx evidence yet, and not every genetic variant has been fully characterized. This is why this report is structured carefully: only gene-drug combinations with established clinical guidelines are presented as actionable recommendations. Everything else is shown transparently, with clear explanations of what is and isn't yet known.

The Different Types of Pharmacogenes

Not all pharmacogenomic genes work the same way. This report includes several types of genes, each of which influences drug response through a different mechanism. Understanding this helps you interpret the phenotype labels you will encounter:

- **1. Drug–Metabolizing Enzymes (CYP genes, DPYD, TPMT, NUDT15, UGT1A1)** : These genes encode enzymes that chemically break down drugs in the liver or gut. Variants affect how quickly a drug is cleared from the body. These are described using the metabolizer phenotype framework (Normal, Intermediate, Poor, Rapid/Ultrarapid Metabolizer).
- **2. Drug Transporter Genes (SLCO1B1, ABCG2)** : These genes encode proteins that physically move drugs across cell membranes. Variants are described as Decreased Function or Poor Function because the drug is not being metabolized differently — it is being transported differently.
- **3. Drug Target / Mechanism Genes (VKORC1, RYR1, CACNA1S)** : These encode the molecular targets or mechanisms relevant to drug response. Results may therefore use labels other than metabolizer phenotypes.
- **4. Immune and Hypersensitivity Genes (HLA-A, HLA-B)** : These genes are involved in immune recognition. Certain alleles are associated with severe immune reactions to specific medications and may require specialized high-resolution typing.
- **5. Response Prediction Genes (IFNL3)** : Some genes help predict the likelihood of treatment response rather than primarily predicting toxicity.

PGx in Everyday Clinical Practice

Pharmacogenomics-guided prescribing is increasingly used in clinical settings. The examples below illustrate how genetic information can contribute to medication decisions.

Drug / Drug Class	How PGx Can Guide Prescribing
Clopidogrel (antiplatelet)	Alternative antiplatelet therapy may be recommended for CYP2C19 intermediate or poor metabolizers, especially in ACS/PCI settings.
Warfarin (blood thinner)	PGx-guided dosing can help estimate the warfarin starting dose when CYP2C9, VKORC1, CYP4F2, and clinical factors are available.

SSRIs (antidepressants)	CYP2C19 and CYP2D6 variants affect how drugs like escitalopram and sertraline are metabolized. Rapid or ultrarapid metabolism may reduce exposure to some antidepressants, which can affect response.
Statins (cholesterol drugs)	Variants in SLCO1B1, ABCG2, and CYP2C9 can affect statin exposure and the risk of statin-associated muscle symptoms; the level of concern differs by statin and dose.
Thiopurines (chemotherapy / immunosuppression)	TPMT and NUDT15 variants affect tolerance to active thiopurine metabolites and can increase the risk of severe myelosuppression.

Important to Know

PGx information is one input into a prescribing decision, not a prescription in itself. Your doctor combines this genetic information with your medical history, current medications, organ function, and clinical judgment. This report is designed to support that conversation.

About This Report

What This Report Contains

This is a pharmacogenomics (PGx) report that examines how genetic variation may influence response to a range of medications. The report is organized into the following sections:

Section	What It Tells You
Detailed Drug Reports	For each drug with an established PGx guideline, you will find the genes analyzed, your genotype and phenotype, a plain-language explanation of what it means for you, and the clinical recommendation from recognized guidelines.
Other Evaluated Medications	Drugs that were evaluated but do not have a specific clinical action or recommendation in this report. This may occur because the relevant genetic result was unavailable or because the available evidence does not produce a patient-specific clinical recommendation.
No Guideline Available	Drugs for which a relevant gene-drug relationship was identified but no formal prescribing recommendation is available for the specific result represented in this report.
Genes Requiring Specialized Testing	Genes that could not be reliably assessed from the submitted DNA data, together with the reason and the medications for which the missing information may be relevant.
Genotype Summary	A consolidated reference of the genes evaluated, reported diplotypes, and resulting phenotypes for this sample.

The Difference Between 'No Guideline Available' and 'Other Evaluated Medications'

These two sections serve different purposes and should not be interpreted as equivalent.

No Guideline Available: A relevant gene-drug relationship was identified, but the available guideline information does not provide a specific patient-management recommendation for the result shown in this report.

Other Evaluated Medications: The medication was evaluated against the available genetic information but is not presented as a detailed actionable drug report. This may occur because the relevant genetic result was unavailable or because the evaluated result does not generate a patient-specific clinical recommendation.

Important

A medication appearing in either of these sections should not be interpreted as safe, unsafe, effective, or ineffective based on genetics alone. Medication decisions should always incorporate the full clinical picture.

What Do the Key Terms Mean?

Drug-specific entries use genetic terminology to describe what was evaluated and how the result was interpreted.

Term	Plain-Language Explanation
Gene	The specific gene being analyzed. Genes provide instructions for producing enzymes, transporters, receptors, and other proteins that can influence medication response.
Diplotype	Your reported combination of genetic alleles for that gene (for example, *1/*2). The exact notation depends on the gene and the allele-definition system used.
Phenotype	The predicted functional interpretation of the genetic result. Examples include Normal Metabolizer, Intermediate Metabolizer, Poor Metabolizer, Normal Function, or other gene-specific functional labels.

About *1 and Reference Results

A *1 or Reference result means that no defining alternative allele was identified at the relevant positions available for analysis. Because raw DNA files may not contain every position used to define an allele, a *1 or Reference result does not mean that the entire gene was sequenced or that all possible variants were excluded. Results should be interpreted within the marker coverage

of the submitted DNA file.

Where Do the Recommendations Come From?

Clinical recommendations and evidence annotations in this report may be derived from the following recognized pharmacogenomic resources:

Source	What It Provides	How It Is Used in This Report
CPIC	Peer-reviewed pharmacogenomic guidelines developed to help clinicians use available genetic test results in prescribing decisions.	Recommendation strength and evidence are shown where available.
DPWG	Evidence-based pharmacogenomic recommendations developed by the Dutch Pharmacogenetics Working Group.	Used where relevant recommendations are available.
ClinPGx	A curated resource of pharmacogenomic gene-drug associations, guideline information, and supporting evidence.	Evidence annotations may be displayed where available.
FDA	Drug-label and pharmacogenomic information published by the U.S. Food and Drug Administration.	FDA testing or actionable-PGx information may be displayed when applicable.

A Note on the GSI Catalog

The GSI catalog is an additional curated reference used by this report-generation pipeline to identify genes that have reported associations with a medication beyond the gene-drug pairs for which a formal prescribing recommendation is shown. These associations may relate to metabolism, transport, toxicity, treatment response, or another pharmacogenomic mechanism.

Genes listed as additional GSI-associated genes are included for informational completeness. Their presence does not mean that PharmCAT analyzed that gene-drug pair or that a clinical prescribing recommendation is available.

How Were Your Results Generated?

Your results were generated from the raw DNA data file submitted for analysis. Available pharmacogenomic markers were processed through the report pipeline, including **PharmCAT v3.2.0**, to derive supported genetic calls and phenotype interpretations for applicable genes.

The resulting genetic information was then matched against available pharmacogenomic recommendations and supporting reference data to determine which medication findings could be presented in this report.

How This Report Is Structured

The report begins with guidance on interpreting pharmacogenomic results, followed by a summary of medication insights and detailed recommendations organized by therapeutic category. Medications relevant to multiple categories are grouped together, with a single shared recommendation, while the Table of Contents lists them under each applicable category and page number. Additional sections include other evaluated medications, genes requiring specialized testing, and a consolidated genotype summary.

Guideline Currency

Pharmacogenomic guidelines and drug-label information can change as new evidence becomes available. The recommendations shown in this report reflect the guideline and reference data used by the analysis pipeline at the time the report was generated. For high-stakes prescribing decisions, clinicians should confirm that they are using the most current applicable guideline or prescribing information.

Technical Limitations

Coverage varies by gene and by the raw DNA platform used. Some pharmacogenes can be assessed when the required markers are present, while others cannot be fully or reliably characterized from standard SNP-array data.

Structurally complex genes, copy-number changes, high-resolution HLA alleles, certain mitochondrial variants, and rare variants may require dedicated clinical testing or sequencing methods.

Any gene that could not be reliably assessed for this sample is identified separately in the **Genes Requiring Specialized Testing** section. A missing or indeterminate result should not be interpreted as a normal genetic result.

How to Understand Your Results

Understanding Metabolizer Phenotypes

Most pharmacogenomics results for drug-metabolizing enzyme genes are expressed as a metabolizer status. This describes how efficiently your enzyme breaks down a particular drug.

Phenotype	What It Means	Common Clinical Implication
Normal (Extensive) Metabolizer	Your enzyme works as expected for the general population. A 'Normal' result does not mean you are immune to side effects – it means genetics are unlikely to cause an unexpected response at standard doses.	Standard doses are generally appropriate.
Intermediate Metabolizer	Your enzyme works at a reduced rate, typically because you carry one fully functional copy and one reduced-function copy of the gene.	Drug levels may be somewhat higher than average; dose adjustments or slower titration may be considered for some drugs.
Poor Metabolizer	Your enzyme has very little or no activity, often because both copies of the gene carry loss-of-function variants.	Drug may accumulate to high levels (increasing side-effect risk) or, if it requires activation by the enzyme, may have reduced effect.
Rapid or Ultrarapid Metabolizer	Your enzyme works faster than normal, clearing the drug more quickly.	Drug may be less effective at standard doses; higher doses or alternative drugs may be needed. For prodrugs, rapid conversion may lead to unexpectedly strong effects.
Indeterminate	The available data are insufficient or ambiguous to assign a clear functional phenotype.	No recommendation is generated from an indeterminate result. Further testing may be appropriate if the drug is clinically important.
Possible [Phenotype]	The diplotype contains a variant with uncertain or incompletely characterized functional impact, preventing a fully definitive assignment (e.g., CYP3A5 *1/*9).	The recommendation is still provided based on the most likely interpretation, but clinicians should be aware of reduced certainty and may wish to use additional clinical monitoring or seek confirmatory testing.

Understanding Transporter Function Phenotypes

Transporter genes such as SLC01B1 and ABCG2 use a different phenotype vocabulary because they move drugs into and out of cells rather than metabolizing them.

Phenotype	What It Means	Common Clinical Implication
Normal Function	Transporter moves drug into target tissue at the expected rate.	Standard drug exposure; no transporter-based dose concern.
Decreased Function	Transporter moves drug less efficiently. This phenotype reflects reduced transporter activity, which may alter drug distribution and increase systemic drug exposure.	Moderately increased drug plasma concentrations; some drugs may require dose adjustment or monitoring.
Poor Function	Transporter has very little or no activity (two loss-of-function copies).	Substantially higher drug exposure; dose reduction or alternative drug is often recommended.

Why Can the Same Genetic Result Lead to Different Recommendations for Different Drugs?

Example: SLCO1B1 and statins. An SLCO1B1 Decreased Function result can increase plasma exposure to several statins. However, the clinical action is not identical for every statin because each medication has a different baseline exposure–response relationship and risk profile. For example, simvastatin and lovastatin may carry greater concern for muscle toxicity at increased exposure than some other statins. This is why the same genetic result can lead to different recommendations for different medications.

Understanding Susceptibility Phenotypes

Some genes in this report do not primarily affect how a drug is metabolized. Instead, certain variants can indicate susceptibility to a physiological or immune-mediated adverse reaction when specific medications are used.

Phenotype	Gene(s)	What It Means	Clinical Implication
Malignant Hyperthermia Susceptibility	RYR1, CACNA1S	Variants in the ryanodine receptor or calcium channel gene create abnormally unstable calcium channels in skeletal muscle. When exposed to triggering agents, these can release massive amounts of calcium, causing a catastrophic hypermetabolic crisis.	Halogenated volatile anesthetics and succinylcholine are relatively contraindicated. This finding must be communicated to any anesthesiologist before any procedure.
Uncertain Susceptibility	CACNA1S	A report may show Uncertain Susceptibility when available data cannot fully exclude clinically relevant CACNA1S variation. This is different from a confirmed susceptibility.	If an RYR1 susceptibility is already confirmed, the same anesthetic precautions already apply. The CACNA1S 'uncertain' result means the standard SNP array cannot rule out structural or rare CACNA1S variants. If independent confirmation is needed, in vitro contracture testing (IVCT) can be requested.
Adverse Reaction Risk	HLA-B	Certain HLA-B alleles are strongly associated with severe cutaneous adverse reactions (Stevens–Johnson syndrome / toxic epidermal necrolysis) with specific drugs.	High-resolution HLA allele status generally cannot be established reliably from a standard consumer SNP-array file. When an HLA-B result is clinically required for prescribing, dedicated validated HLA testing may be necessary.

Variant-Classified Genes (Not Metabolizer-Based)

The following genes do not fit the metabolizer or transporter framework. Their results are reported as specific variant names or genotype calls rather than phenotype categories:

Gene	Phenotype Format	What It Means
VKORC1	Reported as: -1639 G/A (rs9923231 C/T)	VKORC1 is the direct molecular target of warfarin. The -1639 variant reduces VKORC1 expression, making patients more sensitive to vitamin K antagonists (warfarin, acenocoumarol, phenprocoumon). The result is a variant call, not a metabolizer phenotype.
CYP4F2	Reported as diplotype (e.g., *1/*5)	CYP4F2 metabolizes vitamin K1 (phylloquinone). Reduced CYP4F2 activity means less vitamin K is broken down, resulting in higher vitamin K levels and a slightly higher warfarin dose requirement.

IFNL3	Reported as rs12979860 genotype (C/C, C/T, or T/T)	IFNL3 (also called IL28B) does not metabolize drugs. It encodes interferon lambda, and its variant status predicts likelihood of response to interferon-based hepatitis C therapies. C/C = most favorable response; T/T = least favorable; C/T (heterozygous) = intermediate.
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Understanding the DPYD Activity Score

DPYD is the primary enzyme responsible for breaking down fluoropyrimidine drugs (capecitabine, fluorouracil, tegafur, flucytosine). Unlike other genes in this report, DPYD phenotype is expressed using an **Activity Score (AS)** system that reflects the combined functional impact of both gene copies.

Activity Score	Phenotype	Clinical Implication
2.0	Normal Metabolizer	Standard dosing is appropriate.
1.5	Intermediate Metabolizer	Reduce starting dose by 50%; titrate based on clinical response and therapeutic drug monitoring.
1.0	Intermediate Metabolizer	Reduce starting dose by 50%; titrate based on clinical response and therapeutic drug monitoring.
0.5	Poor Metabolizer	Avoid fluoropyrimidines. If no alternative exists, use $\geq 75\%$ dose reduction with intensive monitoring.
0.0	Poor Metabolizer	Fluoropyrimidines are contraindicated. Avoid entirely or use only under exceptional circumstances with substantial dose reduction and intensive monitoring.

DPYD Diplotype Notation

Unlike most genes in this report, which use **star allele notation** (e.g., *1/*5, *1/*9), DPYD diplotypes are reported in **cDNA nucleotide notation** (e.g., c.1057C>T/c.1484A>G). This is the established international convention for DPYD because the gene's variant landscape does not lend itself to the star allele system.

Each 'c.' entry represents a specific nucleotide change at a defined position in the DPYD gene sequence. The diplotype shown lists one variant on each chromosome copy. You do not need to interpret the raw notation — the phenotype (e.g., Poor Metabolizer) and Activity Score (e.g., 0.0) are the clinically actionable result.

Understanding the PRODRUG Label

Some drugs in this report are labeled **PRODRUG**. This label is clinically relevant for pharmacogenomics because the relationship between a metabolizer phenotype and clinical outcome differs:

- For a **standard active drug**, a Poor Metabolizer of the breakdown enzyme means the drug **accumulates** → higher toxicity risk.
- For a **prodrug** that must be converted to an active form, a Poor Metabolizer of the activating enzyme may not generate enough active drug → reduced efficacy.

In this report:

- **Capecitabine and Tegafur** are prodrugs of 5-fluorouracil. They are converted to 5-FU in the body. The DPYD enzyme then breaks down 5-FU. DPYD Poor Metabolizer status means 5-FU accumulates to toxic levels.
- **5-Fluorouracil** is the active cytotoxic agent itself (not a prodrug). It is the active metabolite that capecitabine and tegafur are converted into.
- **Azathioprine, Mercaptopurine, Thioguanine** are converted to active thiopurine metabolites. TPMT and NUDT15 inactivate the toxic metabolites. A Poor Metabolizer here means the toxic metabolites are not cleared quickly enough.

What If a Gene Shows More Than One Diplotype or Phenotype?

For some genes (such as SLCO1B1, NAT2, CYP4F2), your report may show multiple possible diplotypes in a single row. This happens when the analysis cannot definitively phase your variants — that is, it can identify the variants you carry but cannot always determine with certainty which variants sit on the same chromosome copy. Rather than exclude ambiguous results, the report shows all clinically plausible diplotype–phenotype combinations so your clinician has the full picture.

For **CYP4F2 and warfarin specifically**: When more than one plausible diplotype is shown, the possible interpretations may have similar or overlapping clinical implications. Warfarin dosing should not be interpreted from CYP4F2 alone; CYP2C9, VKORC1, available clinical factors, and the completeness of the submitted genetic data should all be considered together.

What If Different Genes Point to Different Outcomes for the Same Drug?

Some drugs are affected by more than one gene. This report presents the recommendation for each gene separately. Your clinician should consider all findings together. The combined effect of multiple gene variants can be additive — so the overall clinical picture may be more nuanced than any single result alone.

How to Read the Clinical Recommendations

Each drug section ends with a Clinical Recommendations block, sourced from CPIC, DPWG, ClinPGx, or the FDA. Here is how to read them:

Recommendation Type	What It Means in Practice
Dose adjustment	The evidence suggests starting at a lower or higher dose, titrating more slowly, or setting a maximum daily dose based on your phenotype.
Alternative drug	A different medication in the same class is preferred for safety or efficacy reasons, given your genetic result.
Monitor / Other guidance	No immediate dose change is required, but additional monitoring, specific laboratory tests, or counselling about risks is recommended.
No action required	Your genetic result does not indicate a need to change the drug or dose, though your doctor still considers all other clinical factors.

What Is the 'Other Evaluated Medications' Section?

This section lists drugs that were assessed as part of the analysis, but for which there is no recognized guideline-level clinical recommendation. These are included because:

- Your result may become clinically relevant as evidence evolves.
- Your doctor may find the genotype or phenotype information useful even without a formal guideline recommendation.
- Transparency matters — knowing what was tested (and what was not) is important context.

These entries should not be used as the basis for prescribing changes on their own, but can be a useful reference for your healthcare team.

What If a Drug You Take Is Not Listed?

If a medication you currently take does not appear anywhere in this report, it most likely means that either (a) there is no established pharmacogenomic guideline linking any of your tested genes to that drug, or (b) the relevant gene markers were not captured in your submitted data file.

The absence of a listing does not mean genetics are irrelevant — it means the evidence has not yet reached the threshold for a guideline recommendation, or the drug falls outside the scope of this analysis.

What 'Not Evaluated by Platform' Means

In some drug entries, you will see a gene listed with the notation **'Not evaluated by platform'** instead of a diplotype and phenotype. This means the gene's relevant variants were not captured in the DNA dataset submitted for analysis. This is a data limitation of consumer-grade SNP array genotyping, not a problem with the gene itself.

Common reasons include: the gene's variants are located in regions not covered by standard SNP arrays (e.g., CYP2D6 structural variants), the gene requires copy-number variant detection, or the variant is rare. 'Not evaluated by platform' means **no conclusion can be drawn**. If the drug is relevant, a dedicated pharmacogenomic test from a clinical laboratory may be ordered.

Reminder

A strong evidence level (e.g., CPIC A, ClinPGx 1A) means the gene-drug relationship is well-established; it does not automatically mean you must change your medication. All prescribing decisions remain at the discretion of your qualified healthcare provider.

Report Scope

This pharmacogenomics report was generated using SNP array-based genotyping for **Hannah**. Results reflect pharmacogenomically relevant genetic variants for the genes listed in the Genotype Summary. Recommendations are derived from CPIC, DPWG, and FDA prescribing information, as applicable. PharmCAT v3.2.0 was used for variant calling and phenotype assignment. Only gene–drug pairs with a defined phenotype and a recognized guideline–linked recommendation are surfaced as clinical recommendations.

Evidence Sources

CPIC (cpicpgx.org) and DPWG guidelines are peer-reviewed and regularly updated by expert working groups. DPWG guidelines are produced by Dutch clinical pharmacogenomics experts and are widely used when CPIC has not yet issued a guideline for a specific gene–drug pair. ClinPGx (formerly PharmGKB, clinpgx.org) curates gene–drug association evidence that underpins guideline development. FDA label annotations reflect current approved prescribing information. Guideline classifications (Strong, Moderate, Optional) are included per source where available.

How to Identify Key Findings

To identify your most critical findings, please refer to the **Summary of Medication Insights** on the following page. That summary provides a prioritized snapshot of all medications evaluated, categorized by the level of clinical action required (e.g., Action Required, Use With Caution).

Guidance on Missing Genes for Key Drug Decisions

CYP2C9 (not callable from this dataset)

CYP2C9 affects multiple clinically important drugs including warfarin, several NSAIDs, glipizide, glyburide, and losartan. For warfarin in particular, CYP2C9 is the dominant metabolizing enzyme and the primary variable in all validated pharmacogenetic dosing algorithms (including the CPIC–recommended WarfarinDosing.org calculator). This report's warfarin findings are incomplete without CYP2C9 data. A dedicated CYP2C9 pharmacogenomic assay from a clinical laboratory is strongly recommended before attempting genotype–guided warfarin dosing.

TPMT (not callable from this dataset)

TPMT and NUDT15 both independently affect thiopurine toxicity (azathioprine, mercaptopurine, thioguanine). Without TPMT data, the risk assessment is incomplete. In patients of European descent, TPMT has historically been the primary predictor of thiopurine myelosuppression; in East Asian populations, NUDT15 is generally more informative. Before initiating thiopurine therapy, consider ordering a dedicated TPMT enzyme activity assay (available from most clinical biochemistry laboratories) or a clinical–grade TPMT genotyping panel. Monitor CBC closely during dose escalation regardless.

CYP2D6 (not callable from this dataset)

CYP2D6 affects over 25% of commonly prescribed drugs, including codeine, tramadol, tamoxifen, many antidepressants (paroxetine, fluoxetine, nortriptyline, venlafaxine), and many antipsychotics. For patients requiring any of these medications — particularly codeine or tramadol (where CYP2D6 poor or ultrarapid metabolizer status carries serious safety implications), or tamoxifen (where CYP2D6 status affects endoxifen generation and treatment efficacy) — a dedicated clinical CYP2D6 genotyping panel using long-read sequencing or a validated clinical assay is recommended.

HLA-B (not callable from this dataset)

HLA-B allele–level typing is required before prescribing abacavir (HLA-B*57:01), allopurinol (HLA-B*58:01), and certain anticonvulsants (HLA-B*15:02) to prevent severe cutaneous adverse reactions. The risk of HLA-B*58:01-associated allopurinol SJS is markedly higher in individuals of Han Chinese, Korean, Thai, and Vietnamese descent. Dedicated high-resolution HLA typing from a clinical immunogenetics laboratory is required.

Technical Interpretation Notes

- Diplotype calling performed using PharmCAT v3.2.0 with standard reference data.
- SNP array coverage may not detect all star alleles; rare or novel variants may be absent.
- Copy number variants (e.g., CYP2D6 gene duplications) require dedicated confirmatory testing.
- HLA alleles (HLA-A, HLA-B) require specialized high-resolution typing and are not included here.
- Unphased results (e.g., CYP4F2) indicate phase–ambiguous variants; all clinically plausible interpretations are reported for completeness.
- DPYD diplotype is reported in cDNA notation (c.XXXN>N) per international convention, not star allele notation.

- The genotypic phenotypes in this report represent predicted drug metabolism based on germline DNA and do not account for phenoconversion due to concomitant medications, renal or hepatic impairment, or other clinical factors.

A Note on Phenoconversion

A patient's genetically predicted phenotype may not reflect their actual functional phenotype if they are concurrently taking medications that inhibit or induce the same enzyme. **Example:** A patient who is a genotypic Normal Metabolizer for CYP2D6 may function as a Poor Metabolizer if they are also taking a strong CYP2D6 inhibitor such as fluoxetine, paroxetine, or bupropion.

Clinicians should always cross-check the patient's medication list for known inhibitors and inducers of any gene flagged in this report. Common phenoconverting agents include:

Gene	Common Strong Inhibitors (may convert to Poor Metabolizer)	Common Inducers (may increase metabolism)
CYP2D6	Fluoxetine, paroxetine, bupropion, terbinafine, quinidine	None
CYP2C19	Fluvoxamine, fluconazole, ticlopidine	Rifampin, carbamazepine
CYP3A4/5	Ketoconazole, itraconazole, clarithromycin, grapefruit juice	Rifampin, carbamazepine, phenytoin, St. John's wort
CYP2C9	Fluconazole, amiodarone, miconazole	Rifampin

This report does not perform phenoconversion analysis. That assessment must be performed by the prescribing clinician.

Appropriate Use

This report is intended to support, not replace, clinical judgment. Decisions about prescribing or discontinuing medications should integrate this report with the patient's full clinical picture — including comorbidities, concomitant medications, organ function, and patient preferences. The pharmacogenomic findings presented here are one input among many in the prescribing decision. This report does not constitute a prescription, a contraindication, or a guarantee of drug response.

Summary of Medication Insights

Based on your genetic profile, **120 medications** were evaluated.

Action Required (15 medications)

Amifampridine NAT2

Citalopram CYP2C19

Dexlansoprazole CYP2C19

Fluvastatin ABCG2, CYP2C9, CYP3A4, CYP3A5, SLCO1B1

Ivacaftor CFTR

Omeprazole CYP2C19

Sertraline CYP2B6, CYP2C19

Tacrolimus CYP3A4, CYP3A5

Amifampridine Phosphate NAT2

Clopidogrel CYP2C19

Escitalopram CYP2C19

Hydralazine NAT2

Lansoprazole CYP2C19

Pantoprazole CYP2C19

Siponimod CYP2C9

Use With Caution / Monitoring (2 medications)

Atazanavir CYP2C19, UGT1A1

Dronabinol CYP2C9

Further Testing Required (16 medications)

Allopurinol ABCG2, HLA-B

Clomipramine CYP2C19, CYP2D6

Doxepin CYP2C19, CYP2D6

Fosphenytoin CYP2C9, HLA-B

Imipramine CYP2C19, CYP2D6

Methoxyflurane CACNA1S, RYR1

Sevoflurane CACNA1S, RYR1

Trimipramine CYP2C19, CYP2D6

Amitriptyline CYP2C19, CYP2D6

Desflurane CACNA1S, RYR1

Enflurane CACNA1S, RYR1

Halothane CACNA1S, RYR1

Isoflurane CACNA1S, RYR1

Phenytoin CYP2C19, CYP2C9, HLA-B

Succinylcholine CACNA1S, RYR1

Warfarin CYP2C9, CYP4F2, RS12777823, VKORC1

Standard Use (39 medications)

Acenocoumarol CYP2C9, VKORC1

Avatrombopag CYP2C9, F5

Brivaracetam CYP2C19

Celecoxib CYP2C9

Dapsone G6PD

Flucytosine DPYD

Flurbiprofen CYP2C9

Isoniazid NAT2

Lovastatin ABCG2, CYP3A4, CYP3A5, SLCO1B1

Meloxicam CYP2C9

Methylene Blue G6PD

Atorvastatin ABCG2, CYP3A4, CYP3A5, SLCO1B1

Azathioprine NUDT15, TPMT

Capecitabine DPYD

Clobazam CYP2C19

Efavirenz CYP2B6

Fluorouracil DPYD

Ibuprofen CYP2C9

Lornoxicam CYP2C9

Mavacamten CYP2C19

Mercaptopurine NUDT15, TPMT

Nitrofurantoin G6PD

Piroxicam CYP2C9	Pitavastatin ABCG2, CYP3A4, CYP3A5, SLCO1B1
Pravastatin ABCG2, CYP3A4, CYP3A5, SLCO1B1	Primaquine G6PD
Procainamide NAT2	Rasburicase G6PD
Rosuvastatin ABCG2, CYP3A4, CYP3A5, SLCO1B1	Simvastatin ABCG2, CYP3A4, CYP3A5, SLCO1B1
Sulfamethoxazole / Trimethoprim G6PD, NAT2	Sulfasalazine G6PD, NAT2
Tafenoquine G6PD	Tenoxicam CYP2C9
Thioguanine NUDT15, TPMT	Toluidine Blue G6PD
Voriconazole CYP2C19	

No Guideline Available (48 medications)

Abrocitinib CYP2C19	Articaine / Epinephrine G6PD
Ascorbic Acid (Vitamin C), Combinations G6PD	Ascorbic Acid (Vitamin C), Plain G6PD
Belinostat UGT1A1	Belzutifan CYP2C19
Bupivacaine G6PD	Carisoprodol CYP2C19
Chloroquine G6PD	Chlorpropamide G6PD
Dabrafenib G6PD	Deuruxolitinib CYP2C9
Diazepam CYP2C19, CYP3A4	Dolutegravir UGT1A1
Elagolix SLCO1B1	Erdafitinib CYP2C9
Esomeprazole CYP2C19	Flibanserin CYP2C19, CYP2C9
Flutamide G6PD	Glimepiride CYP2C9, G6PD
Glipizide G6PD	Glyburide CYP2C9, G6PD
Hydroxychloroquine G6PD	Irinotecan UGT1A1
Lesinurad CYP2C9	Lidocaine / Prilocaine G6PD
Lidocaine And Tetracaine G6PD	Mepivacaine G6PD
Moviprep G6PD	Nalidixic Acid G6PD
Nateglinide CYP2C9	Nilotinib UGT1A1
Oxymetazoline And Tetracaine G6PD	Pazopanib UGT1A1, HLA-B
Peginterferon Alfa-2A	Peginterferon Alfa-2B
Quetiapine CYP3A4	Rabeprazole CYP2C19
Raltegravir UGT1A1	Ribavirin
Ropivacaine G6PD	Sacituzumab Govitecan UGT1A1
Seladelpar CYP2C9	Sodium Ascorbate G6PD
Sodium Nitrite G6PD	Tegafur DPYD
Tolazamide G6PD	Tolbutamide CYP2C9, G6PD

* All prescribing decisions must be made by your healthcare provider.

Coverage Statement

This report evaluates medications across the following therapeutic categories. Categories with a ✓ indicate that relevant medications with pharmacogenomic guidance were identified in your profile.

Status	Therapeutic Category
✓	Musculo-Skeletal System Drugs
✓	Cardiovascular System Drugs
✓	Alimentary Tract And Metabolism Drugs
✓	Nervous System Drugs
✓	Respiratory System Drugs
✓	Anti-infectives For Systemic Use
✓	Genitourinary System And Sex Hormones
✓	Blood And Blood Forming Organ Drugs
✓	Dermatologicals
✓	Antineoplastic And Immunomodulating Agents
✓	Sensory Organ Drugs
✓	Various Drug Classes In ATC
✓	Antiparasitic Products, Insecticides And Repellents
—	Systemic Hormonal Preparations, Excl. Sex Hormones And Insulins
✓	No Group Assigned

* All medications with pharmacogenomic relevance are shown. Actions are indicated where applicable in the subsequent pages.

A dash (—) indicates that no medications in this therapeutic category had an established pharmacogenomic guideline result based on your tested genes. It does not indicate that the category was outside the scope of analysis — all 15 categories listed above are evaluated for every report.

Drug-Specific Recommendations

This section provides detailed pharmacogenomic recommendations for the medications identified in your profile. All medications with pharmacogenomic relevance are shown. *Actions are indicated where applicable.*

MUSCULO-SKELETAL SYSTEM DRUGS

Allopurinol

Genes Analyzed

Gene	Diplotype	Phenotype	Status
ABCG2	rs2231142 reference (G)/rs2231142 reference (G)	Normal Function	✓ Analyzed

Other Associated Genes:

HLA-B is known to affect this drug's metabolism (GSI catalog), but was not analyzed by PharmCAT in this report.

About this medication

Allopurinol (Zyloprim, Aloprim) is a Musculo-Skeletal System drug that inhibits the enzyme xanthine oxidase, thereby reducing the body's production of uric acid. It is the cornerstone long-term treatment for preventing recurrent gout flares and uric acid kidney stones in patients with elevated serum uric acid levels.

What it means for you

Your ABCG2 transporter function is Normal, meaning your body efficiently effluxes Allopurinol through the intestinal and renal pathways as expected. No pharmacogenomic dose adjustment is anticipated based on ABCG2 status alone. Standard dosing guidelines apply, though other clinical factors may influence your treatment. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

DPWG

ABCG2 (normal function)

Recommendation: The guideline does not provide a recommendation for allopurinol in patients with the the ABCG2 rs2231142 GG genotype (c.421CC; p.141QQ)

Implications: The guideline does not provide a description of the impact of the ABCG2 rs2231142 GG genotype (c.421CC; p.141QQ) on allopurinol.

Citations: <https://pubmed.ncbi.nlm.nih.gov/36056234/>

MUSCULO-SKELETAL SYSTEM DRUGS

Celecoxib

Genes Analyzed

Gene	Diplotype	Phenotype	Status
CYP2C9	*1/*11	Intermediate Metabolizer	✓ Analyzed

Other Medications That May Further Influence This Result

Your **CYP2C9** result shows **Intermediate Metabolizer** status and may influence how your body handles **Celecoxib**. The following are examples of FDA-listed medications or substances that affect **CYP2C9** and may further modify drug exposure or response if taken together:

Amiodarone

Fluconazole

Miconazole

Piperine

If you are currently taking any of the above, mention this to your prescriber. This is general reference information based on your genetic result — it is not a personalized check of your current medications.

About this medication

Celecoxib (Celebrex) is a selective COX-2 inhibitor used for osteoarthritis, rheumatoid arthritis, ankylosing spondylitis, and acute musculoskeletal pain. By selectively targeting COX-2, it reduces prostaglandin synthesis at inflammatory sites while largely sparing constitutive COX-1 activity that maintains gastric mucosal integrity — providing a generally more favorable GI tolerability profile than non-selective NSAIDs, though cardiovascular risks remain. CYP2C9 is the primary enzyme responsible for celecoxib clearance.

What it means for you

Your CYP2C9 Intermediate Metabolizer status (Activity Score 1.5, carrying one reduced-function CYP2C9 allele) means you process Celecoxib at a reduced rate, resulting in mildly elevated drug levels compared to normal metabolizers. This may enhance therapeutic effects but also increases adverse effect risk. Your prescriber may recommend dose adjustments or additional monitoring. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

CPIC: A

PharmGKB: 1A

FDA: Actionable PGx

CPIC

CYP2C9 (Intermediate Metabolizer (AS:1.5))

Recommendation: Initiate therapy with recommended starting dose. In accordance with the prescribing information, use the lowest effective dosage for shortest duration consistent with individual patient treatment goals.

Other Considerations: IMs might have a higher than normal risk of adverse events especially in individuals with other factors affecting clearance of these drugs such as hepatic impairment or advanced age. Further caution should be taken with ibuprofen use in individuals carrying the CYP2C9*2 allele as it is in linkage disequilibrium with CYP2C8*3 and ibuprofen is also metabolized by CYP2C8.

Implications: CYP2C9: Mildly reduced metabolism

CPIC:<https://cpicpgx.org/guidelines/cpic-guideline-for-nsaids-based-on-cyp2c9-genotype/> | **Citations:**<https://pubmed.ncbi.nlm.nih.gov/32189324/>

ANTINEOPLASTIC AND IMMUNOMODULATING AGENTS

About this medication

Celecoxib (Celebrex) is a selective COX-2 inhibitor used in the antineoplastic context for the reduction of colorectal adenoma burden in patients with familial adenomatous polyposis (FAP), as an adjunct to standard surveillance and surgery. COX-2 is overexpressed in many colorectal tumors, and its inhibition reduces tumor-promoting prostaglandin E2 production. CYP2C9 is the main enzyme responsible for celecoxib's hepatic clearance, and metabolizer status meaningfully affects drug exposure in this setting.

What it means for you

Your CYP2C9 Intermediate Metabolizer status (Activity Score 1.5, carrying one reduced-function CYP2C9 allele) means you process Celecoxib at a reduced rate, resulting in mildly elevated drug levels compared to normal metabolizers. This may enhance therapeutic effects but also increases adverse effect risk. Your prescriber may recommend dose adjustments or additional monitoring. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

See the full pharmacogenomic recommendation above under **Musculo-Skeletal System Drugs — celecoxib**.

Key genetic finding: CYP2C9 — intermediate metabolizer (AS:1.5)

MUSCULO-SKELETAL SYSTEM DRUGS

Flurbiprofen

Genes Analyzed

Gene	Diplotype	Phenotype	Status
CYP2C9	*1/*11	Intermediate Metabolizer	✓ Analyzed

Other Medications That May Further Influence This Result

Your **CYP2C9** result shows **Intermediate Metabolizer** status and may influence how your body handles **Flurbiprofen**. The following are examples of FDA-listed medications or substances that affect **CYP2C9** and may further modify drug exposure or response if taken together:

Amiodarone

Fluconazole

Miconazole

Piperine

If you are currently taking any of the above, mention this to your prescriber. This is general reference information based on your genetic result – it is not a personalized check of your current medications.

About this medication

Flurbiprofen (Ansaid) is a non-selective NSAID that inhibits both COX-1 and COX-2 to reduce prostaglandin synthesis, producing analgesic and anti-inflammatory effects. It is used orally for osteoarthritis, rheumatoid arthritis, and other musculoskeletal inflammatory conditions. CYP2C9 is the primary enzyme responsible for its hepatic clearance.

What it means for you

Your CYP2C9 Intermediate Metabolizer status (Activity Score 1.5, carrying one reduced-function CYP2C9 allele) means you process Flurbiprofen at a reduced rate, resulting in mildly elevated drug levels compared to normal metabolizers. This may enhance therapeutic effects but also increases adverse effect risk. Your prescriber may recommend dose adjustments or additional monitoring. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

CPIC: A

PharmGKB: 1A

FDA: Actionable PGx

CPIC

CYP2C9 (Intermediate Metabolizer (AS:1.5))

Recommendation: Initiate therapy with recommended starting dose. In accordance with the prescribing information, use the lowest effective dosage for shortest duration consistent with individual patient treatment goals.

Other Considerations: IMs might have a higher than normal risk of adverse events especially in individuals with other factors affecting clearance of these drugs such as hepatic impairment or advanced age. Further caution should be taken with ibuprofen use in individuals carrying the CYP2C9*2 allele as it is in linkage disequilibrium with CYP2C8*3 and ibuprofen is also metabolized by CYP2C8.

Implications: CYP2C9: Mildly reduced metabolism

CPIC:<https://cpicpgx.org/guidelines/cpic-guideline-for-nsaids-based-on-cyp2c9-genotype/> | Citations:<https://pubmed.ncbi.nlm.nih.gov/32189324/>

RESPIRATORY SYSTEM DRUGS

Flurbiprofen

About this medication

Flurbiprofen (Strefen, Diffiam) in the respiratory system context is available in lozenge or spray formulations for the relief of sore throat and oropharyngeal pain. It provides localized anti-inflammatory and analgesic effects in the pharynx by inhibiting prostaglandin synthesis. Systemic absorption from throat lozenges is low but not absent, and CYP2C9 metabolizer status may influence total drug exposure.

What it means for you

Your CYP2C9 Intermediate Metabolizer status (Activity Score 1.5, carrying one reduced-function CYP2C9 allele) means you process Flurbiprofen at a reduced rate, resulting in mildly elevated drug levels compared to normal metabolizers. This may enhance therapeutic effects but also increases adverse effect risk. Your prescriber may recommend dose adjustments or additional monitoring. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

See the full pharmacogenomic recommendation above under **Musculo-Skeletal System Drugs — flurbiprofen**.

Key genetic finding:CYP2C9 — intermediate metabolizer (AS:1.5)

SENSORY ORGAN DRUGS

Flurbiprofen

About this medication

Flurbiprofen (Ocufen) is available as ophthalmic drops used to inhibit intraoperative miosis (pupil constriction) during cataract surgery and as an anti-inflammatory agent for postoperative ophthalmic care. It inhibits COX-mediated prostaglandin synthesis in ocular tissues. Systemic absorption from ophthalmic use is minimal, though CYP2C9 participates in clearing any absorbed fraction.

What it means for you

Your CYP2C9 Intermediate Metabolizer status (Activity Score 1.5, carrying one reduced-function CYP2C9 allele) means you process Flurbiprofen at a reduced rate, resulting in mildly elevated drug levels compared to normal metabolizers. This may enhance therapeutic effects but also increases adverse effect risk. Your prescriber may recommend dose adjustments or additional monitoring. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

See the full pharmacogenomic recommendation above under **Musculo-Skeletal System Drugs — flurbiprofen**.

Key genetic finding:CYP2C9 — intermediate metabolizer (AS:1.5)

MUSCULO-SKELETAL SYSTEM DRUGS

Ibuprofen

Genes Analyzed

Gene	Diplotype	Phenotype	Status
CYP2C9	*1/*11	Intermediate Metabolizer	✓ Analyzed

Other Medications That May Further Influence This Result

Your **CYP2C9** result shows **Intermediate Metabolizer** status and may influence how your body handles **Ibuprofen**. The following are examples of FDA-listed medications or substances that affect **CYP2C9** and may further modify drug exposure or response if taken together:

Amiodarone

Fluconazole

Miconazole

Piperine

If you are currently taking any of the above, mention this to your prescriber. This is general reference information based on your genetic result — it is not a personalized check of your current medications.

About this medication

Ibuprofen (Advil, Motrin, Brufen) is a non-steroidal anti-inflammatory drug (NSAID) that reversibly inhibits both cyclooxygenase-1 (COX-1) and cyclooxygenase-2 (COX-2) enzymes, thereby reducing the synthesis of prostaglandins — the lipid mediators of pain, fever, and inflammation. In the musculoskeletal context, it is widely used for acute and chronic pain and inflammation associated with osteoarthritis, rheumatoid arthritis, ankylosing spondylitis, acute gout flares, muscle injuries, bursitis, tendinitis, and post-surgical orthopaedic pain. CYP2C9 is the primary hepatic enzyme responsible for metabolizing the pharmacologically active S-enantiomer of ibuprofen.

What it means for you

Your CYP2C9 Intermediate Metabolizer status (Activity Score 1.5, carrying one reduced-function CYP2C9 allele) means your liver processes ibuprofen's active form at a mildly reduced rate, resulting in slightly higher drug concentrations compared to normal metabolizers. For musculoskeletal use, this degree of difference is generally modest and often does not require dose adjustment, but using the lowest effective dose for the shortest necessary duration remains prudent. Your prescriber may recommend monitoring with prolonged use. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

CPIC: A

PharmGKB: 1A

CPIC

CYP2C9 (Intermediate Metabolizer (AS:1.5))

Recommendation: Initiate therapy with recommended starting dose. In accordance with the prescribing information, use the lowest effective dosage for shortest duration consistent with individual patient treatment goals.

Other Considerations: IMs might have a higher than normal risk of adverse events especially in individuals with other factors affecting clearance of these drugs such as hepatic impairment or advanced age. Further caution should be taken with ibuprofen use in individuals carrying the CYP2C9*2 allele as it is in linkage disequilibrium with CYP2C8*3 and ibuprofen is also metabolized by CYP2C8.

Implications: CYP2C9: Mildly reduced metabolism

CPIC:<https://cpicpgx.org/guidelines/cpic-guideline-for-nsaids-based-on-cyp2c9-genotype/> | **Citations:**<https://pubmed.ncbi.nlm.nih.gov/32189324/>

CARDIOVASCULAR SYSTEM DRUGS

Ibuprofen

About this medication

Ibuprofen (Advil, Motrin, Brufen) is a non-selective COX-1 and COX-2 inhibitor with significant cardiovascular pharmacological relevance. Regular or high-dose use raises systemic blood pressure and promotes sodium and fluid retention through inhibition of renal prostaglandins. It is associated with an increased risk of myocardial infarction and stroke, especially with long-term use and in patients with pre-existing cardiovascular disease. Critically, ibuprofen can reversibly compete with low-dose aspirin at the COX-1 active site on platelets, attenuating aspirin's irreversible antiplatelet effect — a clinically important interaction for patients on cardioprotective aspirin. Ibuprofen is also used therapeutically for acute pericarditis. CYP2C9 is the main enzyme clearing the active S-enantiomer of ibuprofen from the body.

What it means for you

Your CYP2C9 Intermediate Metabolizer status (Activity Score 1.5, one reduced-function CYP2C9 allele) leads to slightly higher ibuprofen blood levels compared to normal metabolizers. In the cardiovascular context, this mildly elevated exposure may marginally amplify ibuprofen's blood pressure-raising effects and its capacity to blunt low-dose aspirin's antiplatelet cardioprotection. Patients relying on aspirin for cardiovascular prevention should be especially mindful of this interaction at any ibuprofen exposure level. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

See the full pharmacogenomic recommendation above under **Musculo-Skeletal System Drugs — ibuprofen**.

Key genetic finding:CYP2C9 — intermediate metabolizer (AS:1.5)

GENITO URINARY SYSTEM AND SEX HORMONES

Ibuprofen

About this medication

Ibuprofen (Advil, Motrin, Brufen) is a non-selective NSAID that is considered a first-line pharmacological treatment for primary dysmenorrhea (menstrual cramps). During menstruation, the uterine endometrium produces excess prostaglandins — particularly PGE2 and PGF2 α — which drive intense myometrial contractions and uterine vasoconstriction, causing the characteristic cramping pain. By inhibiting COX-1 and COX-2, ibuprofen reduces uterine prostaglandin synthesis, diminishing contractions and pain. It is also used for endometriosis-associated pelvic pain and other genitourinary inflammatory conditions. CYP2C9 is the primary enzyme responsible for hepatic clearance of the active S-ibuprofen enantiomer.

What it means for you

Your CYP2C9 Intermediate Metabolizer status (Activity Score 1.5, one reduced-function allele) leads to mildly elevated ibuprofen blood levels compared to normal metabolizers. When used for dysmenorrhea or genitourinary pain, the slight increase in drug exposure may provide marginally enhanced pain relief but is unlikely to cause significant additional adverse effects with short-term use at standard doses. Using the lowest dose providing adequate relief is still advisable. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

See the full pharmacogenomic recommendation above under **Musculo-Skeletal System Drugs — ibuprofen**.

Key genetic finding: CYP2C9 — intermediate metabolizer (AS:1.5)

MUSCULO-SKELETAL SYSTEM DRUGS

Lornoxicam

Genes Analyzed

Gene	Diplotype	Phenotype	Status
CYP2C9	*1/*11	Intermediate Metabolizer	✓ Analyzed

Other Medications That May Further Influence This Result

Your **CYP2C9** result shows **Intermediate Metabolizer** status and may influence how your body handles **Lornoxicam**. The following are examples of FDA-listed medications or substances that affect **CYP2C9** and may further modify drug exposure or response if taken together:

Amiodarone

Fluconazole

Miconazole

Piperine

If you are currently taking any of the above, mention this to your prescriber. This is general reference information based on your genetic result – it is not a personalized check of your current medications.

About this medication

Lornoxicam (Xefo) is a Musculo-Skeletal System oxicam NSAID with balanced COX-1/COX-2 inhibition used for acute and chronic pain and inflammation.

What it means for you

Your CYP2C9 Intermediate Metabolizer status (Activity Score 1.5, carrying one reduced-function CYP2C9 allele) means you process Lornoxicam at a reduced rate, resulting in mildly elevated drug levels compared to normal metabolizers. This may enhance therapeutic effects but also increases adverse effect risk. Your prescriber may recommend dose adjustments or additional monitoring. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

CPIC: A

PharmGKB: 1A

CPIC

CYP2C9 (Intermediate Metabolizer (AS:1.5))

Recommendation: Initiate therapy with recommended starting dose. In accordance with the prescribing information, use the lowest effective dosage for shortest duration consistent with individual patient treatment goals.

Other Considerations: IMs might have a higher than normal risk of adverse events especially in individuals with other factors affecting clearance of these drugs such as hepatic impairment or advanced age. Further caution should be taken with ibuprofen use in individuals carrying the CYP2C9*2 allele as it is in linkage disequilibrium with CYP2C8*3 and ibuprofen is also metabolized by CYP2C8.

Implications: CYP2C9: Mildly reduced metabolism

CPIC:<https://cpicpgx.org/guidelines/cpic-guideline-for-nsaids-based-on-cyp2c9-genotype/> | **Citations:**<https://pubmed.ncbi.nlm.nih.gov/32189324/>

MUSCULO-SKELETAL SYSTEM DRUGS

Meloxicam

Genes Analyzed

Gene	Diplotype	Phenotype	Status
CYP2C9	*1/*11	Intermediate Metabolizer	✓ Analyzed

Other Medications That May Further Influence This Result

Your **CYP2C9** result shows **Intermediate Metabolizer** status and may influence how your body handles **Meloxicam**. The following are examples of FDA-listed medications or substances that affect **CYP2C9** and may further modify drug exposure or response if taken together:

Amiodarone

Fluconazole

Miconazole

Piperine

If you are currently taking any of the above, mention this to your prescriber. This is general reference information based on your genetic result – it is not a personalized check of your current medications.

About this medication

Meloxicam (Mobic) is a Musculo-Skeletal System preferentially COX-2 selective NSAID used for osteoarthritis, rheumatoid arthritis, and juvenile idiopathic arthritis.

What it means for you

Your CYP2C9 Intermediate Metabolizer status (Activity Score 1.5, carrying one reduced-function CYP2C9 allele) means you process Meloxicam at a reduced rate, resulting in mildly elevated drug levels compared to normal metabolizers. This may enhance therapeutic effects but also increases adverse effect risk. Your prescriber may recommend dose adjustments or additional monitoring. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

CPIC: A

PharmGKB: 1A

FDA: Actionable PGx

CPIC

CYP2C9 (Intermediate Metabolizer (AS:1.5))

Recommendation: Initiate therapy with recommended starting dose. In accordance with the meloxicam prescribing information, use the lowest effective dosage for shortest duration consistent with individual patient treatment goals.

Other Considerations: IMs might have a higher than normal risk of adverse events especially in individuals with other factors affecting clearance of these drugs such as hepatic impairment or advanced age.

Implications: CYP2C9: Mildly reduced metabolism

CPIC:<https://cpicpgx.org/guidelines/cpic-guideline-for-nsaids-based-on-cyp2c9-genotype/> | **Citations:**<https://pubmed.ncbi.nlm.nih.gov/32189324/>

MUSCULO-SKELETAL SYSTEM DRUGS

Pegloticase

Genes Analyzed

Gene	Diplotype	Phenotype	Status
G6PD	B (reference)/B (reference)	Normal	✓ Analyzed

About this medication

Pegloticase (Krystexxa) is a Musculo-Skeletal System PEGylated recombinant uricase used for chronic refractory gout unresponsive to conventional therapy. It converts uric acid to the more soluble allantoin.

What it means for you

Your G6PD Normal status indicates you have sufficient glucose-6-phosphate dehydrogenase enzyme activity in your red blood cells to protect against oxidative stress from drugs like Pegloticase. No elevated risk of drug-induced hemolytic anemia is expected based on your G6PD result alone. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

CPIC: A

PharmGKB: 3

FDA: Testing Required

CPIC

G6PD (normal)

Recommendation: No reason to avoid based on G6PD status

Implications: G6PD: Low risk of acute hemolytic anemia

CPIC:<https://cpicpgx.org/guidelines/cpic-guideline-for-g6pd/> | Citations:<https://pubmed.ncbi.nlm.nih.gov/24787449/>;
<https://pubmed.ncbi.nlm.nih.gov/36049896/>

MUSCULO-SKELETAL SYSTEM DRUGS

Piroxicam

Genes Analyzed

Gene	Diplotype	Phenotype	Status
CYP2C9	*1/*11	Intermediate Metabolizer	✓ Analyzed

Other Medications That May Further Influence This Result

Your **CYP2C9** result shows **Intermediate Metabolizer** status and may influence how your body handles **Piroxicam**. The following are examples of FDA-listed medications or substances that affect **CYP2C9** and may further modify drug exposure or response if taken together:

Amiodarone

Fluconazole

Miconazole

Piperine

If you are currently taking any of the above, mention this to your prescriber. This is general reference information based on your genetic result – it is not a personalized check of your current medications.

About this medication

Piroxicam (Feldene) is a non-selective NSAID with an exceptionally long plasma half-life of approximately 50 hours, permitting convenient once-daily dosing. It inhibits COX-1 and COX-2 to reduce prostaglandin synthesis and is used for the long-term management of osteoarthritis and rheumatoid arthritis. Its long half-life also means drug accumulation occurs over several days of dosing and washout is prolonged, which is clinically relevant for CYP2C9 poor metabolizers. CYP2C9 is its primary metabolic enzyme.

What it means for you

Your CYP2C9 Intermediate Metabolizer status (Activity Score 1.5, carrying one reduced-function CYP2C9 allele) means you process Piroxicam at a reduced rate, resulting in mildly elevated drug levels compared to normal metabolizers. This may enhance therapeutic effects but also increases adverse effect risk. Your prescriber may recommend dose adjustments or additional monitoring. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

CPIC: A

PharmGKB: 1A

FDA: Actionable PGx

CPIC

CYP2C9 (Intermediate Metabolizer (AS:1.5))

Recommendation: Initiate therapy with recommended starting dose. In accordance with the prescribing information, use the lowest effective dosage for shortest duration consistent with individual patient treatment goals.

Other Considerations: IMs might have a higher than normal risk of adverse events especially in individuals with other factors affecting clearance of these drugs such as hepatic impairment or advanced age.

Implications: CYP2C9: Mildly reduced metabolism

CPIC:<https://cpicpgx.org/guidelines/cpic-guideline-for-nsaids-based-on-cyp2c9-genotype/> | Citations:<https://pubmed.ncbi.nlm.nih.gov/32189324/>

SENSORY ORGAN DRUGS

Piroxicam

About this medication

Piroxicam is available in some countries as a topical ophthalmic gel for the management of external ocular inflammation and as an anti-inflammatory adjunct in ophthalmic surgery. Applied to the eye, it delivers local COX-inhibiting anti-inflammatory effects. Systemic absorption from ophthalmic topical use is low compared to oral administration, and CYP2C9 metabolizes any systemically absorbed fraction.

What it means for you

Your CYP2C9 Intermediate Metabolizer status (Activity Score 1.5, carrying one reduced-function CYP2C9 allele) means you process Piroxicam at a reduced rate, resulting in mildly elevated drug levels compared to normal metabolizers. This may enhance therapeutic effects but also increases adverse effect risk. Your prescriber may recommend dose adjustments or additional monitoring. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

See the full pharmacogenomic recommendation above under **Musculo-Skeletal System Drugs — piroxicam**.

Key genetic finding:CYP2C9 — intermediate metabolizer (AS:1.5)

MUSCULO-SKELETAL SYSTEM DRUGS

Succinylcholine

Genes Analyzed

Gene	Diplotype	Phenotype	Status
RYR1	Reference/Reference	Uncertain Susceptibility	✓ Analyzed

About this medication

Succinylcholine (Anectine, Quelicin) is a Musculo-Skeletal System depolarizing neuromuscular blocking agent used to achieve rapid muscle relaxation and facilitate intubation during general anesthesia. It activates nicotinic acetylcholine receptors causing initial fasciculations followed by muscle paralysis.

What it means for you

Your RYR1 variant falls under Uncertain Susceptibility for malignant hyperthermia triggered by Succinylcholine. While your specific variant has not been definitively proven to trigger MH, the RYR1 gene is the most common genetic cause of this life-threatening reaction, and uncertainty does not equate to safety. Your anesthesia team should be informed of this genetic finding, and a cautious anesthetic plan that avoids triggering agents is prudent. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

CPIC: A

PharmGKB: 1A

FDA: Actionable PGx

CPIC

RYR1 (uncertain susceptibility)

Recommendation: No CACNA1S genotype is available. Based on RYR1 status, clinical findings, family history, further genetic testing and other laboratory data should guide use of halogenated volatile anesthetics or depolarizing muscle relaxants.

Implications: RYR1: These results do not eliminate the chance that this patient is susceptible to malignant hyperthermia (MH). The genetic cause of about half of all MH survivors, with MH susceptibility confirmed by contracture test, remains unknown (PMID 28902675).

CPIC:<https://cpicpgx.org/guidelines/cpic-guideline-for-ryr1-and-cacna1s/> | **Citations:**<https://pubmed.ncbi.nlm.nih.gov/30499100/>

MUSCULO-SKELETAL SYSTEM DRUGS

Tenoxicam

Genes Analyzed

Gene	Diplotype	Phenotype	Status
CYP2C9	*1/*11	Intermediate Metabolizer	✓ Analyzed

Other Medications That May Further Influence This Result

Your **CYP2C9** result shows **Intermediate Metabolizer** status and may influence how your body handles **Tenoxicam**. The following are examples of FDA-listed medications or substances that affect **CYP2C9** and may further modify drug exposure or response if taken together:

Amiodarone

Fluconazole

Miconazole

Piperine

If you are currently taking any of the above, mention this to your prescriber. This is general reference information based on your genetic result – it is not a personalized check of your current medications.

About this medication

Tenoxicam (Tilcotil) is a Musculo-Skeletal System oxycam NSAID with a long half-life used for musculoskeletal disorders, osteoarthritis, and rheumatoid arthritis.

What it means for you

Your CYP2C9 Intermediate Metabolizer status (Activity Score 1.5, carrying one reduced-function CYP2C9 allele) means you process Tenoxicam at a reduced rate, resulting in mildly elevated drug levels compared to normal metabolizers. This may enhance therapeutic effects but also increases adverse effect risk. Your prescriber may recommend dose adjustments or additional monitoring. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

CPIC: A

PharmGKB: 1A

CPIC

CYP2C9 (Intermediate Metabolizer (AS:1.5))

Recommendation: Initiate therapy with recommended starting dose. In accordance with the prescribing information, use the lowest effective dosage for shortest duration consistent with individual patient treatment goals.

Other Considerations: IMs might have a higher than normal risk of adverse events especially in individuals with other factors affecting clearance of these drugs such as hepatic impairment or advanced age.

Implications: CYP2C9: Mildly reduced metabolism

CPIC:<https://cpicpgx.org/guidelines/cpic-guideline-for-nsaids-based-on-cyp2c9-genotype/> | **Citations:**<https://pubmed.ncbi.nlm.nih.gov/32189324/>

CARDIOVASCULAR SYSTEM DRUGS

Atorvastatin

Genes Analyzed

Gene	Diplotype	Phenotype	Status
SLCO1B1	*1/*20	Normal Function	✓ Analyzed

Other Associated Genes:

ABCG2, **CYP3A4** and **CYP3A5** are known to affect this drug's metabolism (GSI catalog), but was not analyzed by PharmCAT in this report.

About this medication

Atorvastatin (Lipitor) is a Cardiovascular System drug belonging to the HMG-CoA reductase inhibitor ('statin') class. It lowers LDL cholesterol and triglycerides by blocking cholesterol synthesis in the liver and is widely prescribed to reduce the risk of heart attack, stroke, and coronary artery disease.

What it means for you

Your SLCO1B1 Normal Function status means your hepatic OATP1B1 transporter efficiently clears Atorvastatin from the bloodstream into the liver. No elevated risk of statin-induced myopathy based on SLCO1B1 status alone is expected, and standard statin dosing applies. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

CPIC: A

PharmGKB: 1A

CPIC

SLCO1B1 (normal function)

Recommendation: Prescribe desired starting dose and adjust doses based on disease-specific guidelines.

Other Considerations: The potential for drug-drug interactions and dose limits based on renal and hepatic function and ancestry should be evaluated prior to initiating a statin.

Implications: SLCO1B1: Typical myopathy risk and statin exposure

CPIC:<https://cpicpgx.org/guidelines/cpic-guideline-for-statins/> | **Citations:**<https://pubmed.ncbi.nlm.nih.gov/35152405/>

CARDIOVASCULAR SYSTEM DRUGS

Fluvastatin

Genes Analyzed

Gene	Diplotype	Phenotype	Status
CYP2C9	*1/*11	Intermediate Metabolizer	✓ Analyzed
SLCO1B1	*1/*20	Normal Function	✓ Analyzed

Other Associated Genes:

ABCG2, **CYP3A4** and **CYP3A5** are known to affect this drug's metabolism (GSI catalog), but was not analyzed by PharmCAT in this report.

Other Medications That May Further Influence This Result

Your **CYP2C9** result shows **Intermediate Metabolizer** status and may influence how your body handles **Fluvastatin**. The following are examples of FDA-listed medications or substances that affect **CYP2C9** and may further modify drug exposure or response if taken together:

Amiodarone

Fluconazole

Miconazole

Piperine

If you are currently taking any of the above, mention this to your prescriber. This is general reference information based on your genetic result – it is not a personalized check of your current medications.

About this medication

Fluvastatin (Lescol, Lescol XL) is a Cardiovascular System drug in the statin class that works by competitively inhibiting HMG-CoA reductase to reduce hepatic cholesterol synthesis. It is used to lower elevated LDL cholesterol and reduce the risk of cardiovascular events.

What it means for you

CYP2C9

Your CYP2C9 Intermediate Metabolizer status (Activity Score 1.5, carrying one reduced-function CYP2C9 allele) means you process Fluvastatin at a reduced rate, resulting in mildly elevated drug levels compared to normal metabolizers. This may enhance therapeutic effects but also increases adverse effect risk. Your prescriber may recommend dose adjustments or additional monitoring. Always consult your healthcare provider before starting, stopping, or changing your medication.

SLCO1B1

Your SLCO1B1 Normal Function status means your hepatic OATP1B1 transporter efficiently clears Fluvastatin from the bloodstream into the liver. No elevated risk of statin-induced myopathy based on SLCO1B1 status alone is expected, and standard statin dosing applies. Always consult your healthcare provider before starting, stopping, or changing your medication.

CPIC

CYP2C9 (Intermediate Metabolizer (AS:1.5))

Recommendation: Based on CYP2C9 status, prescribe $\leq 40\text{mg}$ per day as a starting dose and adjust doses of fluvastatin based on disease-specific guidelines. If dose $> 40\text{mg}$ needed for desired efficacy, consider an alternative statin or combination therapy (i.e., fluvastatin plus non-statin guideline directed medical therapy) (PMID: 30423391). SLCO1B1 genotype result is not available.

Other Considerations: The potential for drug-drug interactions and dose limits based on renal and hepatic function should be evaluated prior to initiating a statin. The effects of drug-drug interactions may be more pronounced resulting in a higher risk of myopathy.

Implications: CYP2C9: Increased fluvastatin exposure as compared to normal metabolizer which may translate to increased myopathy risk. SLCO1B1: n/a

SLCO1B1 (normal function)

Recommendation: Based on SLCO1B1 status, prescribe desired starting dose and adjust doses based on disease-specific guidelines. CYP2C9 genotype result is not available.

Other Considerations: The potential for drug-drug interactions and dose limits based on renal and hepatic function and ancestry should be evaluated prior to initiating a statin.

Implications: CYP2C9: n/a SLCO1B1: Typical myopathy risk and statin exposure.

CPIC: <https://cpicpgx.org/guidelines/cpic-guideline-for-statins/> | Citations: <https://pubmed.ncbi.nlm.nih.gov/35152405/>

CARDIOVASCULAR SYSTEM DRUGS

Hydralazine

Genes Analyzed

Gene	Diplotype	Phenotype	Status
NAT2	*5/*6	Poor Metabolizer	✓ Analyzed

About this medication

Hydralazine (Apresoline) is a Cardiovascular System direct vasodilator used for hypertension and heart failure.

What it means for you

NAT2 Slow/Poor Acetylators have reduced activity of the N-acetyltransferase 2 enzyme, which is responsible for inactivating Hydralazine and other drugs that undergo acetylation metabolism. As a result, Hydralazine accumulates to higher blood levels and remains active for longer periods, increasing the risk of dose-dependent adverse effects. For isoniazid, slow acetylators are at higher risk of peripheral neuropathy and hepatotoxicity. For hydralazine and procainamide, slow acetylators are at higher risk of drug-induced lupus. Dose adjustment and close monitoring for drug toxicity are recommended. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

CPIC: A

PharmGKB: 3

FDA: Actionable PGx

CPIC

NAT2 (Poor Metabolizer)

Recommendation: Initiate therapy at a total daily dose of 40 to 75 mg. Carefully titrate dose upward to clinical effect or guideline-recommended dose; use caution with total daily hydralazine doses of 200 mg or more.

Implications: Predicted to have increased hydralazine plasma concentrations compared to NAT2 rapid and intermediate metabolizers, which may lead to both increased efficacy and adverse effects, including drug-induced systemic lupus erythematosus.

CPIC:<https://cpicpgx.org/guidelines/cpic-guideline-for-hydralazine-and-nat2/> | **Citations:**<https://pubmed.ncbi.nlm.nih.gov/40974042/>

CARDIOVASCULAR SYSTEM DRUGS

Lovastatin

Genes Analyzed

Gene	Diplotype	Phenotype	Status
SLCO1B1	*1/*20	Normal Function	✓ Analyzed

Other Associated Genes:

ABCG2, **CYP3A4** and **CYP3A5** are known to affect this drug's metabolism (GSI catalog), but was not analyzed by PharmCAT in this report.

About this medication

Lovastatin (Mevacor, Altoprev) is a naturally derived Cardiovascular System statin drug that inhibits HMG-CoA reductase to decrease cholesterol production in the liver. It is used to lower LDL cholesterol and reduce the risk of coronary heart disease and atherosclerosis.

What it means for you

Your SLCO1B1 Normal Function status means your hepatic OATP1B1 transporter efficiently clears Lovastatin from the bloodstream into the liver. No elevated risk of statin-induced myopathy based on SLCO1B1 status alone is expected, and standard statin dosing applies. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

CPIC: A

PharmGKB: 1A

CPIC

SLCO1B1 (normal function)

Recommendation: Prescribe desired starting dose and adjust doses based on disease-specific guidelines.

Other Considerations: The potential for drug-drug interactions and dose limits based on renal and hepatic function and ancestry should be evaluated prior to initiating a statin.

Implications: SLCO1B1: Typical myopathy risk and statin exposure

CPIC:<https://cpicpgx.org/guidelines/cpic-guideline-for-statins/> | **Citations:**<https://pubmed.ncbi.nlm.nih.gov/35152405/>

CARDIOVASCULAR SYSTEM DRUGS

Mavacamten

Genes Analyzed

Gene	Diplotype	Phenotype	Status
CYP2C19	*1/*2	Intermediate Metabolizer	✓ Analyzed

Other Medications That May Further Influence This Result

Your **CYP2C19** result shows **Intermediate Metabolizer** status and may influence how your body handles **Mavacamten**. The following are examples of FDA-listed medications or substances that affect **CYP2C19** and may further modify drug exposure or response if taken together:

Fluconazole

Fluoxetine

Fluvoxamine

Ticlopidine

If you are currently taking any of the above, mention this to your prescriber. This is general reference information based on your genetic result — it is not a personalized check of your current medications.

About this medication

Mavacamten (Camzyos) is a Cardiovascular System cardiac myosin inhibitor that reduces the number of myosin-actin cross-bridges formed during systole. It is used for symptomatic obstructive hypertrophic cardiomyopathy (oHCM).

What it means for you

Your CYP2C19 Intermediate Metabolizer status means you have reduced — but not absent — enzyme activity for metabolizing Mavacamten. This typically leads to moderately elevated drug levels compared to normal metabolizers, potentially increasing the drug's effects and side effect risk. Your prescriber may recommend monitoring or a moderate dose adjustment. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

DPWG

CYP2C19 (Intermediate Metabolizer)

Recommendation: NO action is required for this gene–drug interaction.

Implications: Although the plasma concentration of mavacamten may increase due to the reduced CYP2C19 activity, this does not appear to lead to an increase in side effects.

FDA

CYP2C19 (Intermediate Metabolizer)

Recommendation: "Results in higher systemic concentrations and may have higher adverse reaction risk (heart failure). Dosage is based on individual response. The dose titration and monitoring schedule accounts for differences due to CYP2C19 genetic variation, so adjustments based on CYP2C19 genotype are not necessary. Refer to FDA labeling for specific dosing recommendations and monitoring."

Affected Subgroup: CYP2C9 intermediate or poor metabolizers

CARDIOVASCULAR SYSTEM DRUGS

Pitavastatin

Genes Analyzed

Gene	Diplotype	Phenotype	Status
SLCO1B1	*1/*20	Normal Function	✓ Analyzed

Other Associated Genes:

ABCG2, **CYP3A4** and **CYP3A5** are known to affect this drug's metabolism (GSI catalog), but was not analyzed by PharmCAT in this report.

About this medication

Pitavastatin (Livalo, Zypitamag) is a Cardiovascular System statin that potently inhibits HMG-CoA reductase to lower LDL cholesterol. Unlike other statins, it undergoes minimal CYP450 metabolism, reducing potential drug interactions.

What it means for you

Your SLCO1B1 Normal Function status means your hepatic OATP1B1 transporter efficiently clears Pitavastatin from the bloodstream into the liver. No elevated risk of statin-induced myopathy based on SLCO1B1 status alone is expected, and standard statin dosing applies. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

CPIC: A

PharmGKB: 1A

CPIC

SLCO1B1 (normal function)

Recommendation: Prescribe desired starting dose and adjust doses based on disease-specific guidelines.

Other Considerations: The potential for drug-drug interactions and dose limits based on renal and hepatic function and ancestry should be evaluated prior to initiating a statin.

Implications: SLCO1B1: Typical myopathy risk and statin exposure

CPIC:<https://cpicpgx.org/guidelines/cpic-guideline-for-statins/> | Citations:<https://pubmed.ncbi.nlm.nih.gov/35152405/>

CARDIOVASCULAR SYSTEM DRUGS

Pravastatin

Genes Analyzed

Gene	Diplotype	Phenotype	Status
SLCO1B1	*1/*20	Normal Function	✓ Analyzed

Other Associated Genes:

ABCG2, **CYP3A4** and **CYP3A5** are known to affect this drug's metabolism (GSI catalog), but was not analyzed by PharmCAT in this report.

About this medication

Pravastatin (Pravachol) is a Cardiovascular System statin drug that inhibits hepatic cholesterol synthesis via HMG-CoA reductase inhibition. It is used for hypercholesterolemia and reducing cardiovascular risk, and is notably hydrophilic compared to other statins.

What it means for you

Your SLCO1B1 Normal Function status means your hepatic OATP1B1 transporter efficiently clears Pravastatin from the bloodstream into the liver. No elevated risk of statin-induced myopathy based on SLCO1B1 status alone is expected, and standard statin dosing applies. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

CPIC: A

PharmGKB: 1A

CPIC

SLCO1B1 (normal function)

Recommendation: Prescribe desired starting dose and adjust doses based on disease-specific guidelines.

Other Considerations: The potential for drug-drug interactions and dose limits based on renal and hepatic function and ancestry should be evaluated prior to initiating a statin.

Implications: SLCO1B1: Typical myopathy risk and statin exposure

CPIC:<https://cpicpgx.org/guidelines/cpic-guideline-for-statins/> | **Citations:**<https://pubmed.ncbi.nlm.nih.gov/35152405/>

CARDIOVASCULAR SYSTEM DRUGS

Procainamide

Genes Analyzed

Gene	Diplotype	Phenotype	Status
NAT2	*5/*6	Poor Metabolizer	✓ Analyzed

About this medication

Procainamide (Procanbid) is a Cardiovascular System Class IA antiarrhythmic drug that slows cardiac conduction by blocking sodium channels. It is used to treat ventricular and supraventricular arrhythmias, though long-term use is limited by side effects including drug-induced lupus.

What it means for you

NAT2 Slow/Poor Acetylators have reduced activity of the N-acetyltransferase 2 enzyme, which is responsible for inactivating Procainamide and other drugs that undergo acetylation metabolism. As a result, Procainamide accumulates to higher blood levels and remains active for longer periods, increasing the risk of dose-dependent adverse effects. For isoniazid, slow acetylators are at higher risk of peripheral neuropathy and hepatotoxicity. For hydralazine and procainamide, slow acetylators are at higher risk of drug-induced lupus. Dose adjustment and close monitoring for drug toxicity are recommended. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

FDA: Actionable PGx

FDA

NAT2 (Poor Metabolizer)

Recommendation: "Alters systemic parent drug and metabolite concentrations. May result in higher adverse reaction risk." Note: The FDA table refers to the gene "Nonspecific (NAT)" which ClinPGx interprets as NAT2.

Affected Subgroup: Nonspecific (NAT) poor metabolizers

CARDIOVASCULAR SYSTEM DRUGS

Rosuvastatin

Genes Analyzed

Gene	Diplotype	Phenotype	Status
ABCG2	rs2231142 reference (G)/rs2231142 reference (G)	Normal Function	✓ Analyzed
SLCO1B1	*1/*20	Normal Function	✓ Analyzed

Other Associated Genes:

CYP3A4 and **CYP3A5** are known to affect this drug's metabolism (GSI catalog), but was not analyzed by PharmCAT in this report.

About this medication

Rosuvastatin (Crestor, Ezallor) is a potent Cardiovascular System statin that inhibits HMG-CoA reductase to markedly reduce LDL cholesterol and triglycerides. It is prescribed for primary hyperlipidemia and atherosclerotic cardiovascular disease risk reduction.

What it means for you

ABCG2

Your ABCG2 transporter function is Normal, meaning your body efficiently effluxes Rosuvastatin through the intestinal and renal pathways as expected. No pharmacogenomic dose adjustment is anticipated based on ABCG2 status alone. Standard dosing guidelines apply, though other clinical factors may influence your treatment. Always consult your healthcare provider before starting, stopping, or changing your medication.

SLCO1B1

Your SLCO1B1 Normal Function status means your hepatic OATP1B1 transporter efficiently clears Rosuvastatin from the bloodstream into the liver. No elevated risk of statin-induced myopathy based on SLCO1B1 status alone is expected, and standard statin dosing applies. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

CPIC: A

PharmGKB: 1A

FDA: Actionable PGx

CPIC

ABCG2 (normal function)

Recommendation: Based on ABCG2 status, prescribe desired starting dose and adjust doses of rosuvastatin based on disease-specific and specific population guidelines. SLCO1B1 genotype result is not available.

Other Considerations: The potential for drug-drug interactions and dose limits based on renal and hepatic function and Asian ancestry should be evaluated prior to initiating rosuvastatin.

Implications: ABCG2: Typical myopathy risk and rosuvastatin exposure SLCO1B1: n/a

SLCO1B1 (normal function)

Recommendation: Based on SLCO1B1 status, prescribe desired starting dose and adjust doses based on disease-specific guidelines. ABCG2 genotype result is not available.

Other Considerations: The potential for drug-drug interactions and dose limits based on renal and hepatic function and ancestry should be evaluated prior to initiating a statin.

Implications: ABCG2: n/a SLCO1B1: Typical myopathy risk and statin exposure

CPIC:<https://cpicpgx.org/guidelines/cpic-guideline-for-statins/> | **Citations:**<https://pubmed.ncbi.nlm.nih.gov/35152405/>

CARDIOVASCULAR SYSTEM DRUGS

Simvastatin

Genes Analyzed

Gene	Diplotype	Phenotype	Status
SLCO1B1	*1/*20	Normal Function	✓ Analyzed

Other Associated Genes:

ABCG2, **CYP3A4** and **CYP3A5** are known to affect this drug's metabolism (GSI catalog), but was not analyzed by PharmCAT in this report.

About this medication

Simvastatin (Zocor, FloLipid) is a Cardiovascular System statin drug derived from lovastatin that reduces cholesterol synthesis by inhibiting HMG-CoA reductase in the liver. It is one of the most widely prescribed medications for lowering LDL cholesterol and preventing cardiovascular events.

What it means for you

Your SLCO1B1 Normal Function status means your hepatic OATP1B1 transporter efficiently clears Simvastatin from the bloodstream into the liver. No elevated risk of statin-induced myopathy based on SLCO1B1 status alone is expected, and standard statin dosing applies. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

CPIC: A

PharmGKB: 1A

CPIC

SLCO1B1 (normal function)

Recommendation: Prescribe desired starting dose and adjust doses based on disease-specific guidelines.

Other Considerations: The potential for drug-drug interactions and dose limits based on renal and hepatic function and ancestry should be evaluated prior to initiating a statin.

Implications: SLCO1B1: Typical myopathy risk and statin exposure

CPIC:<https://cpicpgx.org/guidelines/cpic-guideline-for-statins/> | **Citations:**<https://pubmed.ncbi.nlm.nih.gov/22617227/>;
<https://pubmed.ncbi.nlm.nih.gov/24918167/>;<https://pubmed.ncbi.nlm.nih.gov/35152405/>

NO GROUP ASSIGNED

Amifampridine phosphate

Genes Analyzed

Gene	Diplotype	Phenotype	Status
NAT2	*5/*6	Poor Metabolizer	✓ Analyzed

About this medication

Amifampridine phosphate (Firdapse) is the phosphate salt form of amifampridine, a Nervous System voltage-gated potassium channel blocker used for Lambert-Eaton myasthenic syndrome. The phosphate form provides improved stability.

What it means for you

NAT2 Slow/Poor Acetylators have reduced activity of the N-acetyltransferase 2 enzyme, which is responsible for inactivating Amifampridine Phosphate and other drugs that undergo acetylation metabolism. As a result, Amifampridine Phosphate accumulates to higher blood levels and remains active for longer periods, increasing the risk of dose-dependent adverse effects. For isoniazid, slow acetylators are at higher risk of peripheral neuropathy and hepatotoxicity. For hydralazine and procainamide, slow acetylators are at higher risk of drug-induced lupus. Dose adjustment and close monitoring for drug toxicity are recommended. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

FDA: Actionable PGx

FDA

NAT2 (Poor Metabolizer)

Recommendation: "The recommended starting dosage of FIRDAPSE in known N-acetyltransferase 2 (NAT2) poor metabolizers is 15 mg daily, taken orally in 3 divided doses...patients should be closely monitored for adverse reactions."

NO GROUP ASSIGNED

Methylene blue

Genes Analyzed

Gene	Diplotype	Phenotype	Status
G6PD	B (reference)/B (reference)	Normal	✓ Analyzed

About this medication

Methylene blue is an Antiinfective/Other drug with multiple uses: treatment of methemoglobinemia (by reducing ferric hemoglobin back to functional ferrous form), urinary antiseptic, and as a dye for medical procedures.

What it means for you

Your G6PD Normal status indicates you have sufficient glucose-6-phosphate dehydrogenase enzyme activity in your red blood cells to protect against oxidative stress from drugs like Methylene Blue. No elevated risk of drug-induced hemolytic anemia is expected based on your G6PD result alone. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

CPIC: A

PharmGKB: 3

FDA: Actionable PGx

CPIC

G6PD (normal)

Recommendation: No reason to avoid based on G6PD status

Implications: G6PD: Low risk of acute hemolytic anemia

CPIC:<https://cpicpgx.org/guidelines/cpic-guideline-for-g6pd/> | Citations:<https://pubmed.ncbi.nlm.nih.gov/24787449/>;
<https://pubmed.ncbi.nlm.nih.gov/36049896/>

NO GROUP ASSIGNED

Toluidine blue

Genes Analyzed

Gene	Diplotype	Phenotype	Status
G6PD	B (reference)/B (reference)	Normal	✓ Analyzed

About this medication

Toluidine blue (Tolonium chloride) is a dye used as a medical diagnostic agent for detecting oral and cervical cancers, and as a topical antidote for methemoglobinemia in some formulations.

What it means for you

Your G6PD Normal status indicates you have sufficient glucose-6-phosphate dehydrogenase enzyme activity in your red blood cells to protect against oxidative stress from drugs like Toluidine Blue. No elevated risk of drug-induced hemolytic anemia is expected based on your G6PD result alone. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

CPIC: A

CPIC

G6PD (normal)

Recommendation: No reason to avoid based on G6PD status

Other Considerations: Toluidine blue classification strength is based on extrapolation from methylene blue data

Implications: G6PD: Low risk of acute hemolytic anemia

CPIC:<https://cpicpgx.org/guidelines/cpic-guideline-for-g6pd/> | **Citations:**<https://pubmed.ncbi.nlm.nih.gov/24787449/>;
<https://pubmed.ncbi.nlm.nih.gov/36049896/>

ALIMENTARY TRACT AND METABOLISM DRUGS

Dexlansoprazole

Genes Analyzed

Gene	Diplotype	Phenotype	Status
CYP2C19	*1/*2	Intermediate Metabolizer	✓ Analyzed

Other Medications That May Further Influence This Result

Your **CYP2C19** result shows **Intermediate Metabolizer** status and may influence how your body handles **Dexlansoprazole**. The following are examples of FDA-listed medications or substances that affect **CYP2C19** and may further modify drug exposure or response if taken together:

Fluconazole

Fluoxetine

Fluvoxamine

Ticlopidine

If you are currently taking any of the above, mention this to your prescriber. This is general reference information based on your genetic result — it is not a personalized check of your current medications.

About this medication

Dexlansoprazole (Dexilant) is an Alimentary Tract and Metabolism PPI containing the R-enantiomer of lansoprazole in a dual delayed-release formulation. It is used for erosive esophagitis healing and symptomatic non-erosive GERD.

What it means for you

Your CYP2C19 Intermediate Metabolizer status means you have reduced — but not absent — enzyme activity for metabolizing Dexlansoprazole. This typically leads to moderately elevated drug levels compared to normal metabolizers, potentially increasing the drug's effects and side effect risk. Your prescriber may recommend monitoring or a moderate dose adjustment. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

CPIC: B

PharmGKB: 1A

FDA: Actionable PGx

CPIC

CYP2C19 (Intermediate Metabolizer)

Recommendation: Initiate standard starting daily dose. For chronic therapy (>12 weeks) and efficacy achieved, consider 50% reduction in daily dose and monitor for continued efficacy.

Implications: CYP2C19: Increased plasma concentration of PPI compared to CYP2C19 NMs; increased chance of efficacy and potentially toxicity

CPIC:<https://cpicpgx.org/guidelines/cpic-guideline-for-proton-pump-inhibitors-and-cyp2c19/> | **Citations:**<https://pubmed.ncbi.nlm.nih.gov/32770672/>

ALIMENTARY TRACT AND METABOLISM DRUGS

Dronabinol

Genes Analyzed

Gene	Diplotype	Phenotype	Status
CYP2C9	*1/*11	Intermediate Metabolizer	✓ Analyzed

Other Medications That May Further Influence This Result

Your **CYP2C9** result shows **Intermediate Metabolizer** status and may influence how your body handles **Dronabinol**. The following are examples of FDA-listed medications or substances that affect **CYP2C9** and may further modify drug exposure or response if taken together:

[Amiodarone](#)[Fluconazole](#)[Miconazole](#)[Piperine](#)

If you are currently taking any of the above, mention this to your prescriber. This is general reference information based on your genetic result — it is not a personalized check of your current medications.

About this medication

Dronabinol (Marinol, Syndros) is a Nervous System cannabinoid (synthetic delta-9-tetrahydrocannabinol) used as an antiemetic for chemotherapy-induced nausea unresponsive to other treatments and for appetite stimulation in AIDS-related anorexia.

What it means for you

Your CYP2C9 Intermediate Metabolizer status (Activity Score 1.5, carrying one reduced-function CYP2C9 allele) means you process Dronabinol at a reduced rate, resulting in mildly elevated drug levels compared to normal metabolizers. This may enhance therapeutic effects but also increases adverse effect risk. Your prescriber may recommend dose adjustments or additional monitoring. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

FDA: Actionable PGx**FDA**

CYP2C9 (Intermediate Metabolizer (AS:1.5))

Recommendation: "Monitoring for increased adverse reactions is recommended in patients known to carry genetic variants associated with diminished CYP2C9 function." See label for more information.

ALIMENTARY TRACT AND METABOLISM DRUGS

Lansoprazole

Genes Analyzed

Gene	Diplootype	Phenotype	Status
CYP2C19	*1/*2	Intermediate Metabolizer	✓ Analyzed

Other Medications That May Further Influence This Result

Your **CYP2C19** result shows **Intermediate Metabolizer** status and may influence how your body handles **Lansoprazole**. The following are examples of FDA-listed medications or substances that affect **CYP2C19** and may further modify drug exposure or response if taken together:

Fluconazole

Fluoxetine

Fluvoxamine

Ticlopidine

If you are currently taking any of the above, mention this to your prescriber. This is general reference information based on your genetic result — it is not a personalized check of your current medications.

About this medication

Lansoprazole (Prevacid) is an Alimentary Tract and Metabolism PPI that inhibits the gastric proton pump. It is used for GERD, peptic ulcers, Zollinger–Ellison syndrome, and H. pylori eradication.

What it means for you

Your CYP2C19 Intermediate Metabolizer status means you have reduced — but not absent — enzyme activity for metabolizing Lansoprazole. This typically leads to moderately elevated drug levels compared to normal metabolizers, potentially increasing the drug's effects and side effect risk. Your prescriber may recommend monitoring or a moderate dose adjustment. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

CPIC: A

PharmGKB: 1A

CPIC

CYP2C19 (Intermediate Metabolizer)

Recommendation: Initiate standard starting daily dose. For chronic therapy (>12 weeks) and efficacy achieved, consider 50% reduction in daily dose and monitor for continued efficacy.

Implications: CYP2C19: Increased plasma concentration of PPI compared to CYP2C19 NMs; increased chance of efficacy and potentially toxicity

CPIC:<https://cpicpgx.org/guidelines/cpic-guideline-for-proton-pump-inhibitors-and-cyp2c19/> | **Citations:**<https://pubmed.ncbi.nlm.nih.gov/32770672/>

ALIMENTARY TRACT AND METABOLISM DRUGS

Omeprazole

Genes Analyzed

Gene	Diplotype	Phenotype	Status
CYP2C19	*1/*2	Intermediate Metabolizer	✓ Analyzed

Other Medications That May Further Influence This Result

Your **CYP2C19** result shows **Intermediate Metabolizer** status and may influence how your body handles **Omeprazole**. The following are examples of FDA-listed medications or substances that affect **CYP2C19** and may further modify drug exposure or response if taken together:

Fluconazole

Fluoxetine

Fluvoxamine

Ticlopidine

If you are currently taking any of the above, mention this to your prescriber. This is general reference information based on your genetic result — it is not a personalized check of your current medications.

About this medication

Omeprazole (Prilosec, Losec) is an Alimentary Tract and Metabolism proton pump inhibitor (PPI) that irreversibly inhibits the H⁺/K⁺-ATPase enzyme in gastric parietal cells. It is used for GERD, peptic ulcer disease, Helicobacter pylori eradication, and Zollinger–Ellison syndrome.

What it means for you

Your CYP2C19 Intermediate Metabolizer status means you have reduced — but not absent — enzyme activity for metabolizing Omeprazole. This typically leads to moderately elevated drug levels compared to normal metabolizers, potentially increasing the drug's effects and side effect risk. Your prescriber may recommend monitoring or a moderate dose adjustment. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

CPIC: A

PharmGKB: 1A

FDA: Actionable PGx

CPIC

CYP2C19 (Intermediate Metabolizer)

Recommendation: Initiate standard starting daily dose. For chronic therapy (>12 weeks) and efficacy achieved, consider 50% reduction in daily dose and monitor for continued efficacy.

Implications: CYP2C19: Increased plasma concentration of PPI compared to CYP2C19 NMs; increased chance of efficacy and potentially toxicity

CPIC:<https://cpicpgx.org/guidelines/cpic-guideline-for-proton-pump-inhibitors-and-cyp2c19/> | **Citations:**<https://pubmed.ncbi.nlm.nih.gov/32770672/>

ALIMENTARY TRACT AND METABOLISM DRUGS

Pantoprazole

Genes Analyzed

Gene	Diplotype	Phenotype	Status
CYP2C19	*1/*2	Intermediate Metabolizer	✓ Analyzed

Other Medications That May Further Influence This Result

Your **CYP2C19** result shows **Intermediate Metabolizer** status and may influence how your body handles **Pantoprazole**. The following are examples of FDA-listed medications or substances that affect **CYP2C19** and may further modify drug exposure or response if taken together:

Fluconazole

Fluoxetine

Fluvoxamine

Ticlopidine

If you are currently taking any of the above, mention this to your prescriber. This is general reference information based on your genetic result — it is not a personalized check of your current medications.

About this medication

Pantoprazole (Protonix) is an Alimentary Tract and Metabolism PPI used for GERD, erosive esophagitis, and pathological hypersecretory conditions. It has minimal drug interactions compared to other PPIs.

What it means for you

Your CYP2C19 Intermediate Metabolizer status means you have reduced — but not absent — enzyme activity for metabolizing Pantoprazole. This typically leads to moderately elevated drug levels compared to normal metabolizers, potentially increasing the drug's effects and side effect risk. Your prescriber may recommend monitoring or a moderate dose adjustment. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

CPIC: A

PharmGKB: 1A

FDA: Actionable PGx

CPIC

CYP2C19 (Intermediate Metabolizer)

Recommendation: Initiate standard starting daily dose. For chronic therapy (>12 weeks) and efficacy achieved, consider 50% reduction in daily dose and monitor for continued efficacy.

Implications: CYP2C19: Increased plasma concentration of PPI compared to CYP2C19 NMs; increased chance of efficacy and potentially toxicity

CPIC:<https://cpicpgx.org/guidelines/cpic-guideline-for-proton-pump-inhibitors-and-cyp2c19/> | **Citations:**<https://pubmed.ncbi.nlm.nih.gov/32770672/>

ALIMENTARY TRACT AND METABOLISM DRUGS

Sulfasalazine

Genes Analyzed

Gene	Diplotype	Phenotype	Status
NAT2	*5/*6	Poor Metabolizer	✓ Analyzed

Other Associated Genes:

G6PD is known to affect this drug's metabolism (GSI catalog), but was not analyzed by PharmCAT in this report.

About this medication

Sulfasalazine (Azulfidine) is an Alimentary Tract/Immunological drug combining sulfapyridine and 5-aminosalicylic acid. It is used for inflammatory bowel disease (ulcerative colitis, Crohn's disease) and rheumatoid arthritis.

What it means for you

NAT2 Slow/Poor Acetylators have reduced activity of the N-acetyltransferase 2 enzyme, which is responsible for inactivating Sulfasalazine and other drugs that undergo acetylation metabolism. As a result, Sulfasalazine accumulates to higher blood levels and remains active for longer periods, increasing the risk of dose-dependent adverse effects. For isoniazid, slow acetylators are at higher risk of peripheral neuropathy and hepatotoxicity. For hydralazine and procainamide, slow acetylators are at higher risk of drug-induced lupus. Dose adjustment and close monitoring for drug toxicity are recommended. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

FDA: Actionable PGx

FDA

NAT2 (Poor Metabolizer)

Recommendation: "Results in higher systemic metabolite concentrations and higher adverse reaction risk." Note: The FDA table refers to the gene "Nonspecific (NAT)" which ClinPGx interprets as NAT2.

Affected Subgroup: Nonspecific (NAT) poor metabolizers

NERVOUS SYSTEM DRUGS

Amifampridine

Genes Analyzed

Gene	Diplotype	Phenotype	Status
NAT2	*5/*6	Poor Metabolizer	✓ Analyzed

About this medication

Amifampridine (Firdapse, Ruzurgi) is a Nervous System potassium channel blocker that prolongs action potentials at neuromuscular junctions. It is used for Lambert-Eaton myasthenic syndrome, a rare autoimmune neuromuscular disorder.

What it means for you

NAT2 Slow/Poor Acetylators have reduced activity of the N-acetyltransferase 2 enzyme, which is responsible for inactivating Amifampridine and other drugs that undergo acetylation metabolism. As a result, Amifampridine accumulates to higher blood levels and remains active for longer periods, increasing the risk of dose-dependent adverse effects. For isoniazid, slow acetylators are at higher risk of peripheral neuropathy and hepatotoxicity. For hydralazine and procainamide, slow acetylators are at higher risk of drug-induced lupus. Dose adjustment and close monitoring for drug toxicity are recommended. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

FDA: Actionable PGx

FDA

NAT2 (Poor Metabolizer)

Recommendation: "The recommended starting dosage of RUZURGI in pediatric patients weighing 45 kg or more who are known N-acetyltransferase 2 (NAT2) poor metabolizers is 15 mg daily taken orally in divided doses. The recommended starting dosage in pediatric patients weighing less than 45 kg who are known NAT2 poor metabolizers is 7.5 mg daily taken orally in divided doses...monitor for adverse reactions."

NERVOUS SYSTEM DRUGS

Amitriptyline

Genes Analyzed

Gene	Diplootype	Phenotype	Status
CYP2C19	*1/*2	Intermediate Metabolizer	✓ Analyzed
CYP2D6	—	—	ⓘ Not evaluated by platform

Other Medications That May Further Influence This Result

Your **CYP2C19** result shows **Intermediate Metabolizer** status and may influence how your body handles **Amitriptyline**. The following are examples of FDA-listed medications or substances that affect **CYP2C19** and may further modify drug exposure or response if taken together:

Fluconazole

Fluoxetine

Fluvoxamine

Ticlopidine

If you are currently taking any of the above, mention this to your prescriber. This is general reference information based on your genetic result — it is not a personalized check of your current medications.

About this medication

Amitriptyline (Elavil) is a Nervous System tricyclic antidepressant (TCA) that inhibits the reuptake of serotonin and norepinephrine and has strong anticholinergic properties. It is used for major depressive disorder, neuropathic pain, migraine prevention, and insomnia.

What it means for you

Your CYP2C19 Intermediate Metabolizer status means you have reduced — but not absent — enzyme activity for metabolizing Amitriptyline. This typically leads to moderately elevated drug levels compared to normal metabolizers, potentially increasing the drug's effects and side effect risk. Your prescriber may recommend monitoring or a moderate dose adjustment. Always consult your healthcare provider before starting, stopping, or changing your medication.

This gene is not evaluated by consumer DNA genotyping platforms and was therefore not assessed in this analysis. If genotyping of this gene is clinically relevant, a dedicated pharmacogenomic laboratory test may be requested through your healthcare provider.

Clinical Recommendations

CPIC: A

PharmGKB: 1A

FDA: Actionable PGx

CPIC

CYP2C19 (Intermediate Metabolizer)

Recommendation: Initiate therapy with recommended starting dose.

Other Considerations: Patients may receive an initial low dose of a tricyclic, which is then increased over several days to the recommended steady-state dose. The starting dose in this guideline refers to the recommended steady-state dose.

Implications: CYP2C19: Reduced metabolism of tertiary amines compared to normal metabolizers CYP2D6: n/a

CPIC: <https://cpicpgx.org/guidelines/guideline-for-tricyclic-antidepressants-and-cyp2d6-and-cyp2c19/> | Citations: <https://pubmed.ncbi.nlm.nih.gov/23486447/>; <https://pubmed.ncbi.nlm.nih.gov/27997040/>

NERVOUS SYSTEM DRUGS

Brivaracetam

Genes Analyzed

Gene	Diplotype	Phenotype	Status
CYP2C19	*1/*2	Intermediate Metabolizer	✓ Analyzed

Other Medications That May Further Influence This Result

Your **CYP2C19** result shows **Intermediate Metabolizer** status and may influence how your body handles **Brivaracetam**. The following are examples of FDA-listed medications or substances that affect **CYP2C19** and may further modify drug exposure or response if taken together:

Fluconazole

Fluoxetine

Fluvoxamine

Ticlopidine

If you are currently taking any of the above, mention this to your prescriber. This is general reference information based on your genetic result — it is not a personalized check of your current medications.

About this medication

Brivaracetam (Briviact) is a Nervous System anticonvulsant that binds to synaptic vesicle protein 2A (SV2A), modulating neurotransmitter release. It is used as an adjunctive treatment for partial-onset seizures.

What it means for you

Your CYP2C19 Intermediate Metabolizer status means you have reduced — but not absent — enzyme activity for metabolizing Brivaracetam. This typically leads to moderately elevated drug levels compared to normal metabolizers, potentially increasing the drug's effects and side effect risk. Your prescriber may recommend monitoring or a moderate dose adjustment. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

PharmGKB: 3

FDA: Actionable PGx

FDA

CYP2C19 (Intermediate Metabolizer)

Recommendation: "Results in higher systemic concentrations and higher adverse reaction risk. Consider dosage reductions in poor metabolizers."

Affected Subgroup: CYP2C19 intermediate or poor metabolizers

NERVOUS SYSTEM DRUGS

Citalopram

Genes Analyzed

Gene	Diplotype	Phenotype	Status
CYP2C19	*1/*2	Intermediate Metabolizer	✓ Analyzed

Other Associated Genes:

CYP2D6 is known to affect this drug's metabolism (GSI catalog), but was not analyzed by PharmCAT in this report.

Other Medications That May Further Influence This Result

Your **CYP2C19** result shows **Intermediate Metabolizer** status and may influence how your body handles **Citalopram**. The following are examples of FDA-listed medications or substances that affect **CYP2C19** and may further modify drug exposure or response if taken together:

Fluconazole

Fluoxetine

Fluvoxamine

Ticlopidine

If you are currently taking any of the above, mention this to your prescriber. This is general reference information based on your genetic result — it is not a personalized check of your current medications.

About this medication

Citalopram (Celexa) is a Nervous System SSRI antidepressant with high selectivity for serotonin reuptake inhibition. It is used for major depressive disorder and has an FDA dose limitation due to QT-interval prolongation risk.

What it means for you

Your CYP2C19 Intermediate Metabolizer status means you have reduced — but not absent — enzyme activity for metabolizing Citalopram. This typically leads to moderately elevated drug levels compared to normal metabolizers, potentially increasing the drug's effects and side effect risk. Your prescriber may recommend monitoring or a moderate dose adjustment. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

CPIC: A

PharmGKB: 1A

FDA: Actionable PGx

CPIC

CYP2C19 (Intermediate Metabolizer)

Recommendation: Initiate therapy with recommended starting dose. Consider a slower titration schedule and lower maintenance dose than normal metabolizers.

Other Considerations: Drug-drug interactions and other patient characteristics (e.g., age, renal function, liver function) should be considered when adjusting dose.

Implications: CYP2C19: Reduced metabolism when compared to CYP2C19 normal metabolizers. Higher plasma concentrations may increase the probability of side effects.

CPIC:<https://cpicpgx.org/guidelines/cpic-guideline-for-ssri-and-snri-antidepressants/> | **Citations:**<https://pubmed.ncbi.nlm.nih.gov/25974703/>; <https://pubmed.ncbi.nlm.nih.gov/37032427/>

NERVOUS SYSTEM DRUGS

Clobazam

Genes Analyzed

Gene	Diplotype	Phenotype	Status
CYP2C19	*1/*2	Intermediate Metabolizer	✓ Analyzed

Other Medications That May Further Influence This Result

Your **CYP2C19** result shows **Intermediate Metabolizer** status and may influence how your body handles **Clobazam**. The following are examples of FDA-listed medications or substances that affect **CYP2C19** and may further modify drug exposure or response if taken together:

Fluconazole

Fluoxetine

Fluvoxamine

Ticlopidine

If you are currently taking any of the above, mention this to your prescriber. This is general reference information based on your genetic result — it is not a personalized check of your current medications.

About this medication

Clobazam (Onfi, Sympazan) is a Nervous System benzodiazepine that enhances GABA-A receptor-mediated inhibition. It is used as adjunctive therapy for Lennox-Gastaut syndrome and other refractory epilepsies.

What it means for you

Your CYP2C19 Intermediate Metabolizer status means you have reduced — but not absent — enzyme activity for metabolizing Clobazam. This typically leads to moderately elevated drug levels compared to normal metabolizers, potentially increasing the drug's effects and side effect risk. Your prescriber may recommend monitoring or a moderate dose adjustment. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

PharmGKB: 3

FDA: Actionable PGx

FDA

CYP2C19 (Intermediate Metabolizer)

Recommendation: "Results in higher systemic active metabolite concentrations. Poor metabolism results in higher adverse reaction risk. Dosage adjustment is recommended. Refer to FDA labeling for specific dosing recommendations."

Affected Subgroup: CYP2C19 intermediate or poor metabolizers

NERVOUS SYSTEM DRUGS

Clomipramine

Genes Analyzed

Gene	Diplotype	Phenotype	Status
CYP2C19	*1/*2	Intermediate Metabolizer	✓ Analyzed
CYP2D6	—	—	ⓘ Not evaluated by platform

Other Medications That May Further Influence This Result

Your **CYP2C19** result shows **Intermediate Metabolizer** status and may influence how your body handles **Clomipramine**. The following are examples of FDA-listed medications or substances that affect **CYP2C19** and may further modify drug exposure or response if taken together:

Fluconazole

Fluoxetine

Fluvoxamine

Ticlopidine

If you are currently taking any of the above, mention this to your prescriber. This is general reference information based on your genetic result — it is not a personalized check of your current medications.

About this medication

Clomipramine (Anafranil) is a Nervous System tricyclic antidepressant that potently inhibits serotonin reuptake and is considered the gold standard pharmacological treatment for obsessive-compulsive disorder (OCD). It also has significant norepinephrine reuptake inhibition and anticholinergic effects.

What it means for you

Your CYP2C19 Intermediate Metabolizer status means you have reduced — but not absent — enzyme activity for metabolizing Clomipramine. This typically leads to moderately elevated drug levels compared to normal metabolizers, potentially increasing the drug's effects and side effect risk. Your prescriber may recommend monitoring or a moderate dose adjustment. Always consult your healthcare provider before starting, stopping, or changing your medication.

This gene is not evaluated by consumer DNA genotyping platforms and was therefore not assessed in this analysis. If genotyping of this gene is clinically relevant, a dedicated pharmacogenomic laboratory test may be requested through your healthcare provider.

Clinical Recommendations

CPIC: B

PharmGKB: 1A

FDA: Actionable PGx

CPIC

CYP2C19 (Intermediate Metabolizer)

Recommendation: Initiate therapy with recommended starting dose.

Other Considerations: Patients may receive an initial low dose of a tricyclic, which is then increased over several days to the recommended steady-state dose. The starting dose in this guideline refers to the recommended steady-state dose.

Implications: CYP2C19: Reduced metabolism of tertiary amines compared to normal metabolizers CYP2D6: n/a

CPIC:<https://cpicpgx.org/guidelines/guideline-for-tricyclic-antidepressants-and-cyp2d6-and-cyp2c19/> | Citations:<https://pubmed.ncbi.nlm.nih.gov/23486447/>; <https://pubmed.ncbi.nlm.nih.gov/27997040/>

NERVOUS SYSTEM DRUGS

Desflurane

Genes Analyzed

Gene	Diplotype	Phenotype	Status
RYR1	Reference/Reference	Uncertain Susceptibility	✓ Analyzed

About this medication

Desflurane (Suprane) is a Nervous System halogenated volatile anesthetic used for induction and maintenance of general anesthesia during surgery. It is administered as an inhaled gas and produces rapid onset and offset of anesthesia due to its low blood-gas solubility.

What it means for you

Your RYR1 variant falls under Uncertain Susceptibility for malignant hyperthermia triggered by Desflurane. While your specific variant has not been definitively proven to trigger MH, the RYR1 gene is the most common genetic cause of this life-threatening reaction, and uncertainty does not equate to safety. Your anesthesia team should be informed of this genetic finding, and a cautious anesthetic plan that avoids triggering agents is prudent. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

CPIC: A

PharmGKB: 1A

FDA: Actionable PGx

CPIC

RYR1 (uncertain susceptibility)

Recommendation: No CACNA1S genotype is available. Based on RYR1 status, clinical findings, family history, further genetic testing and other laboratory data should guide use of halogenated volatile anesthetics or depolarizing muscle relaxants.

Implications: RYR1: These results do not eliminate the chance that this patient is susceptible to malignant hyperthermia (MH). The genetic cause of about half of all MH survivors, with MH susceptibility confirmed by contracture test, remains unknown (PMID 28902675).

CPIC:<https://cpicpgx.org/guidelines/cpic-guideline-for-ryr1-and-cacna1s/> | **Citations:**<https://pubmed.ncbi.nlm.nih.gov/30499100/>

NERVOUS SYSTEM DRUGS

Doxepin

Genes Analyzed

Gene	Diplotype	Phenotype	Status
CYP2C19	*1/*2	Intermediate Metabolizer	✓ Analyzed
CYP2D6	—	—	ⓘ Not evaluated by platform

Other Medications That May Further Influence This Result

Your **CYP2C19** result shows **Intermediate Metabolizer** status and may influence how your body handles **Doxepin**. The following are examples of FDA-listed medications or substances that affect **CYP2C19** and may further modify drug exposure or response if taken together:

[Fluconazole](#)
[Fluoxetine](#)
[Fluvoxamine](#)
[Ticlopidine](#)

If you are currently taking any of the above, mention this to your prescriber. This is general reference information based on your genetic result — it is not a personalized check of your current medications.

About this medication

Doxepin (Sinequan, Silenor) is a Nervous System tricyclic antidepressant with potent histamine H1 receptor antagonism. At lower doses it is FDA-approved for chronic insomnia; at higher doses it is used for depression and anxiety disorders.

What it means for you

Your CYP2C19 Intermediate Metabolizer status means you have reduced — but not absent — enzyme activity for metabolizing Doxepin. This typically leads to moderately elevated drug levels compared to normal metabolizers, potentially increasing the drug's effects and side effect risk. Your prescriber may recommend monitoring or a moderate dose adjustment. Always consult your healthcare provider before starting, stopping, or changing your medication.

This gene is not evaluated by consumer DNA genotyping platforms and was therefore not assessed in this analysis. If genotyping of this gene is clinically relevant, a dedicated pharmacogenomic laboratory test may be requested through your healthcare provider.

Clinical Recommendations

CPIC: B PharmGKB: 1A FDA: Actionable PGx

CPIC

CYP2C19 (Intermediate Metabolizer)

Recommendation: Initiate therapy with recommended starting dose.

Other Considerations: Patients may receive an initial low dose of a tricyclic, which is then increased over several days to the recommended steady-state dose. The starting dose in this guideline refers to the recommended steady-state dose.

Implications: CYP2C19: Reduced metabolism of tertiary amines compared to normal metabolizers CYP2D6: n/a

CPIC:<https://cpicpgx.org/guidelines/guideline-for-tricyclic-antidepressants-and-cyp2d6-and-cyp2c19/> | Citations:<https://pubmed.ncbi.nlm.nih.gov/23486447/>; <https://pubmed.ncbi.nlm.nih.gov/27997040/>

NERVOUS SYSTEM DRUGS

Enflurane

Genes Analyzed

Gene	Diplotype	Phenotype	Status
RYR1	Reference/Reference	Uncertain Susceptibility	✓ Analyzed

About this medication

Enflurane (Ethrane) is a halogenated volatile anesthetic belonging to the Nervous System category used for induction and maintenance of general anesthesia. It produces unconsciousness and muscle relaxation through modulation of GABA-A receptors and other ion channels.

What it means for you

Your RYR1 variant falls under Uncertain Susceptibility for malignant hyperthermia triggered by Enflurane. While your specific variant has not been definitively proven to trigger MH, the RYR1 gene is the most common genetic cause of this life-threatening reaction, and uncertainty does not equate to safety. Your anesthesia team should be informed of this genetic finding, and a cautious anesthetic plan that avoids triggering agents is prudent. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

CPIC: A

PharmGKB: 1A

FDA: Actionable PGx

CPIC

RYR1 (uncertain susceptibility)

Recommendation: No CACNA1S genotype is available. Based on RYR1 status, clinical findings, family history, further genetic testing and other laboratory data should guide use of halogenated volatile anesthetics or depolarizing muscle relaxants.

Implications: RYR1: These results do not eliminate the chance that this patient is susceptible to malignant hyperthermia (MH). The genetic cause of about half of all MH survivors, with MH susceptibility confirmed by contracture test, remains unknown (PMID 28902675).

CPIC:<https://cpicpgx.org/guidelines/cpic-guideline-for-ryr1-and-cacna1s/> | **Citations:**<https://pubmed.ncbi.nlm.nih.gov/30499100/>

NERVOUS SYSTEM DRUGS

Escitalopram

Genes Analyzed

Gene	Diplotype	Phenotype	Status
CYP2C19	*1/*2	Intermediate Metabolizer	✓ Analyzed

Other Associated Genes:

CYP2D6 is known to affect this drug's metabolism (GSI catalog), but was not analyzed by PharmCAT in this report.

Other Medications That May Further Influence This Result

Your **CYP2C19** result shows **Intermediate Metabolizer** status and may influence how your body handles **Escitalopram**. The following are examples of FDA-listed medications or substances that affect **CYP2C19** and may further modify drug exposure or response if taken together:

Fluconazole

Fluoxetine

Fluvoxamine

Ticlopidine

If you are currently taking any of the above, mention this to your prescriber. This is general reference information based on your genetic result — it is not a personalized check of your current medications.

About this medication

Escitalopram (Lexapro, Ciprallex) is a Nervous System SSRI and the S-enantiomer of citalopram, offering greater selectivity for the serotonin transporter. It is used for major depressive disorder and generalized anxiety disorder.

What it means for you

Your CYP2C19 Intermediate Metabolizer status means you have reduced — but not absent — enzyme activity for metabolizing Escitalopram. This typically leads to moderately elevated drug levels compared to normal metabolizers, potentially increasing the drug's effects and side effect risk. Your prescriber may recommend monitoring or a moderate dose adjustment. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

CPIC: A

PharmGKB: 1A

FDA: Actionable PGx

CPIC

CYP2C19 (Intermediate Metabolizer)

Recommendation: Initiate therapy with recommended starting dose. Consider a slower titration schedule and lower maintenance dose than normal metabolizers.

Other Considerations: Drug-drug interactions and other patient characteristics (e.g., age, renal function, liver function) should be considered when adjusting dose.

Implications: CYP2C19: Reduced metabolism when compared to CYP2C19 normal metabolizers. Higher plasma concentrations may increase the probability of side effects.

CPIC:<https://cpicpgx.org/guidelines/cpic-guideline-for-ssri-and-snri-antidepressants/> | **Citations:**<https://pubmed.ncbi.nlm.nih.gov/25974703/>; <https://pubmed.ncbi.nlm.nih.gov/37032427/>

NERVOUS SYSTEM DRUGS

Fosphenytoin

Genes Analyzed

Gene	Diplotype	Phenotype	Status
CYP2C9	*1/*11	Intermediate Metabolizer	✓ Analyzed

Other Associated Genes:

HLA-B is known to affect this drug's metabolism (GSI catalog), but was not analyzed by PharmCAT in this report.

Other Medications That May Further Influence This Result

Your **CYP2C9** result shows **Intermediate Metabolizer** status and may influence how your body handles **Fosphenytoin**. The following are examples of FDA-listed medications or substances that affect **CYP2C9** and may further modify drug exposure or response if taken together:

Amiodarone

Fluconazole

Miconazole

Piperine

If you are currently taking any of the above, mention this to your prescriber. This is general reference information based on your genetic result – it is not a personalized check of your current medications.

About this medication

Fosphenytoin (Cerebyx) is a Nervous System water-soluble prodrug of phenytoin used for parenteral administration in status epilepticus and neurosurgical seizure prophylaxis. It is converted to phenytoin in the body.

What it means for you

Your CYP2C9 Intermediate Metabolizer status (Activity Score 1.5, carrying one reduced-function CYP2C9 allele) means you process Fosphenytoin at a reduced rate, resulting in mildly elevated drug levels compared to normal metabolizers. This may enhance therapeutic effects but also increases adverse effect risk. Your prescriber may recommend dose adjustments or additional monitoring. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

CPIC: A

FDA: Actionable PGx

CPIC

CYP2C9 (Intermediate Metabolizer (AS:1.5))

Patients who are phenytoin-naïve

Recommendation: No adjustments needed from typical dosing strategies. Subsequent doses should be adjusted according to therapeutic drug monitoring, response and side effects. An HLA-B*15:02 negative test does not eliminate the risk of phenytoin-induced SJS/TEN and patients should be carefully monitored according to a usual standard.

Implications: CYP2C9: Slightly reduced phenytoin metabolism; however, this does not appear to translate into increased side effects. HLA-B: n/a Patients who have been continuously taking phenytoin or fosphenytoin for longer than 3 months without developing cutaneous reactions

Recommendation: No adjustments needed from typical dosing strategies. Subsequent doses should be adjusted according to therapeutic drug monitoring, response and side effects. An HLA-B*15:02 negative test does not eliminate the risk of phenytoin-induced SJS/TEN and patients should be carefully monitored according to a usual standard.

Implications: CYP2C9: Slightly reduced phenytoin metabolism; however, this does not appear to translate into increased side effects. HLA-B: n/a

CPIC:<https://cpicpgx.org/guidelines/guideline-for-phenytoin-and-cyp2c9-and-hla-b/> | **Citations:**<https://pubmed.ncbi.nlm.nih.gov/25099164/>;<https://pubmed.ncbi.nlm.nih.gov/32779747/>

NERVOUS SYSTEM DRUGS

Halothane

Genes Analyzed

Gene	Diplotype	Phenotype	Status
RYR1	Reference/Reference	Uncertain Susceptibility	✓ Analyzed

About this medication

Halothane (Fluothane) is a halogenated volatile anesthetic used for general anesthesia induction and maintenance. It was once widely used but has been largely replaced by newer agents; it acts on central nervous system ion channels to produce anesthesia and has significant cardiovascular depressant effects.

What it means for you

Your RYR1 variant falls under Uncertain Susceptibility for malignant hyperthermia triggered by Halothane. While your specific variant has not been definitively proven to trigger MH, the RYR1 gene is the most common genetic cause of this life-threatening reaction, and uncertainty does not equate to safety. Your anesthesia team should be informed of this genetic finding, and a cautious anesthetic plan that avoids triggering agents is prudent. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

CPIC: A

PharmGKB: 1A

CPIC

RYR1 (uncertain susceptibility)

Recommendation: No CACNA1S genotype is available. Based on RYR1 status, clinical findings, family history, further genetic testing and other laboratory data should guide use of halogenated volatile anesthetics or depolarizing muscle relaxants.

Implications: RYR1: These results do not eliminate the chance that this patient is susceptible to malignant hyperthermia (MH). The genetic cause of about half of all MH survivors, with MH susceptibility confirmed by contracture test, remains unknown (PMID 28902675).

CPIC:<https://cpicpgx.org/guidelines/cpic-guideline-for-ryr1-and-cacna1s/> | **Citations:**<https://pubmed.ncbi.nlm.nih.gov/30499100/>

NERVOUS SYSTEM DRUGS

Imipramine

Genes Analyzed

Gene	Diplootype	Phenotype	Status
CYP2C19	*1/*2	Intermediate Metabolizer	✓ Analyzed
CYP2D6	—	—	ⓘ Not evaluated by platform

Other Medications That May Further Influence This Result

Your **CYP2C19** result shows **Intermediate Metabolizer** status and may influence how your body handles **Imipramine**. The following are examples of FDA-listed medications or substances that affect **CYP2C19** and may further modify drug exposure or response if taken together:

Fluconazole

Fluoxetine

Fluvoxamine

Ticlopidine

If you are currently taking any of the above, mention this to your prescriber. This is general reference information based on your genetic result — it is not a personalized check of your current medications.

About this medication

Imipramine (Tofranil) is a Nervous System tricyclic antidepressant (TCA) that inhibits the reuptake of both serotonin and norepinephrine. It is used for depression, panic disorder, and childhood enuresis.

What it means for you

Your CYP2C19 Intermediate Metabolizer status means you have reduced — but not absent — enzyme activity for metabolizing Imipramine. This typically leads to moderately elevated drug levels compared to normal metabolizers, potentially increasing the drug's effects and side effect risk. Your prescriber may recommend monitoring or a moderate dose adjustment. Always consult your healthcare provider before starting, stopping, or changing your medication.

This gene is not evaluated by consumer DNA genotyping platforms and was therefore not assessed in this analysis. If genotyping of this gene is clinically relevant, a dedicated pharmacogenomic laboratory test may be requested through your healthcare provider.

Clinical Recommendations

CPIC: B

PharmGKB: 1A

FDA: Actionable PGx

CPIC

CYP2C19 (Intermediate Metabolizer)

Recommendation: Initiate therapy with recommended starting dose.

Other Considerations: Patients may receive an initial low dose of a tricyclic, which is then increased over several days to the recommended steady-state dose. The starting dose in this guideline refers to the recommended steady-state dose.

Implications: CYP2C19: Reduced metabolism of tertiary amines compared to normal metabolizers CYP2D6: n/a

CPIC:<https://cpicpgx.org/guidelines/guideline-for-tricyclic-antidepressants-and-cyp2d6-and-cyp2c19/> | Citations:<https://pubmed.ncbi.nlm.nih.gov/23486447/>; <https://pubmed.ncbi.nlm.nih.gov/27997040/>

NERVOUS SYSTEM DRUGS

Isoflurane

Genes Analyzed

Gene	Diplotype	Phenotype	Status
RYR1	Reference/Reference	Uncertain Susceptibility	✓ Analyzed

About this medication

Isoflurane (Forane) is a halogenated volatile inhalational anesthetic used during surgical procedures. It provides anesthesia through potentiation of inhibitory neurotransmission (GABA-A) and produces dose-dependent cardiovascular and respiratory depression.

What it means for you

Your RYR1 variant falls under Uncertain Susceptibility for malignant hyperthermia triggered by Isoflurane. While your specific variant has not been definitively proven to trigger MH, the RYR1 gene is the most common genetic cause of this life-threatening reaction, and uncertainty does not equate to safety. Your anesthesia team should be informed of this genetic finding, and a cautious anesthetic plan that avoids triggering agents is prudent. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

CPIC: A

PharmGKB: 1A

FDA: Actionable PGx

CPIC

RYR1 (uncertain susceptibility)

Recommendation: No CACNA1S genotype is available. Based on RYR1 status, clinical findings, family history, further genetic testing and other laboratory data should guide use of halogenated volatile anesthetics or depolarizing muscle relaxants.

Implications: RYR1: These results do not eliminate the chance that this patient is susceptible to malignant hyperthermia (MH). The genetic cause of about half of all MH survivors, with MH susceptibility confirmed by contracture test, remains unknown (PMID 28902675).

CPIC:<https://cpicpgx.org/guidelines/cpic-guideline-for-ryr1-and-cacna1s/> | **Citations:**<https://pubmed.ncbi.nlm.nih.gov/30499100/>

NERVOUS SYSTEM DRUGS

Methoxyflurane

Genes Analyzed

Gene	Diplotype	Phenotype	Status
RYR1	Reference/Reference	Uncertain Susceptibility	✓ Analyzed

About this medication

Methoxyflurane (Pentrox) is a volatile halogenated anesthetic primarily used for its analgesic properties in emergency and procedural pain management. At low doses, it provides rapid-onset analgesia without full unconsciousness.

What it means for you

Your RYR1 variant falls under Uncertain Susceptibility for malignant hyperthermia triggered by Methoxyflurane. While your specific variant has not been definitively proven to trigger MH, the RYR1 gene is the most common genetic cause of this life-threatening reaction, and uncertainty does not equate to safety. Your anesthesia team should be informed of this genetic finding, and a cautious anesthetic plan that avoids triggering agents is prudent. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

CPIC: A

PharmGKB: 1A

CPIC

RYR1 (uncertain susceptibility)

Recommendation: No CACNA1S genotype is available. Based on RYR1 status, clinical findings, family history, further genetic testing and other laboratory data should guide use of halogenated volatile anesthetics or depolarizing muscle relaxants.

Implications: RYR1: These results do not eliminate the chance that this patient is susceptible to malignant hyperthermia (MH). The genetic cause of about half of all MH survivors, with MH susceptibility confirmed by contracture test, remains unknown (PMID 28902675).

CPIC:<https://cpicpgx.org/guidelines/cpic-guideline-for-ryr1-and-cacna1s/> | **Citations:**<https://pubmed.ncbi.nlm.nih.gov/30499100/>

NERVOUS SYSTEM DRUGS

Phenytoin

Genes Analyzed

Gene	Diplotype	Phenotype	Status
CYP2C9	*1/*11	Intermediate Metabolizer	✓ Analyzed

Other Associated Genes:

CYP2C19 and **HLA-B** are known to affect this drug's metabolism (GSI catalog), but was not analyzed by PharmCAT in this report.

Other Medications That May Further Influence This Result

Your **CYP2C9** result shows **Intermediate Metabolizer** status and may influence how your body handles **Phenytoin**. The following are examples of FDA-listed medications or substances that affect **CYP2C9** and may further modify drug exposure or response if taken together:

Amiodarone

Fluconazole

Miconazole

Piperine

If you are currently taking any of the above, mention this to your prescriber. This is general reference information based on your genetic result — it is not a personalized check of your current medications.

About this medication

Phenytoin (Dilantin, Phenytek) is a Nervous System anticonvulsant that stabilizes neuronal membranes by blocking voltage-gated sodium channels. It is used for seizure disorders and has a narrow therapeutic index requiring monitoring.

What it means for you

Your CYP2C9 Intermediate Metabolizer status (Activity Score 1.5, carrying one reduced-function CYP2C9 allele) means you process Phenytoin at a reduced rate, resulting in mildly elevated drug levels compared to normal metabolizers. This may enhance therapeutic effects but also increases adverse effect risk. Your prescriber may recommend dose adjustments or additional monitoring. Always consult your healthcare provider before starting, stopping, or changing your medication.

CPIC

CYP2C9 (Intermediate Metabolizer (AS:1.5))

Patients who are phenytoin-naïve

Recommendation: No adjustments needed from typical dosing strategies. Subsequent doses should be adjusted according to therapeutic drug monitoring, response and side effects. An HLA-B*15:02 negative test does not eliminate the risk of phenytoin-induced SJS/TEN and patients should be carefully monitored according to a usual standard.

Implications: CYP2C9: Slightly reduced phenytoin metabolism; however, this does not appear to translate into increased side effects. HLA-B: n/a Patients who have been continuously taking phenytoin or fosphenytoin for longer than 3 months without developing cutaneous reactions

Recommendation: No adjustments needed from typical dosing strategies. Subsequent doses should be adjusted according to therapeutic drug monitoring, response and side effects. An HLA-B*15:02 negative test does not eliminate the risk of phenytoin-induced SJS/TEN and patients should be carefully monitored according to a usual standard.

Implications: CYP2C9: Slightly reduced phenytoin metabolism; however, this does not appear to translate into increased side effects. HLA-B: n/a

CPIC:<https://cpicpgx.org/guidelines/guideline-for-phenytoin-and-cyp2c9-and-hla-b/> | **Citations:**<https://pubmed.ncbi.nlm.nih.gov/25099164/>;<https://pubmed.ncbi.nlm.nih.gov/32779747/>

NERVOUS SYSTEM DRUGS

Sertraline

Genes Analyzed

Gene	Diplotype	Phenotype	Status
CYP2B6	*1/*1	Normal Metabolizer	✓ Analyzed
CYP2C19	*1/*2	Intermediate Metabolizer	✓ Analyzed

Other Associated Genes:

CYP2D6 is known to affect this drug's metabolism (GSI catalog), but was not analyzed by PharmCAT in this report.

Other Medications That May Further Influence This Result

Your **CYP2C19** result shows **Intermediate Metabolizer** status and may influence how your body handles **Sertraline**. The following are examples of FDA-listed medications or substances that affect **CYP2C19** and may further modify drug exposure or response if taken together:

Fluconazole

Fluoxetine

Fluvoxamine

Ticlopidine

If you are currently taking any of the above, mention this to your prescriber. This is general reference information based on your genetic result — it is not a personalized check of your current medications.

About this medication

Sertraline (Zoloft) is a Nervous System SSRI antidepressant that selectively inhibits serotonin reuptake to treat major depressive disorder, OCD, panic disorder, PTSD, social anxiety disorder, and premenstrual dysphoric disorder.

What it means for you

CYP2B6

Your CYP2B6 Normal Metabolizer status indicates your liver metabolizes Sertraline at the typical expected rate, and standard dosing should achieve predictable drug levels. No pharmacogenomic dose adjustment is anticipated based on CYP2B6 status alone. Always consult your healthcare provider before starting, stopping, or changing your medication.

CYP2C19

Your CYP2C19 Intermediate Metabolizer status means you have reduced — but not absent — enzyme activity for metabolizing Sertraline. This typically leads to moderately elevated drug levels compared to normal metabolizers, potentially increasing the drug's effects and side effect risk. Your prescriber may recommend monitoring or a moderate dose adjustment. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

CPIC: B

PharmGKB: 1A

CPIC

CYP2B6 (Normal Metabolizer)

Recommendation: Initiate therapy with recommended starting dose.

Other Considerations: CYP2C19 metabolizer status, drug–drug interactions and other patient characteristics (e.g., age, renal function, liver function) should be also considered.

Implications: CYP2B6: Normal metabolism of sertraline to less active compounds. CYP2C19: n/a

CYP2C19 (Intermediate Metabolizer)

Recommendation: Initiate therapy with recommended starting dose. Consider a slower titration schedule and lower maintenance dose than CYP2C19 normal metabolizers.

Other Considerations: CYP2B6 metabolizer status, drug–drug interactions and other patient characteristics (e.g., age, renal function, liver function) should be also considered.

Implications: CYP2B6: n/a CYP2C19: Reduced metabolism of sertraline to less active compounds when compared to CYP2C19 normal metabolizers.

CPIC:<https://cpicpgx.org/guidelines/cpic-guideline-for-ssri-and-snri-antidepressants/> | **Citations:**<https://pubmed.ncbi.nlm.nih.gov/25974703/>;<https://pubmed.ncbi.nlm.nih.gov/37032427/>

NERVOUS SYSTEM DRUGS

Sevoflurane

Genes Analyzed

Gene	Diplotype	Phenotype	Status
RYR1	Reference/Reference	Uncertain Susceptibility	✓ Analyzed

About this medication

Sevoflurane (Ultane, Sojourn) is a widely used Nervous System halogenated volatile anesthetic for general anesthesia induction and maintenance. It is preferred in pediatric anesthesia for its non-pungent odor and rapid, smooth induction.

What it means for you

Your RYR1 variant falls under Uncertain Susceptibility for malignant hyperthermia triggered by Sevoflurane. While your specific variant has not been definitively proven to trigger MH, the RYR1 gene is the most common genetic cause of this life-threatening reaction, and uncertainty does not equate to safety. Your anesthesia team should be informed of this genetic finding, and a cautious anesthetic plan that avoids triggering agents is prudent. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

CPIC: A

PharmGKB: 1A

FDA: Actionable PGx

CPIC

RYR1 (uncertain susceptibility)

Recommendation: No CACNA1S genotype is available. Based on RYR1 status, clinical findings, family history, further genetic testing and other laboratory data should guide use of halogenated volatile anesthetics or depolarizing muscle relaxants.

Implications: RYR1: These results do not eliminate the chance that this patient is susceptible to malignant hyperthermia (MH). The genetic cause of about half of all MH survivors, with MH susceptibility confirmed by contracture test, remains unknown (PMID 28902675).

CPIC:<https://cpicpgx.org/guidelines/cpic-guideline-for-ryr1-and-cacna1s/> | **Citations:**<https://pubmed.ncbi.nlm.nih.gov/30499100/>

NERVOUS SYSTEM DRUGS

Trimipramine

Genes Analyzed

Gene	Diplotype	Phenotype	Status
CYP2C19	*1/*2	Intermediate Metabolizer	✓ Analyzed
CYP2D6	—	—	ⓘ Not evaluated by platform

Other Medications That May Further Influence This Result

Your **CYP2C19** result shows **Intermediate Metabolizer** status and may influence how your body handles **Trimipramine**. The following are examples of FDA-listed medications or substances that affect **CYP2C19** and may further modify drug exposure or response if taken together:

Fluconazole

Fluoxetine

Fluvoxamine

Ticlopidine

If you are currently taking any of the above, mention this to your prescriber. This is general reference information based on your genetic result — it is not a personalized check of your current medications.

About this medication

Trimipramine (Surmontil) is a Nervous System tricyclic antidepressant with strong sedating properties due to its antihistaminergic and antidopaminergic activity. It is primarily used for depression and anxiety with a prominent insomnia component.

What it means for you

Your CYP2C19 Intermediate Metabolizer status means you have reduced — but not absent — enzyme activity for metabolizing Trimipramine. This typically leads to moderately elevated drug levels compared to normal metabolizers, potentially increasing the drug's effects and side effect risk. Your prescriber may recommend monitoring or a moderate dose adjustment. Always consult your healthcare provider before starting, stopping, or changing your medication.

This gene is not evaluated by consumer DNA genotyping platforms and was therefore not assessed in this analysis. If genotyping of this gene is clinically relevant, a dedicated pharmacogenomic laboratory test may be requested through your healthcare provider.

Clinical Recommendations

CPIC: B

PharmGKB: 1A

FDA: Actionable PGx

CPIC

CYP2C19 (Intermediate Metabolizer)

Recommendation: Initiate therapy with recommended starting dose.

Other Considerations: Patients may receive an initial low dose of a tricyclic, which is then increased over several days to the recommended steady-state dose. The starting dose in this guideline refers to the recommended steady-state dose.

Implications: CYP2C19: Reduced metabolism of tertiary amines compared to normal metabolizers CYP2D6: n/a

CPIC:<https://cpicpgx.org/guidelines/guideline-for-tricyclic-antidepressants-and-cyp2d6-and-cyp2c19/> | Citations:<https://pubmed.ncbi.nlm.nih.gov/23486447/>; <https://pubmed.ncbi.nlm.nih.gov/27997040/>

RESPIRATORY SYSTEM DRUGS

Ivacaftor

Genes Analyzed

Gene	Diplotype	Phenotype	Status
CFTR	Reference/Reference	ivacaftor non-responsive in CF patients	✓ Analyzed

About this medication

Ivacaftor (Kalydeco) is a Respiratory System CFTR potentiator indicated for cystic fibrosis in patients with specific ivacaftor-responsive CFTR mutations. It works by increasing the probability that the CFTR protein channel remains open, improving chloride transport and hydration of airway surfaces.

What it means for you

Your CFTR gene variant is classified as ivacaftor non-responsive, meaning the specific mutation in your CFTR gene does not produce a protein that benefits from ivacaftor-based potentiator therapy. CFTR modulator drugs like ivacaftor require the presence of at least some functional CFTR protein to work; in non-responsive variants, the protein is either absent or so severely misfolded that these drugs cannot improve its function. Other therapeutic approaches and supportive care strategies remain important for managing your condition. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

CPIC: A

PharmGKB: 1A

FDA: Testing Required

CPIC

CFTR (ivacaftor non-responsive in cf patients)

Recommendation: Ivacaftor is not recommended

Implications: CFTR: An individual diagnosed with cystic fibrosis (CF) and negative for a CFTR variant listed in the FDA-approved drug label as being responsive to ivacaftor.

CPIC:<https://cpicpgx.org/guidelines/guideline-for-ivacaftor-and-cftr/> | **Citations:**<https://pubmed.ncbi.nlm.nih.gov/24598717/>

ANTIINFECTIVES FOR SYSTEMIC USE

Atazanavir

Genes Analyzed

Gene	Diplotype	Phenotype	Status
UGT1A1	*1/*1	Normal Metabolizer	✓ Analyzed

Other Associated Genes:

CYP2C19 is known to affect this drug's metabolism (GSI catalog), but was not analyzed by PharmCAT in this report.

About this medication

Atazanavir (Reyataz) is an Antiinfective HIV protease inhibitor that blocks viral maturation by inhibiting HIV-1 aspartyl protease. It is used in combination antiretroviral therapy and requires UGT1A1 pharmacogenomics consideration due to its inhibition of this enzyme.

What it means for you

Your UGT1A1 Normal Metabolizer status indicates your glucuronidation enzyme efficiently processes Atazanavir. Standard dosing guidelines apply without UGT1A1-based dose adjustment. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

CPIC: A

PharmGKB: 1A

CPIC

UGT1A1 (Normal Metabolizer)

Recommendation: There is no need to avoid prescribing of atazanavir based on UGT1A1 genetic test result. Inform the patient that some patients stop atazanavir because of jaundice (yellow eyes and skin), but that this patient's genotype makes this unlikely (less than about a 1 in 20 chance of stopping atazanavir because of jaundice).

Other Considerations: All studies correlating UGT1A1 genotypes with atazanavir adverse events have involved ritonavir boosting. However, concentration-time profiles are equivalent when boosted with either cobicistat or ritonavir (PMID 23532097), and bilirubin-related adverse events including discontinuation of atazanavir occur in a similar percentage of patients prescribed atazanavir with cobicistat or ritonavir (PMID 23532097). Associations between UGT1A1 genotype, bilirubin elevations, and atazanavir/r discontinuation therefore almost certainly translate to atazanavir/cobicistat. "reference" function refers to the UGT1A1 allele to which other alleles are compared.

Implications: UGT1A1: Reference UGT1A1 activity; very low likelihood of bilirubin-related discontinuation of atazanavir.

CPIC:<https://cpicpgx.org/guidelines/guideline-for-atazanavir-and-ugt1a1/> | **Citations:**<https://pubmed.ncbi.nlm.nih.gov/26417955/>

ANTIINFECTIVES FOR SYSTEMIC USE

Dapsone

Genes Analyzed

Gene	Diplotype	Phenotype	Status
G6PD	B (reference)/B (reference)	Normal	✓ Analyzed

About this medication

Dapsone (Aczone, Avlosulfon) is an Antiinfective/Dermatological sulfone drug with antibacterial and anti-inflammatory properties. It is used for leprosy, dermatitis herpetiformis, and various skin conditions, and serves as prophylaxis for *Pneumocystis jirovecii* pneumonia.

What it means for you

Your G6PD Normal status indicates you have sufficient glucose-6-phosphate dehydrogenase enzyme activity in your red blood cells to protect against oxidative stress from drugs like Dapsone. No elevated risk of drug-induced hemolytic anemia is expected based on your G6PD result alone. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

CPIC: A

PharmGKB: 4

FDA: Actionable PGx

CPIC

G6PD (normal)

Recommendation: No reason to avoid based on G6PD status

Implications: G6PD: Low risk of acute hemolytic anemia

CPIC:<https://cpicpgx.org/guidelines/cpic-guideline-for-g6pd/> | **Citations:**<https://pubmed.ncbi.nlm.nih.gov/24787449/>; <https://pubmed.ncbi.nlm.nih.gov/36049896/>

DERMATOLOGICALS

Dapsone

About this medication

Dapsone (Aczone, Avlosulfon) is an Antiinfective/Dermatological sulfone drug with antibacterial and anti-inflammatory properties. It is used for leprosy, dermatitis herpetiformis, and various skin conditions, and serves as prophylaxis for *Pneumocystis jirovecii* pneumonia.

What it means for you

Your G6PD Normal status indicates you have sufficient glucose-6-phosphate dehydrogenase enzyme activity in your red blood cells to protect against oxidative stress from drugs like Dapsone. No elevated risk of drug-induced hemolytic anemia is expected based on your G6PD result alone. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

See the full pharmacogenomic recommendation above under **Antiinfectives For Systemic Use — dapsone**.

Key genetic finding:G6PD — normal

ANTIINFECTIVES FOR SYSTEMIC USE

Efavirenz

Genes Analyzed

Gene	Diplotype	Phenotype	Status
CYP2B6	*1/*1	Normal Metabolizer	✓ Analyzed

About this medication

Efavirenz (Sustiva) is an Antiinfective drug (non-nucleoside reverse transcriptase inhibitor, NNRTI) used in combination antiretroviral therapy for HIV infection. It blocks HIV reverse transcriptase, preventing viral replication, and is known for central nervous system side effects including vivid dreams and dizziness.

What it means for you

Your CYP2B6 Normal Metabolizer status indicates your liver metabolizes Efavirenz at the typical expected rate, and standard dosing should achieve predictable drug levels. No pharmacogenomic dose adjustment is anticipated based on CYP2B6 status alone. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

CPIC: A

PharmGKB: 1A

FDA: Actionable PGx

CPIC

CYP2B6 (Normal Metabolizer)

Recommendation: Initiate efavirenz with standard dosing (600 mg/day)

Other Considerations: The ENCORE study showed that in treatment-naïve patients randomized to initiate efavirenz-based regimens (combined with tenofovir and emtricitabine), 400 mg/day was non-inferior to 600 mg/day regardless of CYP2B6 genotype (PMID 24522178).

Implications: CYP2B6: Normal efavirenz metabolism

CPIC:<https://cpicpgx.org/guidelines/cpic-guideline-for-efavirenz-based-on-cyp2b6-genotype/> | **Citations:**<https://pubmed.ncbi.nlm.nih.gov/31006110/>

ANTIINFECTIVES FOR SYSTEMIC USE

Flucytosine

Genes Analyzed

Gene	Diplotype	Phenotype	Status
DPYD	c.85T>C (*9A)/c.1627A>G (*5)	Normal Metabolizer	✓ Analyzed

About this medication

Flucytosine (Ancobon) is an Antiinfective antifungal agent that is converted intracellularly to 5-fluorouracil, inhibiting fungal RNA and DNA synthesis. It is used in combination with amphotericin B for cryptococcal meningitis and candidiasis.

What it means for you

Your DPYD Normal Metabolizer status indicates you have adequate DPD enzyme activity to break down Flucytosine at the expected rate. Standard dosing protocols apply, though other patient-specific factors such as age, renal function, and performance status also influence fluoropyrimidine tolerability. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

DPWG

DPYD (Normal Metabolizer (AS:2.0))

Recommendation: The guideline does not provide a recommendation for flucytosine in patients with a DPYD activity score of 2.

Implications: The guideline does not provide a description of the impact of a DPYD activity score of 2 on flucytosine.

DERMATOLOGICALS

Flucytosine

About this medication

Flucytosine (Ancobon) is an Antiinfective antifungal agent that is converted intracellularly to 5-fluorouracil, inhibiting fungal RNA and DNA synthesis. It is used in combination with amphotericin B for cryptococcal meningitis and candidiasis.

What it means for you

Your DPYD Normal Metabolizer status indicates you have adequate DPD enzyme activity to break down Flucytosine at the expected rate. Standard dosing protocols apply, though other patient-specific factors such as age, renal function, and performance status also influence fluoropyrimidine tolerability. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

See the full pharmacogenomic recommendation above under **Antiinfectives For Systemic Use — flucytosine**.

Key genetic finding:DPYD — normal metabolizer (AS:2.0)

ANTIINFECTIVES FOR SYSTEMIC USE

Isoniazid

Genes Analyzed

Gene	Diplotype	Phenotype	Status
NAT2	*5/*6	Poor Metabolizer	✓ Analyzed

About this medication

Isoniazid (INH, Nydrazid) is an Antiinfective drug that is a first-line agent for tuberculosis treatment and prophylaxis. It inhibits mycolic acid synthesis in Mycobacterium tuberculosis cell walls and requires activation by the bacterial enzyme KatG.

What it means for you

NAT2 Slow/Poor Acetylators have reduced activity of the N-acetyltransferase 2 enzyme, which is responsible for inactivating Isoniazid and other drugs that undergo acetylation metabolism. As a result, Isoniazid accumulates to higher blood levels and remains active for longer periods, increasing the risk of dose-dependent adverse effects. For isoniazid, slow acetylators are at higher risk of peripheral neuropathy and hepatotoxicity. For hydralazine and procainamide, slow acetylators are at higher risk of drug-induced lupus. Dose adjustment and close monitoring for drug toxicity are recommended. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

PharmGKB: 1B

FDA

NAT2 (Poor Metabolizer)

Recommendation: "May result in higher systemic concentrations and adverse reaction risk." Note: The FDA table refers to the gene "Nonspecific (NAT)" which ClinPGx interprets as NAT2.

Affected Subgroup: Nonspecific (NAT) poor metabolizers

ANTIINFECTIVES FOR SYSTEMIC USE

Nitrofurantoin

Genes Analyzed

Gene	Diplotype	Phenotype	Status
G6PD	B (reference)/B (reference)	Normal	✓ Analyzed

About this medication

Nitrofurantoin (Macrobid, Macrodantin) is an Antiinfective nitrofuran antibiotic used specifically for urinary tract infections. It is reduced by bacterial enzymes to reactive intermediates that damage multiple bacterial targets including DNA, ribosomes, and cell wall synthesis.

What it means for you

Your G6PD Normal status indicates you have sufficient glucose-6-phosphate dehydrogenase enzyme activity in your red blood cells to protect against oxidative stress from drugs like Nitrofurantoin. No elevated risk of drug-induced hemolytic anemia is expected based on your G6PD result alone. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

CPIC: A

PharmGKB: 3

FDA: Actionable PGx

CPIC

G6PD (normal)

Recommendation: No reason to avoid based on G6PD status

Implications: G6PD: Low risk of acute hemolytic anemia

CPIC:<https://cpicpgx.org/guidelines/cpic-guideline-for-g6pd/>

ANTIINFECTIVES FOR SYSTEMIC USE

Sulfamethoxazole / trimethoprim

Genes Analyzed

Gene	Diplotype	Phenotype	Status
NAT2	*5/*6	Poor Metabolizer	✓ Analyzed

Other Associated Genes:

G6PD is known to affect this drug's metabolism (GSI catalog), but was not analyzed by PharmCAT in this report.

About this medication

Sulfamethoxazole/trimethoprim (Bactrim, Septra) is an Antiinfective sulfonamide/antifolate combination that sequentially blocks two steps in folate synthesis essential for bacterial growth. It is used for urinary tract infections, Pneumocystis jirovecii pneumonia, and many other bacterial infections.

What it means for you

NAT2 Slow/Poor Acetylators have reduced activity of the N-acetyltransferase 2 enzyme, which is responsible for inactivating Sulfamethoxazole / Trimethoprim and other drugs that undergo acetylation metabolism. As a result, Sulfamethoxazole / Trimethoprim accumulates to higher blood levels and remains active for longer periods, increasing the risk of dose-dependent adverse effects. For isoniazid, slow acetylators are at higher risk of peripheral neuropathy and hepatotoxicity. For hydralazine and procainamide, slow acetylators are at higher risk of drug-induced lupus. Dose adjustment and close monitoring for drug toxicity are recommended. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

PharmGKB: 3

FDA: Actionable PGx

FDA

NAT2 (Poor Metabolizer)

Recommendation: "May result in higher adverse reaction risk." Note: The FDA table refers to the gene "Nonspecific (NAT)" which ClinPGx interprets as NAT2.

Affected Subgroup: Nonspecific (NAT) poor metabolizers

ANTIINFECTIVES FOR SYSTEMIC USE

Voriconazole

Genes Analyzed

Gene	Diplotype	Phenotype	Status
CYP2C19	*1/*2	Intermediate Metabolizer	✓ Analyzed

Other Medications That May Further Influence This Result

Your **CYP2C19** result shows **Intermediate Metabolizer** status and may influence how your body handles **Voriconazole**. The following are examples of FDA-listed medications or substances that affect **CYP2C19** and may further modify drug exposure or response if taken together:

Fluconazole

Fluoxetine

Fluvoxamine

Ticlopidine

If you are currently taking any of the above, mention this to your prescriber. This is general reference information based on your genetic result — it is not a personalized check of your current medications.

About this medication

Voriconazole (Vfend) is an Antiinfective broad-spectrum triazole antifungal that inhibits ergosterol synthesis via CYP51 inhibition. It is the drug of choice for invasive aspergillosis and is also used for Candida and other fungal infections. CYP2C19 is a major metabolic pathway.

What it means for you

Your CYP2C19 Intermediate Metabolizer status means you have reduced — but not absent — enzyme activity for metabolizing Voriconazole. This typically leads to moderately elevated drug levels compared to normal metabolizers, potentially increasing the drug's effects and side effect risk. Your prescriber may recommend monitoring or a moderate dose adjustment. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

CPIC: A

PharmGKB: 1A

FDA: Actionable PGx

CPIC

CYP2C19 (Intermediate Metabolizer)

Recommendation: Adult patients

Recommendation: Initiate therapy with recommended standard of care dosing

Other Considerations: Further dose adjustments or selection of alternative therapy may be necessary due to other clinical factors, such as drug interactions, hepatic function, renal function, species, site of infection, therapeutic drug monitoring, and comorbidities.

Implications: CYP2C19: Higher dose-adjusted trough concentrations of voriconazole compared with normal metabolizers

CPIC:<https://cpicpgx.org/guidelines/guideline-for-voriconazole-and-cyp2c19/> | Citations:<https://pubmed.ncbi.nlm.nih.gov/27981572/>

BLOOD AND BLOOD FORMING ORGAN DRUGS

Acenocoumarol

Genes Analyzed

Gene	Diplotype	Phenotype	Status
VKORC1	rs9923231 C/C	-1639 G/G	✓ Analyzed

Other Associated Genes:

CYP2C9 is known to affect this drug's metabolism (GSI catalog), but was not analyzed by PharmCAT in this report.

About this medication

Acenocoumarol (Sintrom) is a Cardiovascular System coumarin-type oral anticoagulant with a mechanism similar to warfarin, inhibiting vitamin K-dependent clotting factor synthesis. It is used for thromboembolism prevention and treatment, and is more commonly used in Europe.

What it means for you

VKORC1 encodes vitamin K epoxide reductase, the enzyme that recycles vitamin K — the essential cofactor for producing blood clotting factors II, VII, IX, and X. VKORC1 is the direct molecular target of warfarin and related anticoagulants. With your rs9923231 reference (C)/rs9923231 reference (C) genotype, you carry the reference genotype (C/C) for rs9923231, which is associated with normal VKORC1 expression. Standard dose considerations apply for Acenocoumarol with respect to this gene. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

DPWG

VKORC1 (rs9923231 reference (c)/rs9923231 reference (c))

Recommendation: The guideline does not provide a recommendation for acenocoumarol in patients with the VKORC1 rs9923231 CC genotype (-1639 GG genotype).

Implications: The guideline does not provide a description of the impact of the -1639 GG genotype on acenocoumarol.

Submitted Genotype: VKORC1: rs9923231 reference (C)/rs9923231 reference (C)

Citations: <https://pubmed.ncbi.nlm.nih.gov/21412232/>

BLOOD AND BLOOD FORMING ORGAN DRUGS

Avatrombopag

Genes Analyzed

Gene	Diplotype	Phenotype	Status
CYP2C9	*1/*11	Intermediate Metabolizer	✓ Analyzed

Other Associated Genes:

F2 and **F5** are known to affect this drug's metabolism (GSI catalog), but was not analyzed by PharmCAT in this report.

Other Medications That May Further Influence This Result

Your **CYP2C9** result shows **Intermediate Metabolizer** status and may influence how your body handles **Avatrombopag**. The following are examples of FDA-listed medications or substances that affect **CYP2C9** and may further modify drug exposure or response if taken together:

Amiodarone

Fluconazole

Miconazole

Piperine

If you are currently taking any of the above, mention this to your prescriber. This is general reference information based on your genetic result – it is not a personalized check of your current medications.

About this medication

Avatrombopag (Doptelet) is an Alimentary Tract and Metabolism (Hematology) thrombopoietin receptor agonist that stimulates platelet production. It is used for thrombocytopenia in chronic liver disease and immune thrombocytopenia.

What it means for you

Your CYP2C9 Intermediate Metabolizer status (Activity Score 1.5, carrying one reduced-function CYP2C9 allele) means you process Avatrombopag at a reduced rate, resulting in mildly elevated drug levels compared to normal metabolizers. This may enhance therapeutic effects but also increases adverse effect risk. Your prescriber may recommend dose adjustments or additional monitoring. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

FDA

CYP2C9 (Intermediate Metabolizer (AS:1.5))

Recommendation: "Results in higher systemic concentrations."

Affected Subgroup: CYP2C9 intermediate or poor metabolizers

BLOOD AND BLOOD FORMING ORGAN DRUGS

Clopidogrel

PRODRUG

Genes Analyzed

Gene	Diplotype	Phenotype	Status
CYP2C19	*1/*2	Intermediate Metabolizer	✓ Analyzed

Other Medications That May Further Influence This Result

Your **CYP2C19** result shows **Intermediate Metabolizer** status and may influence how your body handles **Clopidogrel**. The following are examples of FDA-listed medications or substances that affect **CYP2C19** and may further modify drug exposure or response if taken together:

Fluconazole

Fluoxetine

Fluvoxamine

Ticlopidine

If you are currently taking any of the above, mention this to your prescriber. This is general reference information based on your genetic result – it is not a personalized check of your current medications.

About this medication

Clopidogrel (Plavix) is a Cardiovascular System thienopyridine antiplatelet prodrug that requires CYP2C19-mediated hepatic activation to form its pharmacologically active thiol metabolite, which then irreversibly binds and blocks the platelet P2Y₁₂ ADP receptor, preventing platelet aggregation and clot formation. Patients who are CYP2C19 poor or intermediate metabolizers convert significantly less prodrug to the active form, substantially reducing the antiplatelet effect. It is used to reduce the risk of myocardial infarction, stroke, and thrombotic cardiovascular events in patients with acute coronary syndrome or recent cardiovascular events.

What it means for you

Clopidogrel requires CYP2C19 to convert it to its active antiplatelet form. As an Intermediate Metabolizer, you produce a reduced amount of the active metabolite, resulting in diminished platelet inhibition compared to normal metabolizers. Clinical data suggest modestly increased cardiovascular risk, particularly in high-risk settings like after stenting. Your cardiologist may consider alternative antiplatelet agents or closer monitoring. Always consult your healthcare provider before starting, stopping, or changing your medication.

CPIC

CYP2C19 (Intermediate Metabolizer)

Patients with cardiovascular indications of acute coronary syndrome (ACS) and/or percutaneous coronary intervention (PCI)

Recommendation: Avoid standard dose (75 mg) clopidogrel if possible. Use prasugrel or ticagrelor at standard dose if no contraindication.

Other Considerations: For cardiovascular indications of acute coronary syndrome (ACS) and/or percutaneous coronary intervention (PCI). ACS and/or PCI includes patients undergoing PCI for an ACS or non-ACS (elective) indication.

Implications: CYP2C19: Reduced clopidogrel active metabolite formation; increased on-treatment platelet reactivity; increased risk for adverse cardiac and cerebrovascular events

Patients with neurovascular indications

Recommendation: Consider an alternative P2Y12 inhibitor at standard dose if clinically indicated and no contraindication.

Other Considerations: For neurovascular indications. Neurovascular disease includes acute ischemic stroke or transient ischemic attack, secondary prevention of stroke, or prevention of thromboembolic events following neurointerventional procedures such as carotid artery stenting and stent-assisted coiling of intracranial aneurysms. Alternative P2Y12 inhibitors not impacted by CYP2C19 genetic variants include ticagrelor and ticlopidine. Prasugrel is contraindicated in patients with a history of stroke or TIA. Given limited outcomes data for genotype-guided anti-platelet therapy for neurovascular indications, selection of therapy should depend on individual patient treatment goals and risks for adverse events.

Implications: CYP2C19: Reduced clopidogrel active metabolite formation; increased on-treatment platelet reactivity; increased risk for adverse cardiac and cerebrovascular events

Patients with non-acute coronary syndrome (non-ACS) and non-percutaneous coronary intervention (non-PCI) cardiovascular indications

Other Considerations: For non-acute coronary syndrome (non-ACS) and non-percutaneous coronary intervention (non-PCI) cardiovascular indications. Non-ACS, non-PCI cardiovascular indications include peripheral arterial disease and stable coronary artery disease following a recent myocardial infarction outside the setting of PCI.

Implications: CYP2C19: Reduced clopidogrel active metabolite formation; increased on-treatment platelet reactivity; increased risk for adverse cardiac and cerebrovascular events

CPIC:<https://cpicpgx.org/guidelines/guideline-for-clopidogrel-and-cyp2c19/> | **Citations:**<https://pubmed.ncbi.nlm.nih.gov/21716271/>;<https://pubmed.ncbi.nlm.nih.gov/23698643/>;<https://pubmed.ncbi.nlm.nih.gov/35034351/>

BLOOD AND BLOOD FORMING ORGAN DRUGS

Phenprocoumon

Genes Analyzed

Gene	Diplotype	Phenotype	Status
VKORC1	rs9923231 C/C	-1639 G/G	✓ Analyzed

Other Associated Genes:

CYP2C9 is known to affect this drug's metabolism (GSI catalog), but was not analyzed by PharmCAT in this report.

About this medication

Phenprocoumon (Marcumar, Falithrom) is a Cardiovascular System long-acting coumarin oral anticoagulant used primarily in Europe for venous thromboembolism treatment and stroke prevention in atrial fibrillation. It has a significantly longer half-life than warfarin.

What it means for you

VKORC1 encodes vitamin K epoxide reductase, the enzyme that recycles vitamin K — the essential cofactor for producing blood clotting factors II, VII, IX, and X. VKORC1 is the direct molecular target of warfarin and related anticoagulants. With your rs9923231 reference (C)/rs9923231 reference (C) genotype, you carry the reference genotype (C/C) for rs9923231, which is associated with normal VKORC1 expression. Standard dose considerations apply for Phenprocoumon with respect to this gene. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

DPWG

VKORC1 (rs9923231 reference (c)/rs9923231 reference (c))

Recommendation: The guideline does not provide a recommendation for phenprocoumon in patients with the VKORC1 rs9923231 CC genotype (-1639 GG genotype).

Implications: The guideline does not provide a description of the impact of the VKORC1 rs9923231 CC genotype (-1639 GG genotype) on phenprocoumon.

Submitted Genotype: VKORC1: rs9923231 reference (C)/rs9923231 reference (C)

Citations: <https://pubmed.ncbi.nlm.nih.gov/21412232/>

BLOOD AND BLOOD FORMING ORGAN DRUGS

Warfarin

Genes Analyzed

Gene	Diplotype	Phenotype	Status
CYP2C9	*1/*11	Intermediate Metabolizer	✓ Analyzed
CYP4F2	*1/*1	*1/*1	✓ Analyzed
VKORC1	rs9923231 C/C	-1639 G/G	✓ Analyzed

Other Associated Genes:

rs12777823 is known to affect this drug's metabolism (GSI catalog), but was not analyzed by PharmCAT in this report.

Other Medications That May Further Influence This Result

Your **CYP2C9** result shows **Intermediate Metabolizer** status and may influence how your body handles **Warfarin**. The following are examples of FDA-listed medications or substances that affect **CYP2C9** and may further modify drug exposure or response if taken together:

Amiodarone

Fluconazole

Miconazole

Piperine

If you are currently taking any of the above, mention this to your prescriber. This is general reference information based on your genetic result – it is not a personalized check of your current medications.

About this medication

Warfarin (Coumadin, Jantoven) is a Cardiovascular System anticoagulant ('blood thinner') that works by inhibiting vitamin K epoxide reductase, blocking the synthesis of vitamin K-dependent clotting factors (II, VII, IX, X). It is used to prevent and treat deep vein thrombosis, pulmonary embolism, and stroke in patients with atrial fibrillation or mechanical heart valves.

What it means for you

CYP2C9

Your CYP2C9 Intermediate Metabolizer status (Activity Score 1.5, carrying one reduced-function CYP2C9 allele) means you clear Warfarin at a reduced rate, leading to mildly higher drug exposure and anticoagulant effect than seen in normal metabolizers. Standard dosing guidance applies, though careful INR monitoring during initiation is advisable. Always consult your healthcare provider before starting, stopping, or changing your medication.

CYP4F2

CYP4F2 is a liver enzyme that metabolizes vitamin K1 (phylloquinone), the essential cofactor for the production of clotting factors. Genetic variations in CYP4F2 affect how quickly your body clears vitamin K, which in turn influences the dose of vitamin K antagonist anticoagulants needed to achieve the target anticoagulation level. With your *1/*1 genotype, you carry the common *1/*1 genotype, which is associated with normal vitamin K1 metabolism. Standard dose calculations for Warfarin apply with respect to this gene. Always consult your healthcare provider before starting, stopping, or changing your medication.

VKORC1

VKORC1 encodes vitamin K epoxide reductase, the enzyme that recycles vitamin K – the essential cofactor for producing blood clotting factors II, VII, IX, and X. VKORC1 is the direct molecular target of warfarin and related anticoagulants. With your rs9923231 reference (C)/rs9923231 reference (C) genotype, you carry the reference genotype (C/C) for rs9923231, which is associated with normal VKORC1 expression. Standard dose considerations apply for Warfarin with respect to this gene. Always consult your healthcare provider before starting, stopping, or changing your medication.

CPIC

CYP2C9 (Intermediate Metabolizer (AS:1.5))

Recommendation: CPIC Guideline for Pharmacogenetics–Guided Warfarin Dosing: 2017 Update [Article: 28198005] recommends that pharmacogenetic warfarin dosing be accomplished through the use of one of the pharmacogenetic dosing algorithms, as summarized in Figure 2. See CPIC guideline annotation for details.

VKORC1 (rs9923231 reference (c)/rs9923231 reference (c))

Recommendation: CPIC Guideline for Pharmacogenetics–Guided Warfarin Dosing: 2017 Update [Article: 28198005] recommends that pharmacogenetic warfarin dosing be accomplished through the use of one of the pharmacogenetic dosing algorithms, as summarized in Figure 2. See CPIC guideline annotation for details.

Submitted Genotype: VKORC1: rs9923231 reference (C)/rs9923231 reference (C)

Warfarin Dosing Algorithm

The CPIC guidelines reference a specific dosing algorithm to calculate Warfarin dosage based on CYP2C9 and VKORC1 variants, as well as patient age, weight, and clinical factors. Doctors can access the validated, interactive dosing calculator directly at [WarfarinDosing.org](https://www.warfarindosing.org).

CPIC:<https://cpicpgx.org/guidelines/guideline-for-warfarin-and-cyp2c9-and-vkorc1/> | **Citations:**<https://pubmed.ncbi.nlm.nih.gov/21900891/>;<https://pubmed.ncbi.nlm.nih.gov/28198005/>

DERMATOLOGICALS

Tacrolimus

Genes Analyzed

Gene	Diplotype	Phenotype	Status
CYP3A5	*1/*3	Intermediate Metabolizer	✓ Analyzed

Other Associated Genes:

CYP3A4 is known to affect this drug's metabolism (GSI catalog), but was not analyzed by PharmCAT in this report.

Other Medications That May Further Influence This Result

Your **CYP3A5** result shows **Intermediate Metabolizer** status and may influence how your body handles **Tacrolimus**. The following are examples of FDA-listed medications or substances that affect **CYP3A** and may further modify drug exposure or response if taken together:

Adagrasib

Ceritinib

Clarithromycin

Cobicistat

If you are currently taking any of the above, mention this to your prescriber. This is general reference information based on your genetic result – it is not a personalized check of your current medications.

About this medication

Tacrolimus (Prograf, Advagraf, Astagraf) is an Immunosuppressant (Antiinfective/Systemic) calcineurin inhibitor that blocks T-cell activation by inhibiting calcineurin-mediated transcription of interleukin-2. It is used to prevent organ rejection after transplantation and for atopic dermatitis.

What it means for you

As a CYP3A5 Intermediate Metabolizer, you express one functional CYP3A5 allele, resulting in intermediate enzyme activity. Tacrolimus clearance is moderate, and dosing requirements tend to fall between those of poor and normal metabolizers. Regular trough level monitoring is essential for optimal immunosuppression. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

CPIC: A

PharmGKB: 1A

CPIC

CYP3A5 (Intermediate Metabolizer)

Recommendation: Increase starting dose 1.5 to 2 times recommended starting dose. Total starting dose should not exceed 0.3 mg/kg/day. Use therapeutic drug monitoring to guide dose adjustments.

Other Considerations: This recommendation includes the use of tacrolimus in kidney, heart, lung and hematopoietic stem cell transplant patients, and liver transplant patients where the donor and recipient genotypes are identical. Further dose adjustments or selection of alternative therapy may be necessary due to other clinical factors (e.g., medication interactions, or hepatic function). Typically with other CYP enzymes, a normal metabolizer would be classified as having normal metabolism, and therefore, the drug dose would not change based on the patient's genotype. However, in the case of CYP3A5 and tacrolimus, a CYP3A5 expresser (i.e., CYP3A5 normal metabolizer or intermediate metabolizer) would require a higher recommended starting dose and the CYP3A5 non-expresser (i.e., poor metabolizer) would require the standard recommended starting dose.

Implications: CYP3A5: Lower dose-adjusted trough concentrations of tacrolimus and decreased chance of achieving target tacrolimus concentrations.

ANTINEOPLASTIC AND IMMUNOMODULATING AGENTS

Tacrolimus

About this medication

Tacrolimus (Prograf, Advagraf, Astagraf) is an Immunosuppressant (Antiinfective/Systemic) calcineurin inhibitor that blocks T-cell activation by inhibiting calcineurin-mediated transcription of interleukin-2. It is used to prevent organ rejection after transplantation and for atopic dermatitis.

What it means for you

Your CYP3A4 Normal Metabolizer status means Tacrolimus is metabolized at the standard expected rate, and standard dosing guidelines apply. Always consult your healthcare provider before starting, stopping, or changing your medication.

As a CYP3A5 Intermediate Metabolizer, you express one functional CYP3A5 allele, resulting in intermediate enzyme activity. Tacrolimus clearance is moderate, and dosing requirements tend to fall between those of poor and normal metabolizers. Regular trough level monitoring is essential for optimal immunosuppression. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

See the full pharmacogenomic recommendation above under **Dermatologicals — tacrolimus**.

Key genetic finding: CYP3A4 — normal metabolizer; CYP3A5 — intermediate metabolizer

ANTINEOPLASTIC AND IMMUNOMODULATING AGENTS

Azathioprine

PRODRUG

Genes Analyzed

Gene	Diplotype	Phenotype	Status
NUDT15	*1/*1	Normal Metabolizer	✓ Analyzed
TPMT	*1/*1	Normal Metabolizer	✓ Analyzed

About this medication

Azathioprine (Imuran, Azasan) is an Immunosuppressant prodrug converted to 6-mercaptopurine (6-MP) that inhibits purine synthesis, suppressing T and B lymphocyte proliferation. It is used for autoimmune diseases (rheumatoid arthritis, Crohn's disease, lupus) and transplant rejection prevention.

What it means for you

NUDT15

Your NUDT15 Normal Metabolizer status indicates your enzyme efficiently inactivates the toxic thiopurine metabolites generated from Azathioprine. You are not at elevated NUDT15-mediated risk for thiopurine myelosuppression. TPMT pharmacogenomics should also be considered, as both genes independently affect thiopurine toxicity risk. Always consult your healthcare provider before starting, stopping, or changing your medication.

TPMT

Your TPMT Normal Metabolizer status means your thiopurine methyltransferase enzyme efficiently inactivates Azathioprine's toxic metabolites. You are not at elevated TPMT-driven risk for thiopurine myelotoxicity. Note that NUDT15 pharmacogenomics should also be checked, as both genes independently contribute to thiopurine toxicity risk, particularly in Asian populations. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

CPIC: A

PharmGKB: 1A

FDA: Testing Recommended

CPIC

NUDT15 (Normal Metabolizer)

Recommendation: Based on NUDT15, initiate therapy with standard starting dose (e.g., 2 mg/kg/day for autoimmune diseases). During therapy, adjust doses of azathioprine based on disease-specific guidelines. It usually takes at least 2 weeks to reach steady state after each dose adjustment.

Other Considerations: Standard starting doses may vary by patient characteristics and/or treatment regimens and according to local clinical practice.

Implications: Normal risk of thiopurine-related leukopenia, neutropenia and myelosuppression.

CPIC:<https://cpicpgx.org/guidelines/guideline-for-thiopurines-and-tpmt/> | **Citations:**<https://pubmed.ncbi.nlm.nih.gov/21270794/>;<https://pubmed.ncbi.nlm.nih.gov/23422873/>;<https://pubmed.ncbi.nlm.nih.gov/30447069/>;<https://pubmed.ncbi.nlm.nih.gov/41618934/>

ANTINEOPLASTIC AND IMMUNOMODULATING AGENTS

Capecitabine

PRODRUG

Genes Analyzed

Gene	Diplotype	Phenotype	Status
DPYD	c.85T>C (*9A)/c.1627A>G (*5)	Normal Metabolizer	✓ Analyzed

About this medication

Capecitabine (Xeloda) is an Antineoplastic oral prodrug that is enzymatically converted to 5-fluorouracil (5-FU) preferentially in tumor tissue. It is used for breast cancer, colorectal cancer, and gastric cancer.

What it means for you

Your DPYD Normal Metabolizer status indicates you have adequate DPD enzyme activity to break down Capecitabine at the expected rate. Standard dosing protocols apply, though other patient-specific factors such as age, renal function, and performance status also influence fluoropyrimidine tolerability. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

CPIC: A

PharmGKB: 1A

FDA: Testing Recommended

CPIC

DPYD (Normal Metabolizer (AS:2.0))

Recommendation: Based on genotype, there is no indication to change dose or therapy. Use label-recommended dosage and administration.

Implications: DPYD: Normal DPD activity and "normal" risk for fluoropyrimidine toxicity

CPIC:<https://cpicpgx.org/guidelines/guideline-for-fluoropyrimidines-and-dpyd/> | **Citations:**<https://pubmed.ncbi.nlm.nih.gov/23988873/>;<https://pubmed.ncbi.nlm.nih.gov/29152729/>

ANTINEOPLASTIC AND IMMUNOMODULATING AGENTS

Fluorouracil

Genes Analyzed

Gene	Diplotype	Phenotype	Status
DPYD	c.85T>C (*9A)/c.1627A>G (*5)	Normal Metabolizer	✓ Analyzed

About this medication

Fluorouracil (5-FU, Aducril, Efudex) is an Antineoplastic fluoropyrimidine drug that inhibits thymidylate synthase, blocking DNA synthesis. It is used for colorectal, breast, head and neck, and skin cancers.

What it means for you

Your DPYD Normal Metabolizer status indicates you have adequate DPD enzyme activity to break down Fluorouracil at the expected rate. Standard dosing protocols apply, though other patient-specific factors such as age, renal function, and performance status also influence fluoropyrimidine tolerability. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

CPIC: A

PharmGKB: 1A

FDA: Actionable PGx

CPIC

DPYD (Normal Metabolizer (AS:2.0))

Recommendation: Based on genotype, there is no indication to change dose or therapy. Use label-recommended dosage and administration.

Implications: DPYD: Normal DPD activity and "normal" risk for fluoropyrimidine toxicity

CPIC:<https://cpicpgx.org/guidelines/guideline-for-fluoropyrimidines-and-dpyd/> | **Citations:**<https://pubmed.ncbi.nlm.nih.gov/23988873/>;<https://pubmed.ncbi.nlm.nih.gov/29152729/>

ANTINEOPLASTIC AND IMMUNOMODULATING AGENTS

Mercaptopurine

PRODRUG

Genes Analyzed

Gene	Diplotype	Phenotype	Status
NUDT15	*1/*1	Normal Metabolizer	✓ Analyzed
TPMT	*1/*1	Normal Metabolizer	✓ Analyzed

About this medication

Mercaptopurine (Purinethol, Purixan) is an Antineoplastic/Immunosuppressant thiopurine that acts as a prodrug: it is converted intracellularly through a multi-step enzymatic process to cytotoxic thioguanine nucleotides (TGNs) — the pharmacologically active metabolites responsible for its therapeutic effect. TGNs are incorporated into DNA and RNA, inhibiting purine synthesis and disrupting DNA replication in rapidly dividing cells. TPMT and NUDT15 are key enzymes that influence thioguanine nucleotide accumulation; reduced activity of either enzyme leads to dangerously elevated TGN levels and risk of severe myelosuppression. Mercaptopurine is used for acute lymphoblastic leukemia (ALL) and inflammatory bowel disease.

What it means for you

NUDT15

Your NUDT15 Normal Metabolizer status indicates your enzyme efficiently inactivates the toxic thiopurine metabolites generated from Mercaptopurine. You are not at elevated NUDT15-mediated risk for thiopurine myelosuppression. TPMT pharmacogenomics should also be considered, as both genes independently affect thiopurine toxicity risk. Always consult your healthcare provider before starting, stopping, or changing your medication.

TPMT

Your TPMT Normal Metabolizer status means your thiopurine methyltransferase enzyme efficiently inactivates Mercaptopurine's toxic metabolites. You are not at elevated TPMT-driven risk for thiopurine myelotoxicity. Note that NUDT15 pharmacogenomics should also be checked, as both genes independently contribute to thiopurine toxicity risk, particularly in Asian populations. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

CPIC: A

PharmGKB: 1A

FDA: Testing Recommended

CPIC

NUDT15 (Normal Metabolizer)

Recommendation: Based on NUDT15, initiate therapy with standard starting dose of mercaptopurine (e.g., 75 mg/m²/day for malignancy or 1.5 mg/kg/day for nonmalignancy). During therapy, adjust the doses of myelosuppressive agents, as per standard clinical practice. It usually takes at least 2 weeks of stable dosing to reach steady state after each dose adjustment.

Other Considerations: Standard starting doses may vary by patient characteristics and/or treatment regimens and according to local clinical practice.

Implications: Normal risk of thiopurine-related leukopenia, neutropenia and myelosuppression.

TPMT (Normal Metabolizer)

Recommendation: Based on TPMT, initiate therapy with standard starting dose of mercaptopurine (e.g., 75 mg/m²/day for malignancy or 1.5 mg/kg/day for nonmalignancy). During therapy, adjust the doses of myelosuppressive agents, as per standard clinical practice. It usually takes at least 2 weeks of stable dosing to reach steady state after each dose adjustment.

Other Considerations: Standard starting doses may vary by patient characteristics and/or treatment regimens and according to local clinical practice.

Implications: Normal risk of thiopurine-related leukopenia, neutropenia and myelosuppression. TPMT NMs have lower erythrocyte concentrations of TGN metabolites and higher concentrations of MeMPNs compared to TPMT IMs and TPMT PMs. This is the 'normal' pattern.

CPIC:<https://cpicpgx.org/guidelines/guideline-for-thiopurines-and-tpmt/> | **Citations:**<https://pubmed.ncbi.nlm.nih.gov/21270794/>;<https://pubmed.ncbi.nlm.nih.gov/23422873/>;<https://pubmed.ncbi.nlm.nih.gov/30447069/>;<https://pubmed.ncbi.nlm.nih.gov/41618934/>

ANTINEOPLASTIC AND IMMUNOMODULATING AGENTS

Siponimod

Genes Analyzed

Gene	Diplotype	Phenotype	Status
CYP2C9	*1/*11	Intermediate Metabolizer	✓ Analyzed

Other Medications That May Further Influence This Result

Your **CYP2C9** result shows **Intermediate Metabolizer** status and may influence how your body handles **Siponimod**. The following are examples of FDA-listed medications or substances that affect **CYP2C9** and may further modify drug exposure or response if taken together:

Amiodarone

Fluconazole

Miconazole

Piperine

If you are currently taking any of the above, mention this to your prescriber. This is general reference information based on your genetic result – it is not a personalized check of your current medications.

About this medication

Siponimod (Mayzent) is a Nervous System sphingosine-1-phosphate (S1P) receptor modulator (S1P1 and S1P5 selective) used for secondary progressive multiple sclerosis. It sequesters lymphocytes in lymph nodes, reducing CNS inflammation.

What it means for you

Your CYP2C9 Intermediate Metabolizer status (Activity Score 1.5, carrying one reduced-function CYP2C9 allele) means you process Siponimod at a reduced rate, resulting in mildly elevated drug levels compared to normal metabolizers. This may enhance therapeutic effects but also increases adverse effect risk. Your prescriber may recommend dose adjustments or additional monitoring. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

PharmGKB: 1A

FDA: Testing Required

DPWG

CYP2C9 (Intermediate Metabolizer (AS:1.5))

Recommendation: Use 50% of the normal maintenance dose. Reconsider the choice and the potential benefit of siponimod if the patient is also using a moderate CYP3A4 inducer, such as modafinil, or a strong CYP3A4 inducer. For the comparable gene variation *1/*3, the moderate or strong CYP3A4 inducer results in a reduction in the exposure of siponimod by 49%, according to a pharmacokinetic model.

Implications: Theoretically, the risk of adverse reactions is increased, as the gene variation results in higher plasma concentrations of siponimod.

FDA

CYP2C9 (Intermediate Metabolizer (AS:1.5))

Recommendation: "Results in higher systemic concentrations. Adjust dosage based on genotype. Do not use in patients with CYP2C9 *3/*3 genotype. Refer to FDA labeling for specific dosing recommendations."

Affected Subgroup: CYP2C9 intermediate or poor metabolizers

ANTINEOPLASTIC AND IMMUNOMODULATING AGENTS

Thioguanine

PRODRUG

Genes Analyzed

Gene	Diplotype	Phenotype	Status
NUDT15	*1/*1	Normal Metabolizer	✓ Analyzed
TPMT	*1/*1	Normal Metabolizer	✓ Analyzed

About this medication

Thioguanine (Tabloid) is an Antineoplastic thiopurine prodrug that requires intracellular conversion to thioguanine nucleotides (TGNs) — the pharmacologically active metabolites — by HPRT (hypoxanthine phosphoribosyltransferase) and subsequent enzymatic steps. These TGNs are incorporated into DNA and RNA, causing strand breaks, mis-pair mutations, and cell death through disruption of nucleic acid function. TPMT and NUDT15 are critical enzymes governing TGN accumulation; variants that reduce their activity lead to excessive TGN build-up and life-threatening myelosuppression. Thioguanine is used in the treatment of acute leukemias.

What it means for you

NUDT15

Your NUDT15 Normal Metabolizer status indicates your enzyme efficiently inactivates the toxic thiopurine metabolites generated from Thioguanine. You are not at elevated NUDT15-mediated risk for thiopurine myelosuppression. TPMT pharmacogenomics should also be considered, as both genes independently affect thiopurine toxicity risk. Always consult your healthcare provider before starting, stopping, or changing your medication.

TPMT

Your TPMT Normal Metabolizer status means your thiopurine methyltransferase enzyme efficiently inactivates Thioguanine's toxic metabolites. You are not at elevated TPMT-driven risk for thiopurine myelotoxicity. Note that NUDT15 pharmacogenomics should also be checked, as both genes independently contribute to thiopurine toxicity risk, particularly in Asian populations. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

CPIC: A

PharmGKB: 3

FDA: Testing Recommended

CPIC

NUDT15 (Normal Metabolizer)

Recommendation: Based on NUDT15, initiate therapy with standard starting dose of thioguanine (e.g., 40 mg/m²/day for malignancy). During therapy, adjust the doses of myelosuppressive agents, as per standard clinical practice. It usually takes at least 2 weeks of stable dosing to reach steady state after each dose adjustment.

Other Considerations: Standard starting doses may vary by patient characteristics and/or treatment regimens and according to local clinical practice.

Implications: Normal risk of thiopurine-related leukopenia, neutropenia and myelosuppression.

TPMT (Normal Metabolizer)

Recommendation: Based on TPMT, initiate therapy with standard starting dose of thioguanine (e.g., 40 mg/m²/day for malignancy). During therapy, adjust the doses of myelosuppressive agents, as per standard clinical practice. It usually takes at least 2 weeks of stable dosing to reach steady state after each dose adjustment.

Other Considerations: Standard starting doses may vary by patient characteristics and/or treatment regimens and according to local clinical practice.

Implications: Normal risk of thiopurine-related leukopenia, neutropenia and myelosuppression. TPMT NMs have lower erythrocyte concentrations of TGN metabolites compared to TPMT IMs and TPMT PMs. This is the 'normal' pattern.

CPIC:<https://cpicpgx.org/guidelines/guideline-for-thiopurines-and-tpmt/> | **Citations:**<https://pubmed.ncbi.nlm.nih.gov/21270794/>;<https://pubmed.ncbi.nlm.nih.gov/23422873/>;<https://pubmed.ncbi.nlm.nih.gov/30447069/>;<https://pubmed.ncbi.nlm.nih.gov/41618934/>

VARIOUS DRUG CLASSES IN ATC

Rasburicase

Genes Analyzed

Gene	Diplotype	Phenotype	Status
G6PD	B (reference)/B (reference)	Normal	✓ Analyzed

About this medication

Rasburicase (Elitek) is a recombinant uricase enzyme used intravenously to rapidly reduce uric acid in patients with hematologic malignancies receiving chemotherapy. G6PD screening is mandatory because hydrogen peroxide produced during the reaction can cause severe hemolysis.

What it means for you

Your G6PD Normal status indicates you have sufficient glucose-6-phosphate dehydrogenase enzyme activity in your red blood cells to protect against oxidative stress from drugs like Rasburicase. No elevated risk of drug-induced hemolytic anemia is expected based on your G6PD result alone. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

CPIC: A

PharmGKB: 1A

FDA: Testing Required

CPIC

G6PD (normal)

Recommendation: No reason to avoid based on G6PD status

Implications: G6PD: Low risk of acute hemolytic anemia

CPIC:<https://cpicpgx.org/guidelines/cpic-guideline-for-g6pd/> | **Citations:**<https://pubmed.ncbi.nlm.nih.gov/24787449/>;
<https://pubmed.ncbi.nlm.nih.gov/36049896/>

ANTIPARASITIC PRODUCTS, INSECTICIDES AND REPELLENTS

Primaquine

Genes Analyzed

Gene	Diplotype	Phenotype	Status
G6PD	B (reference)/B (reference)	Normal	✓ Analyzed

About this medication

Primaquine is an Antiinfective 8-aminoquinoline antimalarial drug that is the only agent capable of eliminating dormant liver-stage Plasmodium vivax and P. ovale parasites (hypnozoites). It requires G6PD testing before use due to hemolysis risk.

What it means for you

Your G6PD Normal status indicates you have sufficient glucose-6-phosphate dehydrogenase enzyme activity in your red blood cells to protect against oxidative stress from drugs like Primaquine. No elevated risk of drug-induced hemolytic anemia is expected based on your G6PD result alone. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

CPIC: A

PharmGKB: 3

FDA: Testing Required

CPIC

G6PD (normal)

Recommendation: No reason to avoid based on G6PD status

Implications: G6PD: Low risk of acute hemolytic anemia

CPIC:<https://cpicpgx.org/guidelines/cpic-guideline-for-g6pd/> | Citations:<https://pubmed.ncbi.nlm.nih.gov/36049896/>

ANTIPARASITIC PRODUCTS, INSECTICIDES AND REPELLENTS

Tafenoquine

Genes Analyzed

Gene	Diplotype	Phenotype	Status
G6PD	B (reference)/B (reference)	Normal	✓ Analyzed

About this medication

Tafenoquine (Krintafel, Arakoda) is an Antiinfective long-acting 8-aminoquinoline antimalarial drug for radical cure of Plasmodium vivax malaria. Like primaquine, G6PD testing is mandatory before use due to hemolytic anemia risk.

What it means for you

Your G6PD Normal status indicates you have sufficient glucose-6-phosphate dehydrogenase enzyme activity in your red blood cells to protect against oxidative stress from drugs like Tafenoquine. No elevated risk of drug-induced hemolytic anemia is expected based on your G6PD result alone. Always consult your healthcare provider before starting, stopping, or changing your medication.

Clinical Recommendations

CPIC: A

FDA: Testing Required

CPIC

G6PD (normal)

Recommendation: No reason to avoid based on G6PD status

Other Considerations: Tafenoquine's safety has been established for a G6PD enzyme activity $\geq 70\%$ of normal. (Inclusion criteria for clinical trials involving tafenoquine included G6PD activity $\geq 70\%$.)

Implications: G6PD: Low risk of acute hemolytic anemia

CPIC:<https://cpicpgx.org/guidelines/cpic-guideline-for-g6pd/> | **Citations:**<https://pubmed.ncbi.nlm.nih.gov/24787449/>;
<https://pubmed.ncbi.nlm.nih.gov/36049896/>

No Guideline Available

The medications listed below are associated with genes analysed in this report. However, current clinical pharmacogenomics guidelines do not yet include dosing recommendations for these specific combinations.

Medication	
Drug Class / Category	
Ascorbic Acid (Vitamin C), Combinations	ALIMENTARY TRACT AND METABOLISM DRUGS
Ascorbic Acid (Vitamin C), Plain	ALIMENTARY TRACT AND METABOLISM DRUGS
Abrocitinib	DERMATOLOGICALS
Articaine / Epinephrine	NERVOUS SYSTEM DRUGS
Belinostat	ANTINEOPLASTIC AND IMMUNOMODULATING AGENTS
Belzutifan	ANTINEOPLASTIC AND IMMUNOMODULATING AGENTS
Bupivacaine	NERVOUS SYSTEM DRUGS
Carisoprodol	MUSCULO-SKELETAL SYSTEM DRUGS
Chloroquine	ANTIPARASITIC PRODUCTS, INSECTICIDES AND REPELLENTS
Chlorpropamide	ALIMENTARY TRACT AND METABOLISM DRUGS
Dabrafenib	ANTINEOPLASTIC AND IMMUNOMODULATING AGENTS
Deuruxolitinib	NO GROUP ASSIGNED
Diazepam	NERVOUS SYSTEM DRUGS
Dolutegravir	ANTIINFECTIVES FOR SYSTEMIC USE
Elagolix	SYSTEMIC HORMONAL PREPARATIONS, EXCL. SEX HORMONES AND INSULINS
Erdafitinib	ANTINEOPLASTIC AND IMMUNOMODULATING AGENTS
Esomeprazole	ALIMENTARY TRACT AND METABOLISM DRUGS
Flibanserin	GENITO URINARY SYSTEM AND SEX HORMONES
Flutamide	ANTINEOPLASTIC AND IMMUNOMODULATING AGENTS
Glimepiride	ALIMENTARY TRACT AND METABOLISM DRUGS
Glipizide	ALIMENTARY TRACT AND METABOLISM DRUGS
Glyburide	ALIMENTARY TRACT AND METABOLISM DRUGS
Hydroxychloroquine	ANTIPARASITIC PRODUCTS, INSECTICIDES AND REPELLENTS
Irinotecan	ANTINEOPLASTIC AND IMMUNOMODULATING AGENTS
Lesinurad	MUSCULO-SKELETAL SYSTEM DRUGS

Lidocaine / Prilocaine	NO GROUP ASSIGNED
Lidocaine And Tetracaine	NERVOUS SYSTEM DRUGS
Mepivacaine	NERVOUS SYSTEM DRUGS
Moviprep	ALIMENTARY TRACT AND METABOLISM DRUGS
Nalidixic Acid	ANTIINFECTIVES FOR SYSTEMIC USE
Nateglinide	ALIMENTARY TRACT AND METABOLISM DRUGS
Nilotinib	ANTINEOPLASTIC AND IMMUNOMODULATING AGENTS
Oxymetazoline And Tetracaine	NERVOUS SYSTEM DRUGS
Pazopanib	ANTINEOPLASTIC AND IMMUNOMODULATING AGENTS
Quetiapine	NERVOUS SYSTEM DRUGS
Rabeprazole	ALIMENTARY TRACT AND METABOLISM DRUGS
Raltegravir	ANTIINFECTIVES FOR SYSTEMIC USE
Ropivacaine	NERVOUS SYSTEM DRUGS
Sacituzumab Govitecan	ANTINEOPLASTIC AND IMMUNOMODULATING AGENTS
Seladelpar	ALIMENTARY TRACT AND METABOLISM DRUGS
Sodium Ascorbate	NO GROUP ASSIGNED
Sodium Nitrite	VARIOUS DRUG CLASSES IN ATC
Tegafur	ANTINEOPLASTIC AND IMMUNOMODULATING AGENTS
Tolazamide	ALIMENTARY TRACT AND METABOLISM DRUGS
Tolbutamide	ALIMENTARY TRACT AND METABOLISM DRUGS

Other Evaluated Medications

These medications were evaluated using your genetic profile but do not have a patient-specific clinical recommendation in this report. To make this section easier to review, medications are grouped below by the gene and genetic result involved in their evaluation.

CYP2D6 — Unknown/Unknown

This gene could not be confidently determined from your uploaded raw DNA file.

Acetaminophen / Caffeine / Dihydrocodeine, Amoxapine, Amphetamine, Aripiprazole, Aripiprazole Lauroxil, Atomoxetine, Brexpiprazole, Carvedilol, Cevimeline, Clozapine, Codeine, Darifenacin, Desipramine, Deutetrabenazine, Dextromethorphan / Quinidine, Dextromethorphan Hydrobromide / Bupropion Hydrochloride, Donepezil, Eliglustat, Fesoterodine, Flecainide, Fluoxetine, Fluoxetine / Olanzapine, Fluvoxamine, Galantamine, Gefitinib, Haloperidol, Hydrocodone, Iloperidone, Lofexidine, Meclizine, Metoclopramide, Metoprolol, Mirabegron, Nebivolol, Nortriptyline, Oliceridine, Ondansetron, Paroxetine, Perphenazine, Pimozide, Pitolisant, Propafenone, Propranolol, Protriptyline, Risperidone, Tamoxifen, Tamsulosin, Tetrabenazine, Thioridazine, Tolterodine, Tramadol, Tropisetron, Valbenazine, Venlafaxine, Viloxazine, Vortioxetine, Zuclopenthixol

57 medications affected

G6PD — B (reference)/B (reference) — Normal

Aminosalicylic Acid, Aspirin, Ceftriaxone, Chloramphenicol, Chloroprocaine, Ciprofloxacin, Dimercaprol, Doxorubicin, Erythromycin Ethylsuccinate / Sulfisoxazole Acetyl, Furazolidone, Glliclazide, Mafenide, Mesalazine, Metoclopramide, Moxifloxacin, Nicorandil, Nitrofurazone, Norfloxacin, Ofloxacin, Phenazopyridine, Probenecid, Quinacrine, Quinine, Sodium Nitrate, Succimer, Sulfacetamide, Sulfadiazine, Sulfamethazine, Sulfamethoxazole, Sulfanilamide, Sulfisoxazole, Trametinib, Vitamin C, Vitamin K And Analogues

34 medications affected

CYP2C9 — *1/*11 — Intermediate Metabolizer

Aceclofenac, Aficamten, Aspirin, Diclofenac, Dipyrrone, Etrasimod, Glliclazide, Indomethacin, Lumiracoxib, Nabumetone, Naproxen, Ospemifene, Prasugrel, Rimegepant

14 medications affected

CYP2C19 — *1/*2 — Intermediate Metabolizer

Axitinib, Drospirenone / Ethinyl Estradiol, Fluvoxamine, Formoterol, Lacosamide, Mirtazapine, Moclobemide, Nelfinavir, Prasugrel, Ticagrelor, Vonoprazan

11 medications affected

MT-RNR1 — Unknown

This gene could not be confidently determined from your uploaded raw DNA file.

Amikacin, Dibekacin, Gentamicin, Kanamycin, Neomycin, Netilmicin, Paromomycin, Plazomicin, Ribostamycin, Streptomycin, Tobramycin

11 medications affected

F5 — Unknown

This gene could not be confidently determined from your uploaded raw DNA file.

Eltrombopag, Ethinyl Estradiol / Norelgestromin, Hormonal Contraceptives For Systemic Use, Lusutrombopag, Relugolix / Estradiol / Norethindrone Acetate, Rivaroxaban, Tamoxifen

7 medications affected

UGT1A1 — *1/*1 — Normal Metabolizer

Arformoterol, Axitinib, Binimetinib, Cabotegravir / Rilpivirine, Imlunestrant, Indacaterol

6 medications affected

HLA-B — Unknown/Unknown — Indeterminate

Abacavir, Carbamazepine, Flucloxacillin, Lamotrigine, Oxcarbazepine

5 medications affected

CFTR — Reference/Reference — ivacaftor non-responsive in CF patients

Elexacaftor / Tezacaftor / Ivacaftor, Ivacaftor / Lumacaftor, Ivacaftor / Tezacaftor, Vanzacaftor / Tezacaftor / Deutivacaftor

4 medications affected

CYP2B6 — *1/*1 — Normal Metabolizer

Efavirenz / Emtricitabine / Tenofovir Disoproxil, Methadone, Ospemifene, Prasugrel

4 medications affected

ABCG2 — rs2231142 reference (G)/rs2231142 reference (G) — Normal Function

Resmetirom, Sepiapterin

2 medications affected

CYP3A4 — *1/*1 — Normal Metabolizer

Nelfinavir, Paclitaxel

2 medications affected

CYP3A5 — *1/*3 — Intermediate Metabolizer

Nelfinavir, Prasugrel

2 medications affected

NAT2 — *5/*6 — Poor Metabolizer

Hydralazine / Isosorbide Dinitrate, Isoniazid / Pyrazinamide / Rifampin

2 medications affected

HLA-A — Unknown/Unknown — Indeterminate

Carbamazepine

1 medication affected

SLCO1B1 — *1/*20 — Normal Function

Viloxazine

1 medication affected

TPMT — *1/*1 — Normal Metabolizer

Cisplatin

1 medication affected

Genes Requiring Specialized Testing

CACNA1S

Cannot Be Called

A reliable result for this gene could not be determined from your DNA analysis. This may be due to technical limitations, a complex gene structure, or data quality in this region. Your clinician can advise on whether additional specialized testing is available or needed.

FOR CLINICIANS — Genotyping for this gene was indeterminate from this sample. Review raw variant data and sequencing quality metrics; an alternative targeted assay or clinical genetics consultation may be required for actionable phenotype assignment.

Analysis finding: Missing or unphased variants prevent secure diplotype assignment

AFFECTED MEDICATIONS

MUSCULO-SKELETAL SYSTEM DRUGS

Succinylcholine

NERVOUS SYSTEM DRUGS

Desflurane, Enflurane, Halothane, Isoflurane, Methoxyflurane, Sevoflurane

CYP2D6

Cannot Be Called

CYP2D6 is a highly variable gene whose structure can include extra copies, deletions, or rearrangements that standard DNA tests cannot reliably detect. Because these structural changes directly determine how your body processes many common medications, a specialized test is required for a conclusive result.

FOR CLINICIANS — CYP2D6 structural variation — copy number variation, gene deletion/duplication, and hybrid alleles (e.g., CYP2D6*36+*10, *68+*4) — cannot be reliably resolved from standard short-read WGS without dedicated CNV analysis. Long-read sequencing or a dedicated CYP2D6 genotyping panel is recommended for definitive diplotype and activity score assignment.

Analysis finding: Complex gene structure and copy-number variation prevent reliable diplotype calling

AFFECTED MEDICATIONS

ALIMENTARY TRACT AND METABOLISM DRUGS

Eliquis, Metoclopramide, Ondansetron, Palonosetron, Tropisetron, Vonoprazan

ANTINEOPLASTIC AND IMMUNOMODULATING AGENTS

Gefitinib, Rucaparib, Tamoxifen, Upadacitinib

ANTIPARASITIC PRODUCTS, INSECTICIDES AND REPELLENTS

Quinine

CARDIOVASCULAR SYSTEM DRUGS

Acebutolol, Amiodarone, Atenolol, Betaxolol, Bisoprolol, Carvedilol, Clonidine, Dextromethorphan / Quinidine, Disopyramide, Flecainide, Metoprolol, Nebivolol, Propafenone, Propranolol, Quinidine, Sotalol

DERMATOLOGICALS

Terbinafine

GENITO URINARY SYSTEM AND SEX HORMONES

Darifenacin, Fesoterodine, Flibanserin, Mirabegron, Tamsulosin, Tolterodine

NERVOUS SYSTEM DRUGS

Acetaminophen / Caffeine / Dihydrocodeine, Amitriptyline, Amoxapine, Amphetamine, Aripiprazole, Aripiprazole Lauroxil, Atomoxetine, Brexpiprazole, Bupropion, Cariprazine, Cevimeline, Citalopram, Clomipramine, Clonidine, Clozapine, Codeine, Desipramine, Desvenlafaxine, Deutetrabenazine, Dextromethorphan / Quinidine, Dextromethorphan Hydrobromide / Bupropion Hydrochloride, Donepezil, Doxepin, Duloxetine, Escitalopram, Fluoxetine,

Flupenthixol, Fluphenazine, Fluvoxamine, Galantamine, Haloperidol, Iloperidone, Imipramine, Lofexidine, Methadone, Methylphenidate, Mirtazapine, Modafinil, Nefazodone, Nortriptyline, Olanzapine, Oliceridine, Oxycodone, Paliperidone, Paroxetine, Perphenazine, Pimozide, Pitolisant, Protriptyline, Quetiapine, Risperidone, Sertraline, Tetrabenazine, Thioridazine, Tramadol, Trimipramine, Valbenazine, Venlafaxine, Viloxazine, Vortioxetine, Zuclopenthixol

NO GROUP ASSIGNED

Fluoxetine / Olanzapine, Xanomeline / Trospium

RESPIRATORY SYSTEM DRUGS

Arformoterol, Codeine, Formoterol, Hydrocodone, Meclizine, Tiotropium, Umeclidinium

SENSORY ORGAN DRUGS

Betaxolol, Clonidine

HLA-A

Cannot Be Called

Your HLA-A gene type requires a specialized high-resolution test. HLA genes direct how your immune system reacts to certain medications. Standard DNA analysis cannot provide the level of detail needed — a dedicated HLA typing assay is required for an accurate result.

FOR CLINICIANS — HLA allele-level resolution (e.g., HLA-A*31:01) requires dedicated high-resolution HLA typing. Standard SNP microarray or short-read WGS cannot reliably phase HLA haplotypes to the 4-digit allele level required for CPIC hypersensitivity risk assessment (carbamazepine, oxcarbazepine).

Analysis finding: HLA typing requires a specialized high-resolution assay beyond standard sequencing

AFFECTED MEDICATIONS

ANTINEOPLASTIC AND IMMUNOMODULATING AGENTS

Ipilimumab, Tebentafusp

NERVOUS SYSTEM DRUGS

Carbamazepine

HLA-B

Cannot Be Called

Your HLA-B gene type requires a specialized high-resolution test. Certain HLA-B variants are associated with serious medication reactions (such as severe skin reactions to carbamazepine or abacavir). A dedicated HLA typing assay is needed for a conclusive result.

FOR CLINICIANS — HLA allele-level resolution (e.g., HLA-B*57:01, HLA-B*15:02, HLA-B*58:01) requires dedicated high-resolution HLA typing. Standard SNP microarray or short-read WGS cannot reliably phase HLA haplotypes to 4-digit allele resolution required for CPIC SJS/TEN and hypersensitivity reaction risk assessment.

Analysis finding: HLA typing requires a specialized high-resolution assay beyond standard sequencing

AFFECTED MEDICATIONS

ANTIINFECTIVES FOR SYSTEMIC USE

Abacavir, Flucloxacillin, Ribavirin

ANTINEOPLASTIC AND IMMUNOMODULATING AGENTS

Pazopanib

MUSCULO-SKELETAL SYSTEM DRUGS

Allopurinol

NERVOUS SYSTEM DRUGS

Carbamazepine, Fosphenytoin, Lamotrigine, Oxcarbazepine, Phenytoin

NO GROUP ASSIGNED

Cabotegravir / Rilpivirine

MT-RNR1

Cannot Be Called

This gene is located in your mitochondrial DNA — a small, separate set of genetic instructions inherited from your mother. Analyzing it for medication-relevant variants (particularly aminoglycoside antibiotic sensitivity) requires specialized mtDNA sequencing not included in standard DNA analysis.

FOR CLINICIANS — MT-RNR1 is mitochondrial; heteroplasmy detection for m.1555A>G and related variants requires specialized mtDNA sequencing with sufficient allele-fraction resolution. Standard nuclear DNA pipelines do not reliably quantify mitochondrial variant heteroplasmy for aminoglycoside ototoxicity risk assessment.

Analysis finding: Mitochondrial DNA variant not resolved in this pipeline

AFFECTED MEDICATIONS

ALIMENTARY TRACT AND METABOLISM DRUGS

Kanamycin, Neomycin, Paromomycin, Streptomycin

ANTIINFECTIVES FOR SYSTEMIC USE

Amikacin, Dibekacin, Gentamicin, Kanamycin, Neomycin, Netilmicin, Plazomicin, Ribostamycin, Streptomycin, Tobramycin

BLOOD AND BLOOD FORMING ORGAN DRUGS

Neomycin

DERMATOLOGICALS

Amikacin, Gentamicin, Neomycin

RESPIRATORY SYSTEM DRUGS

Neomycin

SENSORY ORGAN DRUGS

Amikacin, Gentamicin, Kanamycin, Neomycin, Netilmicin, Tobramycin

Clinical Note: The genes listed above could not be reliably genotyped from this sample. Drug-specific recommendations for these medications therefore cannot be personalized based on your DNA. Consult your clinician or a pharmacogenomics specialist for guidance — additional specialized testing (e.g., high-resolution HLA assay, long-read sequencing for CYP2D6, enzyme activity testing for G6PD) may be required.

Genotype Summary

Gene	Diplotype(s)	Phenotype
ABCG2	rs2231142 reference (G)/rs2231142 reference (G)	Normal Function
CACNA1S	Unknown/Unknown	No data available
CFTR	Reference/Reference	ivacaftor non-responsive in CF patients
CYP2B6	*1/*1	Normal Metabolizer
CYP2C19	*1/*2	Intermediate Metabolizer
CYP2C9	*1/*11	Intermediate Metabolizer (Activity Score: 1.5)
CYP2D6	Unknown/Unknown	No data available
CYP3A4	*1/*1	Normal Metabolizer
CYP3A5	*1/*3	Intermediate Metabolizer
CYP4F2	*1/*1	*1/*1
DPYD	c.85T>C (*9A)/c.1627A>G (*5)	Normal Metabolizer (Activity Score: 2.0)
G6PD	B (reference)/B (reference)	Normal
HLA-A	Unknown/Unknown	Indeterminate
HLA-B	Unknown/Unknown	Indeterminate
IFNL3	rs12979860 C/C	C/C
MT-RNR1	Unknown	No data available
NAT2	*5/*6	Poor Metabolizer
NUDT15	*1/*1	Normal Metabolizer
RYR1	Reference/Reference	Uncertain Susceptibility
SLCO1B1	*1/*20	Normal Function
TPMT	*1/*1	Normal Metabolizer
UGT1A1	*1/*1	Normal Metabolizer
VKORC1	rs9923231 C/C	-1639 G/G

Analytical Notes

The genes below produced specific analytical notes during diplotype calling (uncalled variants, ambiguous matches, etc.). These do not change the recommendations above but may be relevant for clinical follow-up.

Gene	Note
CACNA1S	Uncalled variants: Reference; c.520C>T; c.3257G>A
CFTR	Uncalled variants: 711+3A->G; 3272-26A->G; 3849+10kbC->T; A120T; A234D; A349V; A455E; A1067T; D110E; D110H; D192G; D579G; D924N; D1152H; E5
CYP2B6	Uncalled variants: *2; *3; *5; *8; *10; *11; *14; *17; *23; *24; *25; *31; *32; *33; *44; *45; *46; *47; *49

CYP2C19	Uncalled variants: *16; *34
CYP2C9	Uncalled variants: *4; *7; *13; *14; *17; *22; *23; *24; *25; *26; *28; *30; *32; *34; *37; *38; *39; *40; *41; *42; *43; *44; *48; *49; *5
CYP3A4	Uncalled variants: *4; *6; *7; *8; *9; *10; *11; *12; *13; *21; *23; *24; *26; *28; *29; *30; *31; *32; *33; *34; *35; *39; *40; *41; *42;
CYP4F2	Uncalled variants: *2; *5; *6; *7; *8; *9; *10; *11; *12; *13; *15; *17; *18; *19; *20; *21; *23
DPYD	Uncalled variants: c.62G>A; c.295_298delTCAT (*7); c.343A>G; c.525G>A; c.1003G>T (*11); c.1156G>T (*12); c.1278G>T; c.1358C>G; c.1371C>T; c
G6PD	Uncalled variants: Aachen; Abeno; Alhambra; Amazonia; Amiens; Amsterdam; Anadia; Andalus; Arakawa; Asahikawa; Aveiro; Bajo Maumere; Bangkok
NAT2	Unphased NAT2 variants resulted in multiple calls. PharmCAT is picking a single call based on frequency data. Please consult the documentation for details. · Uncalled variants: *41; *43; *45; *48; *51; *71
NUDT15	Uncalled variants: *6; *7; *8; *9; *10; *11; *12; *13; *14; *15; *16; *17; *18; *19; *20; *21
RYR1	Uncalled variants: c.38T>G; c.51_53del; c.97A>G; c.103T>C; c.119G>C; c.130C>T; c.131G>A; c.152C>A; c.178G>A; c.178G>T; c.190T>C; c.212C>A;
SLCO1B1	Uncalled variants: *6; *10; *36; *50; *51; *58
TPMT	Uncalled variants: *6; *7; *8; *10; *12; *16; *17; *18; *19; *20; *21; *22; *23; *25; *26; *27; *28; *29; *30; *32; *33; *34; *35; *36; *37
UGT1A1	Uncalled variants: *27; *28; *36; *37

Note

Genotypes are determined based on the variants in your submitted DNA file, matched against PharmCAT allele definitions. Always consult a qualified healthcare provider before making any medication decisions.

Disclaimer

Purpose and Limitations of This Report

This pharmacogenomics report is provided for informational purposes only. It is intended to be used as a supplementary tool to support, not replace, clinical decision-making by qualified healthcare professionals. It does not constitute medical advice, a diagnosis, or a treatment recommendation.

Not a Substitute for Medical Care

The findings in this report must be interpreted in the full context of your individual health situation by a licensed physician, pharmacist, or other qualified healthcare provider. Genetic information is one factor among many that influences medication safety and efficacy. Other critical factors include your age, weight, kidney and liver function, other medications you take (including over-the-counter drugs and supplements), and underlying medical conditions.

Never start, stop, or change any medication based solely on the results of this report. Always consult your doctor or pharmacist before making any change to your treatment.

Scope of Analysis

This report covers only the gene-drug combinations for which established pharmacogenomic guidelines exist at the time of analysis. It does not capture all possible genetic influences on drug response, and it does not assess genes or variants outside the scope of the submitted data file and analysis pipeline. The absence of a finding does not mean a drug is safe or appropriate for you — it means genetic guidance is not yet available for that specific combination.

Evolving Evidence

Pharmacogenomics is a rapidly developing field. Guidelines and evidence levels change as new research emerges. The recommendations in this report reflect the state of published evidence at the time of analysis. Xcode Life does not guarantee that the information in this report remains current indefinitely, and we recommend periodic review with your healthcare provider, particularly if you are starting a new medication for which PGx guidance has recently been updated.

Analytical Limitations

Genotype calls are derived from the raw DNA file submitted by the user, analyzed using the PharmCAT v3.2.0 pipeline. Accuracy depends on the quality of the submitted data. Certain genes cannot be reliably typed using standard VCF-based analysis and are therefore excluded from clinical recommendations. Results for genes with complex structural variation may include multiple possible diplotype assignments where phasing is uncertain.

No Guarantee of Outcomes

Genetic information provides probabilistic guidance, not certainty. Having a particular metabolizer phenotype does not guarantee that you will or will not experience a specific response or side effect. Individual drug responses are influenced by many non-genetic factors that this report cannot account for.

Privacy

Your genetic data is sensitive personal health information. Xcode Life handles all data in accordance with applicable privacy laws and our privacy policy. This report is intended for your personal use and, where relevant, for sharing with your healthcare providers. Your raw DNA data file is processed solely to generate this report and is securely deleted from our active servers within 24 hours of report generation.

In Summary

This report is a tool to help you and your doctor have a more informed conversation about your medications. Use it alongside professional medical guidance, not instead of it.

